
SmallFloats.jl

0.2.0

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2026-07-30T18:23Z

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Chapter 1

SmallFloats.jl

A conforming, performance-oriented Julia implementation of the IEEE P3109 draft standard — arithmetic formats for machine learning at bitwidths 3–16.

Bit-exact defined results on every default path; one projection engine as the single write path into a code point; approximation only behind an explicitly named, exhaustively measured κ registry; **≈ 35 million verified units across 13 gates and tiers**, each labelled exhaustive or sampled in a roll-call printed at the end of every run; table-gather kernels at fractions of a nanosecond per element.

```
julia> using SmallFloats

julia> Binary8p4se(1.6) + Binary8p4se(0.25)
Binary8p4se(1.875  $\equiv$  0x47)

julia> Binary5p3sf(0x08) == Binary5p3sf(1.0)    # an Unsigned is a CODE POINT
true
```

1.1 Two things to know before you start

Most format names are opt-in. using SmallFloats exports the **120 aliases at $K \leq 8$** . The other 384 arrive with using SmallFloats.Formats, and are reachable without it as SmallFloats.Binary16p6se or format(16, 6, true, true).

Binary16p11se is not Float16 (nor is Binary16p8se a BFloat16). The exponent biases differ by one, so every *code point denotes a different value*. Converting between them is a conversion, never a reinterpretation — the [User Guide](#) has the full comparison.

1.2 Documentation map

- **Introduction** — what the package is, the design pillars, a thirty-second tour, installation.
- **Cheat Sheet** — one-page lookup for format names, conversion, projections, operations, arrays, blocks, packed storage, and common traps.
- **User Guide** — the complete public API in usage order: formats, values, projection specifications, the two operation registers, arrays and sorting, blocks, packed storage, conformance and κ , performance guidance.
- **User Examples** — runnable examples in three tiers: basic, machine learning, deep learning.
- **Julia Compatibility** — the Base register: which Base functions work on Binary values, what they map to, and what deliberately isn't mapped.

- **Technical Guide** — internals: the encoding and projection engine, the oracle's rigor classes, tables and kernels, the block layer's exactness filters, the verification and benchmark doctrines.
- **Technical Examples** — internals-level recipes: pipeline introspection, exhaustive verification of custom code, κ measurement, doctrine-compliant benchmarking.
- **Adding Operations** — how to add new scalar (unary through quaternary) and block (scaled, elementwise, reductive) operations with every guarantee intact: registry mechanics, ω -semantics duties, worked examples.
- **External Reference** — docstrings for the documented public surface.
- **Internal Reference** — docstrings for the documented unexported machinery.

Chapter 2

Introduction

SmallFloats.jl is a conforming, performance-oriented Julia implementation of the **IEEE P3109 draft standard** — *Arithmetic Formats for Machine Learning* — covering every binary format the draft defines at bitwidths 3 through 16, every projection (rounding × saturation) mode including the three stochastic families, the full scalar operation catalog, block-scaled ("MX-style") operations, and the draft's conformance and k-approximation machinery.

2.1 Why this package exists

Small floating-point formats are easy to implement *approximately* and surprisingly hard to implement *exactly*. An FP8 Exp differs from a bit-exact one only in a handful of code points near rounding boundaries — invisible in a demo, decisive in a conformance suite, and quietly corrosive in research that compares number formats. SmallFloats.jl takes the position that for value sets this small there is no excuse for approximation:

- **Bit-exact defined results on every default path.** Every operation's result is the correct projection of the mathematically exact value, established through a rigorous oracle (exact arithmetic, correctly-rounded Float128 where IEEE mandates it, and MPFR directed-rounding enclosures with automatic precision escalation everywhere else).
- **One projection engine.** RoundToPrecision → Saturate → Encode is the *single* write path into a code point. There is no second, "fast but slightly different" rounding routine anywhere in the package.
- **Approximation is opt-in, named, and measured.** Faster-but-inexact implementations live behind an explicit registry ([register_approx!](#)); each registration's deviation bound k is *measured by exhaustive enumeration* at registration time, and understated declarations are rejected. Nothing approximate is reachable from the default API.
- **Enumerated where enumeration is affordable, and labelled where it is not.** The code lattice is swept **exhaustively at every bitwidth** — all 504 formats, 7 602 160 code points — and the projection engine is checked against an independent Rational{BigInt} reference over every realized (P, B) cell. Where a space is too large to enumerate (the ordered-pair cross-product above $K = 8$ is 4.3×10^9 pairs for a single format), the suite says "**sampled**" in those words rather than rounding up. Every run ends with a coverage roll-call naming each gate, what it compared, and which of the two it was: ≈ 35 million verified units across 13 gates and tiers.
- **Fast where it matters.** Pure-mode elementwise work runs through precomputed lookup tables (sub-nanosecond per element); the scalar path is fully specialized and allocation-free (decode ≈ 1.2 ns, project ≈ 6.6 ns, a complete Add ≈ 9 ns including projection); sub-byte packed storage, integer-keyed ordering with an $O(n)$ counting sort, and a mask-based rounding core round out the performance story. A reproducible Chairmarks benchmark suite ships in benchmarking/, and every number here comes from it.

2.2 Thirty-second tour

```
julia> using SmallFloats

julia> x = Binary8p4se(1.6)           # construct = project under the default spec
Binary8p4se(1.625 ≈ 0x45)

julia> x + Binary8p4se(0.25)         # Base operators use the format's default spec
Binary8p4se(1.875 ≈ 0x47)

julia> Add(Binary8p4se, RTP_SN, x, Binary8p4se(0.25))
Binary8p4se(1.875 ≈ 0x47)           # spec-named register: any mode, explicitly

julia> σ = RSA_SN();                 # stochastic rounding, default N = 8 bits

julia> Add(Binary8p4se, σ, Binary8p4se(2.0), Binary8p4se(0.03125); R = 255)
Binary8p4se(2.25 ≈ 0x49)           # explicit draw R makes it reproducible

julia> Exp(Binary8p4se, RNE_SN, Binary8p4se.([-1.0, 0.5, 2.0])) # via a 256-byte table
3-element Vector{Binary8p4se}:
 Binary8p4se(0.375 ≈ 0x34)
 Binary8p4se(1.625 ≈ 0x45)
 Binary8p4se(7.5 ≈ 0x57)
```

2.3 The formats

A format is the type `Binary{K,P,SGN,EXT}`: bitwidth $K \in 3:16$, precision $P \in 1:K$ (with the signedness constraint), signed/unsigned `SGN`, and extended (has $\pm\text{Inf}$) or finite `EXT`. All **504** draft formats carry their draft names as aliases. The **120 at $K \leq 8$ are exported** by using `SmallFloats`; the other 384 are opt-in via `using SmallFloats.Formats`, and are always reachable as `SmallFloats.Binary16p6se` or programmatically as `format(16, 6, true, true)`.

`Binary8p4se` represents `Binary{8,4,true,true}` (" $K = 8$, $P = 4$, signed, extended"); the trailing letters are s/u for signedness and e/f for extended/finite. Note that the two are related by `<:`, not `===` — `Binary` is abstract (a format is "a datum set and an encoding"), and the alias names its concrete *representation*.

Values are immutable wrappers around their code point: one byte at $K \leq 8$, two above it. Every format has a single NaN and no negative zero. `Float64` is the exact interchange carrier for the 432 formats whose exponent range it holds; the remaining 72 use `Float128` or an exact dyadic carrier, selected by the format and never by the caller.

2.4 Where to go next

- **User Guide** — the complete tour of the public API: formats and values, projection specifications, the two operation registers, arrays and sorting, blocks and scaled operations, packed storage, conversion, conformance, and performance guidance.
- **User Examples** — worked, runnable examples in three tiers: basic usage, machine-learning quantization workflows, and deep-learning block/kernel patterns.
- **Technical Guide** — the internals: the encoding and projection engine, the oracle's evaluation protocol and its two rigor classes, tables and kernels, the block layer's exactness filters, and the testing and benchmarking doctrine.
- **Technical Examples** — internals-level recipes: pipeline introspection, verifying custom kernels against the oracle, κ measurement, differential builds, and doctrine-compliant benchmarking.

2.5 Installation

The package is not yet registered, so install it by URL:

```
pkg> add https://github.com/JeffreySarnoff/SmallFloats.jl
```

Append `#main` (or a tag/commit) to pin a revision. For an editable install, `pkg> dev https://github.com/JeffreySarnoff/SmallFloats.jl` clones into `~/.julia/dev`, or `pkg> dev path/to/SmallFloats.jl` uses an existing clone.

Requires Julia 1.12. Quadmath (libquadmath's `Float128`) is a hard dependency used only inside the oracle and exact-fallback paths; on platforms where it misbehaves, `ENV["SmallFloats_Float128"] = "disable"` before loading selects the pure-MPFR configuration with **identical results** (this equivalence is itself part of the test suite).

2.6 Status and conformance

`conformance()` returns the live, serializable conformance declaration (formats, operations, modes, instantiated table specializations, and `κ` registrations); `SmallFloats.draft_revision()` names the implemented draft revision. Interpretations made where the draft text under-determined behavior are marked `[interp]` in the source comments.

Chapter 3

Cheat Sheet

A one-page reference for the SmallFloats.jl operations you are most likely to write. For explanations and semantics, see the [User Guide](#); for implementation details, see the [Technical Guide](#).

3.1 Start here

```
using SmallFloats
using Random: Xoshiro

T = Binary8p4se
ρ = RNE_SN

x = T(1.6)
y = T(0.25)
z = Add(T, ρ, x, y)
```

The explicit operation shape is always:

```
Op(result_format, projection_spec, operands...; rng, R)
```

For same-format operands under the default projection, ordinary Julia syntax is available:

```
z == x + y
exp(x)
fma(x, y, z)
```

3.2 Format names

Binary K p P (s|u) (e|f)

└─ extended (Inf) or finite domain
 └─ signed or unsigned
 └─ significand precision, including the implicit bit
 └─ total bitwidth, 3 through 16

Examples:

Name	Meaning
Binary8p4se	8-bit, precision 4, signed, extended
Binary8p4sf	8-bit, precision 4, signed, finite
Binary6p3ue	6-bit, precision 3, unsigned, extended
Binary5p5uf	5-bit, precision 5, unsigned, finite
Binary16p6se	16-bit, precision 6, signed, extended (opt-in export)
Binary16p1uf	16-bit, precision 1, unsigned, finite — the MX scale shape

The alias is the *representation* of the parametric format, related by `<`: and **not** by `===`:

```
Binary8p4se <: Binary{8, 4, true, true} # true
Binary8p4se === Binary{8, 4, true, true} # FALSE — `Binary` is abstract
format(16, 6, true, true) # the alias, from runtime parameters
```

using `SmallFloats` exports the 120 aliases at $K \leq 8$. For the other 384: using `SmallFloats.Formats`, or `SmallFloats.Binary16p6se`, or `format(...)`.

Useful format queries:

```
bitwidth(T)      # K
precision(T)     # P
issigned(T)
isextended(T)
expbias(T)
expbitwidth(T)
trailingsigbits(T)

MaxFiniteOf(T)
MinFiniteOf(T)
MinPositiveOf(T)
MinNormalOf(T)
MaxSubnormalOf(T)
```

Every query above also takes a **value** instead of a type — `bitwidth(x) ≡ bitwidth(typeof(x))` — and folds to the same constant:

```
x = Binary8p4se(1.6)
bitwidth(x)      # 8
MaxFiniteOf(x)   # Binary8p4se(224.0 ≡ 0x7e)
```

3.3 Values and code points

```
x = T(1.6)           # numeric value: project into T
x = Convert(T, p, 1.6) # explicit projection

c = codepoint(x)      # the code point, in the format's storage unit
x = T(c)              # an Unsigned means validated CODE POINT
x = rawvalue(T, c)    # unchecked code-point construction

d = decode(x)         # the exact datum, on the format's own carrier
```

'Unsigned' means code point — at every width

`T(0x02)` constructs code point `0x02`; `T(2)` projects the numeric value two. Signedness of the integer type is what distinguishes them, so the rule holds for `UInt8`, `UInt16` and any other Unsigned: `Binary16p6se(0x0002)` is code point 2, not the value 2.0. Use `Convert` when intent should be unmistakable.

`codepoint` returns the format's *storage unit* — `UInt8` at $K \leq 8$, `UInt16` above — and that is `codeunit_type(T)`. A code-point argument is range-checked against 2^K whatever its width, so passing a wider Unsigned than necessary is lossless and safe.

`convert` is the deliberate exception and is value-preserving: `Binary8p4se(0x02)` is code point 2, while `convert(Binary8p4se, 0x02)` is the value 2.0. Promotion, similar and fill depend on that.

Random values (uniform [0,1) floor-projected; normal round-to-nearest, tails clamped to `MaxFinite`; `randn` signed formats only):

```
rand(T); rand(T, dims); rand!(A)      # uniform on [0, 1)
randn(T); randn(T, dims); randn!(A)  # standard normal, always finite
rand(Xoshiro{1}, T)                  # explicit rng as usual

rand(rng, T; projection=RTP_SN)      # scalar forms: land under any ProjSpec
randn(rng, T; projection=RTZ_SF)     # (stochastic p draws R from same rng)
```

Classification and stepping:

```
isnan(x); isinf(x); isfinite(x); iszero(x)
signbit(x); issubnormal(x)
Class(x)
NextGreaterThan(x); NextLessThan(x)
nextfloat(x); prevfloat(x)
TotalOrder(x, y)
```

3.4 Projection specifications

A projection specification is (rounding mode, saturation mode):

```
p = ProjSpec(TowardPositive(), SatFinite())
roundingmode(p)
saturationmode(p)
```

Deterministic projections

Rounding	SatFinite	SatPropagate	SatNone
nearest, ties even	RNE_SF	RNE_SP	RNE_SN
nearest, ties away	RNA_SF	RNA_SP	RNA_SN
toward $+\infty$	RTP_SF	RTP_SP	RTP_SN
toward $-\infty$	RTN_SF	RTN_SP	RTN_SN
toward zero	RTZ_SF	RTZ_SP	RTZ_SN
round to odd	RTO_SF	RTO_SP	RTO_SN

RNE_SN is the package-wide default. Saturation modes mean:

Mode	Out-of-range behavior
SatFinite	clamp everything to the finite range
SatPropagate	preserve representable infinities; clamp other overflow
SatNone	apply the draft's domain-, signedness-, and direction-dependent rows

Stochastic projections

Constructors are grouped by stochastic variant:

Variant	SatFinite	SatPropagate	SatNone
A	RSA_SF(N)	RSA_SP(N)	RSA_SN(N)
B	RSB_SF(N)	RSB_SP(N)	RSB_SN(N)
C	RSC_SF(N)	RSC_SP(N)	RSC_SN(N)

N is the random-bit budget, $1 \leq N \leq 60$. Omitting it uses $N = 8$.

```

σ = RSA_SN(8)

Add(T, σ, x, y; rng=Xoshiro(1))  # reproducible stream
Add(T, σ, x, y; R=17)            # exact draw, ideal for tests

isstochastic(σ)                  # true
nrandbits(σ)                    # 8

```

For an N-bit mode, explicit R must be in $0:(2^N - 1)$.

Session defaults

Read with `DefaultX()`, set with `DefaultX!(v)`:

```

DefaultType()           # Binary8p2se      DefaultType!(Binary8p4se)
DefaultReturnType()     # Binary8p2se      DefaultReturnType!(Binary8p3se)
DefaultRoundingMode()   # NearestTiesToEven()
DefaultSaturationMode() # SatNone()
DefaultProjection()      # RNE_SN
DefaultRNG()            # Xoshiro      DefaultRNG!(Xoshiro(42))
DefaultRbits()          # 8           DefaultRbits!(16)

```

Setting a rounding/saturation component rebuilds `DefaultProjection`; setting `DefaultProjection!` directly updates both components. Always: `DefaultProjection() == ProjSpec(DefaultRoundingMode(), DefaultSaturationMode())`.

The convenience methods (`a + b`, `Exp(x)`, `T(2.1)`) **do** consult `DefaultProjection`, so changing it changes their results globally. They read it through a speculation guard: allocation-free while the default holds its initial value, one dispatch per call after you change it. Explicit forms (`Add(T, ρ, x, y)`) are unaffected by the session default.

Consume a default via the combinators — never by computing on a bare `DefaultX()` read. No dispatch while the default is unchanged, one barrier dispatch after a change; zero-alloc when `f`'s result type doesn't depend on the default (a default-typed result boxes once at escape):

```
with_default_type((T, x) -> T(x), 1.5)           # Binary8p2se(1.5 ≡ 0x41)
with_default_projection((ρ, x, y) -> Add(T, ρ, x, y), x, y)
with_default_returntype(f, args...)              # f(DefaultReturnType(), args...)
```

3.5 Scalar operation catalog

```
# explicit result format and projection
Add(T, ρ, x, y)
FMA(T, ρ, x, y, z)
Convert(T, ρ, external_value)

# same-format default-projection convenience
Add(x, y)
FMA(x, y, z)
```

Arity	Operations
Unary	Abs, Negate, Sqrt, RSqrt, Recip, Exp, Exp2, ExpMinusOne, Log, Log2, LogOnePlus, Softplus, Sin, Cos, Tan, ArcSin, ArcCos, ArcTan, Sinh, Cosh, Tanh, ArcSinh, ArcCosh, ArcTanh, SinPi, CosPi, TanPi, ArcSinPi, ArcCosPi, ArcTanPi
Binary	CopySign, Add, Subtract, Multiply, Divide, Hypot, ArcTan2, ArcTan2Pi, Maximum, Minimum, MaximumNumber, MinimumNumber, MaximumMagnitude, MinimumMagnitude, MaximumMagnitudeNumber, MinimumMagnitudeNumber, MinimumFinite, MaximumFinite
Ternary	FMA, FAA, Clamp
Con- ver- sion	Convert

Common Base spellings under RNE_SN:

```
x + y; x - y; x * y; x / y
-x; abs(x); inv(x); sqrt(x)
exp(x); exp2(x); expm1(x)
log(x); log2(x); log1p(x)
sin(x); cos(x); tan(x)
asin(x); acos(x); atan(x); atan(y, x)
sinh(x); cosh(x); tanh(x)
min(x, y); max(x, y); clamp(x, y, z)
fma(x, y, z); muladd(x, y, z)
```

3.6 Arrays

Every registered operation has elementwise array methods:

```
A = T.([1.0, 1.5, 2.0])
B = T.([0.25, 0.5, 0.75])

C = Add(T, ρ, A, B)
E = Exp(T, ρ, A)
F = FMA(T, ρ, A, B, C)

C = vmap(:Add, T, ρ, A, B)
vmap!(C, Val(:Add), T, ρ, A, B)
```

Operands and destination must have matching axes. Deterministic unary/binary operations use cached result tables; affordable ternary signatures may also use tables. Stochastic operations compute each element and consume one draw per projection.

```
table_bytes()
empty_tables!()
```

3.7 Blocks and scaled operations

```

FS = Binary8pluf
FE = Binary8p4se

b = Block(FS(4.0), FE(1.5), FE(-0.75), FE(2.0), FE(0.5))

blocksize(b)
scaleformat(b)
elemformat(b)

ConvertFromBlock(T, ρ, b)
BlockReduceAdd(T, ρ, b)
BlockReduceMultiply(T, ρ, b)
BlockDotProduct(T, ρ, b, b)

```

Every scalar operation except Convert has generated block and scaled forms:

```

BlockAdd(T, ρ, b1, b2, result_scale)
BlockExp(T, ρ, b, result_scale)

ScaledAdd(T, ρ, scale1, x1, scale2, x2)
ScaledExp(T, ρ, scale, x)

```

Quantize against a supplied or automatically selected scale:

```

ConvertToBlock(FS, FE, ρ, values_tuple, scale)
ConvertToBlockMaxAbsFinite(FS, FE, scale_ρ, element_ρ, values_tuple)

```

BlockVector stores many equal-shape blocks in a structure-of-arrays layout.

3.8 Packed storage

```

v = Binary5p2se.([0.5, 1.0, 1.5, 2.0])
pv = PackedVector(v)

pv[2]
pv[2] = Binary5p2se(0.75)
collect(pv)

out = vmap(:Exp, Binary5p2se, ρ, pv)

```

PackedVector stores each code point in exactly `bitwidth(F)` bits. Computation unpacks tiles internally; packed arithmetic is deliberately not in-place.

3.9 Conformance and approximations

```

conformance()
conformance_dict()
conformance_report()
draft_revision()

κ, exhaustive = measure_kappa(fn, :Exp, T, (T,), ρ)
register_approx!(:fast_exp, :Exp, T, (T,), ρ, fn; κ)
impl = approx(:fast_exp)
kappa(impl)
kappa_measured(impl)
list_approx()
unregister_approx!(:fast_exp)

```

Approximate implementations are never substituted into the default API.

3.10 Common traps

Trap	Correct pattern
Treating an Unsigned as a number	$T(\text{Int}(c))$ for a numeric integer; $T(c::\text{Unsigned})$ for a code point — at every width
$\text{Vector}\{\text{Binary}\{K,P,\Sigma,\Delta\}, n\}$	Silently boxes every element — the abstract format is not isbits. Use $\text{Vector}\{\text{Binary8p4se}\}$ or $\text{Vector}\{\text{format}(K,P,\Sigma,\Delta)\}$. similar normalizes; $\text{Vector}\{\dots\}(\text{undef}, n)$ cannot be intercepted
Silently mixing formats	Convert explicitly: $\text{Convert}(T, \rho, x)$
Assuming SatNone always clamps	Choose SatFinite when clamping is required
Expecting IEEE division-by-zero	P3109 semantics here define $x / 0 \rightarrow \text{NaN}$
Expecting negative zero	Every format has one zero; $T(-0.0) === \text{zero}(T)$
Passing a Rational	Convert explicitly to an exact supported carrier or knowingly to Float64
Reproducibility with stochastic rounding	Supply a seeded rng, or an explicit R in tests
Using rawvalue on unchecked input	Prefer validated $T(c::\text{Unsigned})$ outside kernels

3.11 Performance checklist

- Keep format types and projection specs in const bindings, function arguments, or type parameters.
- Put code using runtime-selected formats behind a function barrier.
- Expect the first deterministic array call for a specialization to build a table; benchmark warm calls separately.
- Use PackedVector when storage bandwidth matters more than direct byte access.
- Use `table_bytes()` to inspect cache footprint and `empty_tables!()` to reset it.
- Do not replace explicit projection semantics with `@fastmath`.

For worked applications, continue to [User Examples](#). For correctness and benchmark methodology, continue to [Technical Examples](#).

Chapter 4

User Guide

Everything you need to use SmallFloats.jl, in the order you'll need it. All transcripts in this guide are captured from real sessions against the shipped package.

4.1 Formats and format queries

The parametric type `Binary{K,P,SGN,EXT}` defines a format: bitwidth $K \in 3:16$, precision P (significand bits including the implicit bit), `SGN::Bool` for signed/unsigned, `EXT::Bool` for extended (with $\pm\text{Inf}$) versus finite. Every legal combination — **504 formats** — has a draft-named alias. using `SmallFloats` exports the **120 at $K \leq 8$** ; the rest are one opt-in away:

```
julia> Binary8p4se <: Binary{8,4,true,true}      # `<:` , not `===` — see below
true

julia> formatname(Binary{6,3,true,false})
:Binary6p3sf

julia> SmallFloats.Binary16p6se                  # always reachable, unexported
Binary16p6se

julia> format(16, 6, true, true)                  # programmatic, runtime params
Binary16p6se

julia> using SmallFloats.Formats                  # ...or bring all 504 into scope

julia> Binary16p6se
Binary16p6se
```

`Binary8p4se === Binary{8,4,true,true}` is **false**, and the distinction is the draft's own: a *format* is a datum set and an encoding, while the width of the machine word carrying a code point is a representation choice below the standard's level of description. `Binary` is abstract; `Binary8p4se` is its concrete representation and is what an array element type or a constructor target needs. Use the alias, or `format(K, P,  $\Sigma$ ,  $\Delta$ )` when the parameters are runtime values.

The suffix reads: p4 precision 4, s/u signed/unsigned, e/f extended/finite.

Format introspection follows the draft's Group M, with both Julia-style and draft-named accessors:

```
julia> bitwidth(Binary8p4se), precision(Binary8p4se), expbias(Binary8p4se)
(8, 4, 8)

julia> MaxFiniteOf(Binary8p4se)
Binary8p4se(224.0 ≡ 0x7e)

julia> MinPositiveOf(Binary8p4se)
Binary8p4se(0.0009765625 ≡ 0x01)
```

Also available: `MinFiniteOf`, `MinNormalOf`, `MaxSubnormalOf`, `expbitwidth`, `trailingsigbits`, `issigned`, `isextended`, and the draft-named forms (`BitwidthOf`, `PrecisionOf`, `ExponentBiasOf`, ...).

Each of these accepts a **value** as well as a type — the answer is a pure function of the format's type parameters, so `bitwidth(x)` and `bitwidth(typeof(x))` are the same query and both fold to a literal:

```
julia> x = Binary8p4se(1.6);

julia> BitwidthOf(x), PrecisionOf(x), SignednessOf(x), DomainOf(x)
(8, 4, true, true)

julia> MaxFiniteOf(x)
Binary8p4se(224.0 ≡ 0x7e)
```

4.2 Binary16p11se is not Float16, and Binary16p8se is not BFloat16

The $K \leq 16$ grid contains two formats that look like the IEEE-754 binary16 and the bfloat16 used in machine learning. **They are not those types**, and the differences are the kind that produce correct-looking numbers rather than errors.

	Binary16p11se	Float16	Binary16p8se	BFloat16
precision P	11	11	8	8
exponent bias	16	15	128	127
largest finite	65472	65504	3.3762e38	3.3895e38
NaN encodings	1	2046	1	2046
negative zero	no	yes	no	yes
domain is a parameter	yes (se/sf)	no	yes	no

The bias is the trap. P3109 defines the bias so that the exponent range is symmetric about the format's midpoint; IEEE-754 defines it as $2^{(E-1)} - 1$. The two differ by one, so **every datum in the format is a factor of two away from its IEEE counterpart** — Binary16p11se and Float16 share a significand layout and disagree on the value of every code point.

That is why the largest finites differ: 65472 is $(2^{11} - 1) \cdot 2^5$, one ulp short of Float16's 65504, because the top code point of an extended format is Inf rather than a finite value. Converting between them is a conversion, never a reinterpretation:

```
julia> using SmallFloats.Formats

julia> Convert(Binary16p11se, RNE_SF, Float16(1.5))    # a conversion: 1.5 ↦ 1.5
Binary16p11se(1.5 ≡ 0x4200)

julia> reinterpret(Binary16p11se, Float16(1.5))        # the SAME BITS: 1.5 ↦ 0.75
Binary16p11se(0.75 ≡ 0x3e00)
```

The second line is the trap in one expression. `Float16(1.5)` has the bit pattern `0x3e00`; read as a `Binary16p11se` those same bits denote **0.75**, because the exponent bias differs by one. Nothing errors, nothing warns, and the answer is off by a factor of two. Always convert.

The remaining rows are draft decisions rather than accidents. A single NaN encoding means there is no payload to propagate and no quiet/signalling distinction; the 2045 code points IEEE spends on NaN payloads are datums here. No negative zero means `-0.0` and `0.0` are the same code point, so a sign-symmetric format has an odd number of finite values. And the domain parameter — `e` for extended (with $\pm\text{Inf}$), `f` for finite — has no IEEE analogue at all: `Binary16p11sf` spends the two infinity code points on finite values instead.

If you want IEEE semantics, use `Float16` and `BFloat16`; the package converts to and from both, exactly where the datum sets permit (`datumsexact`).

4.3 Values

A value is an immutable wrapper around its code point — **one byte at $K \leq 8$, two above it**. The storage unit is `codeunit_type(T)`, a property of the *representation*; the code point occupies the low K bits and the high bits are maintained zero. Four ways in, two ways out:

```
julia> x = Binary8p4se(1.6)           # construct from a Real: projects (rounds)
Binary8p4se(1.625 == 0x45)

julia> Binary8p4se(0x45)             # construct from a UInt8 CODE POINT (validated)
Binary8p4se(1.625 == 0x45)

julia> rawvalue(Binary8p4se, 0x45)   # same, unchecked — the kernel-internal route
Binary8p4se(1.625 == 0x45)

julia> Convert(Binary8p4se, RNE_SN, 3) # explicit conversion, any mode
Binary8p4se(3.0 == 0x4c)

julia> decode(x)                     # the exact datum, on the format's carrier
1.625

julia> codepoint(x)                  # the code point (extends Base.codepoint)
0x45
```

`decode` is **always exact**, but the type it returns is a property of the format, not of the package: `Float64` for the 432 formats whose exponent range it holds, `Float128` or an exact dyadic carrier for the other 72. `datumcarrier(T)` names it; `Float64(x)` always returns a `Float64` and rounds where the datum does not fit.

UInt8 means code point; every other number means value

`UInt8` is the *only* argument type with code-point semantics — mirroring `Char(0x41)`. Every other `Real` (including other `Integers`) constructs by projecting the numeric value, and `Convert` is numeric for **all** integers:

```
julia> Binary8p4se(0x02), Binary8p4se(2)
(Binary8p4se(0.001953125 == 0x02), Binary8p4se(2.0 == 0x48))
```

Out-of-range codes throw for $K < 8$ (`Binary5p3sf(0x20)` is an error); the range check costs nothing measurable — 2.1 ns, identical to unchecked `rawvalue`. Round-tripping is `T(codepoint(x)) == x`.

`Convert` accepts `Binary` values (any format), `Float16/32/64`, `Float128`, `Integer`, and `BigFloat` — types whose values it can project *exactly*. `Float128` projects directly, preserving all 113 significand bits: a value that sits a hair above a rounding midpoint converts correctly where staging through `Float64` would round it onto the midpoint first and then break the tie the wrong way. For other `Reals`, convert explicitly first (e.g. `Binary8p4se(Float64( $\pi$ ))`) and own the double rounding; `Rational` inputs throw with that guidance rather than double-round silently.

Every format has exactly one NaN and no negative zero; extended formats add $\pm\text{Inf}$:

```
julia> Binary8p4se(1e9)           # overflow under the default spec → +Inf
Binary8p4se(Inf ≡ 0x7f)

julia> Binary8p4se(-0.0)          # no -0: projects to the single zero
Binary8p4se(0.0 ≡ 0x00)

julia> isnan(Binary8p4se(NaN)), isfinite(Binary8p4se(2.0))
(true, true)
```

Standard predicates (`isnan`, `isinf`, `isfinite`, `iszero`, `signbit`, `issubnormal`), `zero/one/eps/typemin/typemax`, and stepping via `NextGreaterThan` / `NextLessThan` (the draft's `Next` operations; `nextfloat/prevfloat` alias them) all work:

```
julia> NextGreaterThan(Binary8p4se(1.0))
Binary8p4se(1.125 ≡ 0x41)

julia> Class(Binary8p4se(0.01))   # draft classification
ClassPosNormal::FPClass = 0x06
```

4.4 Random values

The Random API works on every format, in every setup Julia programmers expect.

What the draws mean. `rand` draws the real uniform on $[0, 1)$ at `Float64` and floor-projects it onto the format's grid, so each code point receives exactly the real measure of its interval; results are always in $[0, 1)$. `randn` projects a standard-normal `Float64` draw round-to-nearest with `SatFinite`, so tail draws beyond `MaxFiniteOf(T)` clamp to the extremal finite datum — `randn` never returns $\pm\text{Inf}$ or `NaN`, which matters for tiny-range formats (`Binary3p1se` has `MaxFinite = 1.0`). `randn` requires a signed format; an unsigned format throws:

```
julia> randn(Binary6p3ue)
ERROR: ArgumentError: randn requires a signed format; Binary6p3ue cannot represent negative draws
```

Setup 1 — quick and implicit. No `rng` argument: draws come from Julia's task-local default generator (a Xoshiro), and `Random.seed!` controls them exactly as for any other type:

```
julia> Random.seed!(1234); rand(Binary8p4se)
Binary8p4se(0.3125 ≡ 0x32)

julia> Random.seed!(1234); rand(Binary8p4se)   # same seed, same draw
Binary8p4se(0.3125 ≡ 0x32)
```

Setup 2 — an explicit rng stream. Pass any `AbstractRNG` first, for reproducible streams independent of global state:

```
julia> rand(Xoshiro(1), Binary8p4se)
Binary8p4se(0.0703125 ≡ 0x21)

julia> rand(Xoshiro(8), Binary8p4se, 4)
4-element Vector{Binary8p4se}:
 Binary8p4se(0.40625 ≡ 0x35)
 Binary8p4se(0.021484375 ≡ 0x13)
 Binary8p4se(0.75 ≡ 0x3c)
 Binary8p4se(0.28125 ≡ 0x31)
```

Setup 3 — arrays and in-place. `rand(T, dims...)` / `randn(T, dims...)` build `Array{T}`; `rand!(A)` / `randn!(A)` fill an existing array (one byte per element at $K \leq 8$, two above — cheap to preallocate and reuse):

```
julia> A = Vector{Binary8p4se}(undef, 3); randn!(Xoshiro(2), A)
3-element Vector{Binary8p4se}:
 Binary8p4se(-0.005859375 ≡ 0x86)
 Binary8p4se(1.75 ≡ 0x46)
 Binary8p4se(-1.0 ≡ 0xc0)

julia> randn(Xoshiro(3), Binary8p4se, 2, 3) |> typeof
Matrix{Binary8p4se}
```

Feed array draws straight into the storage layers when that is the goal: `PackedVector(rand(Binary4p2se, n))`.

Setup 4 — choosing the projection. The scalar `:Type` forms take a projection keyword to land the draw under any `ProjSpec`:

```
julia> rand(Xoshiro(2), Binary8p4se; projection = RTP_SN) # ceiling
Binary8p4se(0.0029296875 ≡ 0x03)

julia> randn(Xoshiro(6), Binary8p4se; projection = RTZ_SF) # toward zero
Binary8p4se(-1.875 ≡ 0xc7)

julia> rand(Xoshiro(4), Binary8p4se; projection = RSA_SN(8)) # stochastic
Binary8p4se(0.8125 ≡ 0x3d)
```

A stochastic projection draws its random bits from the *same* rng, so seeded streams stay reproducible. The defaults are the contract-keepers (floor for `rand`, nearest + `SatFinite` for `randn`); opting out can produce `1.0` from `rand` or `±Inf/NaN` from `randn`. The array and `!` forms always use the defaults — for arrays under another projection, draw scalars: `[rand(rng, T; projection = ρ) for _ in 1:n]`. Worked transcripts, including the $K = 3$ tail-clamp comparison: [User Examples](#).

4.5 Projection specifications

Every rounding decision in the package is governed by a `ProjSpec`, the pair of a **rounding mode** and a **saturation mode**. Both are zero-size singleton types, so a `ProjSpec` costs nothing and fully specializes every call it reaches.

Rounding modes: `NearestTiesToEven`, `NearestTiesToAway`, `TowardPositive`, `TowardNegative`, `TowardZero`, `ToOdd`, and the stochastic families `StochasticA{N}`, `StochasticB{N}`, `StochasticC{N}` with a random-bit budget $1 \leq N \leq 60$. Saturation modes: `SatFinite`, `SatPropagate`, `SatNone`.

```
julia> ρ = ProjSpec(TowardPositive(), SatFinite())
(TowardPositive, SatFinite)

julia> RTP_SF === ρ # every pairing is also predefined
true
```

Predefined specs

Every deterministic (rounding, saturation) pairing is exported as a constant, named `R<mode>_Sat<mode>`:

	SatFinite	SatPropagate	SatNone
NearestTiesToEven	RNE_SF	RNE_SP	RNE_SN
NearestTiesToAway	RNA_SF	RNA_SP	RNA_SN
TowardPositive	RTP_SF	RTP_SP	RTP_SN
TowardNegative	RTN_SF	RTN_SP	RTN_SN
TowardZero	RTZ_SF	RTZ_SP	RTZ_SN
ToOdd	RT0_SF	RT0_SP	RT0_SN

The stochastic families are parameterized by the random-bit budget `N`, so their predefined forms are *constructors* rather than constants: `RSA_*` for `StochasticA`, `RSB_*` for `StochasticB`, `RSC_*` for `StochasticC`, each crossed with the three saturation modes. Call them with the budget — or without, for the default `N = 8` (`SmallFloats.DEFAULT_RBITS`):

```
julia> RSA_SN()                                     # StochasticA with the default N = 8
(StochasticA{8}, SatNone)

julia> RSC_SF(16) == ProjSpec(StochasticC{16}(), SatFinite())
true
```

`RNE_SN` is the package's *initial* default spec. `default_projspec` reads the session default (see **Session defaults** below), so every Base-register operator, the same-format convenience methods, and `T(x::Real)` construction follow `DefaultProjection()` — `RNE_SN` until you change it.

Rounding chooses the neighbor; saturation decides what out-of-range results become. Watch all three interact on an overflowing product in `Binary8p4se` (`MaxFinite = 224`):

```
julia> w, two = Binary8p4se(200.0), Binary8p4se(2.0);

julia> Multiply(Binary8p4se, RNE_SN, w, two)      # overflow → ±Inf
Binary8p4se(Inf ≡ 0x7f)

julia> Multiply(Binary8p4se, RNE_SF, w, two)      # overflow → MaxFinite
Binary8p4se(224.0 ≡ 0x7e)

julia> Multiply(Binary8p4se, RTZ_SN, w, two)
Binary8p4se(224.0 ≡ 0x7e) # SatNone + directed-toward-zero clamps finite
```

Note

`SatNone` is not "no saturation handling": it is the draft's row set in which nearest modes overflow to $\pm\text{Inf}$ (or `NaN` for finite formats) while directed modes that point away from the overflow clamp to the extremal finite value.

Stochastic rounding

Stochastic modes consume N random bits per projection. You control the source three ways — implicitly (task-local RNG), by passing an `rng`, or by fixing the draw R itself, which makes single projections exactly reproducible and is the right tool for tests:

```
julia> σ = RSA_SN();                               # ≡ ProjSpec(StochasticA{8}(), SatNone())

julia> Add(Binary8p4se, σ, Binary8p4se(2.0), Binary8p4se(0.03125); rng = Xoshiro(1))
Binary8p4se(2.0 ≡ 0x48)

julia> Add(Binary8p4se, σ, Binary8p4se(2.0), Binary8p4se(0.03125); R = 0)
Binary8p4se(2.0 ≡ 0x48)      # smallest draw: rounds down

julia> Add(Binary8p4se, σ, Binary8p4se(2.0), Binary8p4se(0.03125); R = 255)
Binary8p4se(2.25 ≡ 0x49)      # largest draw: rounds up
```

The exact fraction here is $1/8$ of an ulp, so over all 256 draws exactly 32 round up — `StochasticA` is unbiased in expectation. R must lie in $0:2^N-1$.

Session defaults

Six session-wide defaults are readable as `DefaultX()` and settable as `DefaultX!(v)`:

default	initial value	setter accepts
<code>DefaultType</code>	<code>Binary8p2se</code>	any fully-parameterized Binary type
<code>DefaultReturnType</code>	<code>Binary8p2se</code>	any fully-parameterized Binary type
<code>DefaultRoundingMode</code>	<code>NearestTiesToEven()</code>	mode instance or type
<code>DefaultSaturationMode</code>	<code>SatNone()</code>	mode instance or type
<code>DefaultProjection</code>	<code>RNE_SN</code>	a <code>ProjSpec</code> , or <code>(mode, sat)</code>
<code>DefaultRNG</code>	the Xoshiro type	RNG type or instance
<code>DefaultRbits</code>	8	Int in 1:60

The projection default and its components are kept coherent in both directions: setting `DefaultRoundingMode!` or `DefaultSaturationMode!` rebuilds `DefaultProjection` around the change, and setting `DefaultProjection!` directly decomposes it back into both components — so `DefaultProjection() == ProjSpec(DefaultRoundingMode(), DefaultSaturationMode())` always holds.

```
julia> DefaultRoundingMode!(TowardZero())
TowardZero()

julia> DefaultProjection()                # followed the component
(TowardZero, SatNone)

julia> DefaultProjection!(RNA_SF)
(NearestTiesToAway, SatFinite)

julia> DefaultRoundingMode(), DefaultSaturationMode() # followed the projection
(NearestTiesToAway(), SatFinite())

julia> DefaultProjection!(RNE_SN)        # restore: these setters are PROCESS-wide
(NearestTiesToEven, SatNone)
```

The session defaults are process-global and they stay set

`DefaultProjection!` and its siblings mutate state shared by every module in the process. Nothing scopes them for you: set one in a script, a notebook cell, or a REPL session and every later `T(x::Real)`, `a + b` and `Exp(x)` follows it until something sets it back.

Prefer `with_default_projection` (and `with_default_type`, `with_default_returntype`), which restore on exit even if the body throws:

```
with_default_projection(RTZ_SF) do
    Binary8p4se(1e9)          # clamps, inside this block only
end
```

Every example below assumes the pristine `(NearestTiesToEven, SatNone)`, which is why each demonstration above restores it explicitly.

`DefaultProjection` is not merely advisory: `default_projspec` reads it, so the same-format convenience methods (`a + b`, `Exp(x)`, ...), the Base-register operators, and `T(x::Real)` construction all follow it.


```
julia> x, y = Binary8p4se(200.0), Binary8p4se(2.0);

julia> x * y                                     # RNE_SN: overflow → +Inf
Binary8p4se(Inf ≡ 0x7f)

julia> DefaultProjection!(RTZ_SF);

julia> x * y                                     # now clamps to MaxFinite
Binary8p4se(224.0 ≡ 0x7e)

julia> DefaultProjection!(RNE_SN)               # restore before moving on – see below
(NearestTiesToEven, SatNone)

julia> Multiply(Binary8p4se, RNE_SN, x, y)      # explicit p is unaffected
Binary8p4se(Inf ≡ 0x7f)
```

Changing the default is a global semantic change

Every caller of the convenience forms — including code in other packages — sees it. Library code that needs a specific projection should name it explicitly rather than rely on the session default.

Following the default costs nothing while it holds its initial value: the convenience forms consume it through the same speculation guard as the `with_default_*` combinators, so they compile against the constant and stay allocation-free with concretely inferred results (pinned in the test suite). Once you change the default they cross a function barrier instead — one dynamic dispatch per call, everything inside still specialized. The explicit forms `Add(T, p, x, y)` are unaffected either way.

To *consume* a default in your own code without paying dynamic-dispatch costs, go through the `with_default_*` combinators — `with_default_type`, `with_default_returntype`, `with_default_projection` — which call `f(default, args...)`:

There is no accumulator default

`DefaultAccumulatorType` existed through v0.1.0 and was removed. The wide-precision accumulator in a reduction is not a preference: the span filter measures the operands and picks the cheapest carrier that is *provably exact* for them, so both branches return the defined result. A knob overriding that would make `BlockDotProduct` inexact from the default API. To trade speed for nothing else, set `ENV["SmallFloats_Float128"] = "disable"`, which is bit-identical by construction.

```
julia> with_default_type((T, x) -> T(x), 1.5)
Binary8p2se(1.5 ≡ 0x41)
```

While a default still holds its initial value, the combinator's call is statically compiled against that constant — no dynamic dispatch. After the default is changed it crosses a function barrier: one dynamic dispatch at entry, everything inside fully specialized.

Allocation contract: when `f`'s result type does not depend on the default — `with_default_projection` with the formats fixed by the caller is the normal shape — the call is zero-allocation with a concretely inferred result (pinned in the test suite). When the result's type *is* the default (`with_default_type` used as a constructor), the value is computed on the specialized path but boxes once where it escapes — the irreducible cost of a runtime-chosen type.

4.6 Scalar operations: the two registers

The **spec-named register** exposes every draft operation under its draft name with an explicit result format and `spec: Op(fr, p, operands...)`. Operands may be any formats; the result format is the first argument. The catalog:

- **30 unary** — Abs, Negate, Sqrt, RSqrt, Recip, Exp, Exp2, ExpMinusOne, Log, Log2, LogOnePlus, Softplus, the trig/hyperbolic families, and the π -scaled families;
- **18 binary** — Add, Subtract, Multiply, Divide, CopySign, Hypot, ArcTan2, ArcTan2Pi, and the eight Minimum/Maximum variants plus the Finite variants;
- **3 ternary** — FMA, FAA, Clamp;
- **Convert** — the one operation that also accepts non-Binary operands.

The **Base register** makes each format an ordinary Julia number under its *default* spec (RNE_SN): + - * /, fma, exp, log, sqrt, min, max, abs, atan(y, x), sinpi, inv, comparisons, and friends — same-format operands only, no silent cross-format promotion (mixing formats promotes to Float64 explicitly).

```
julia> exp(Binary8p4se(0.25))           # Base register
Binary8p4se(1.25 ≡ 0x42)

julia> Exp(Binary8p4se, RNE_SN, Binary8p4se(0.25)) # identical, explicit
Binary8p4se(1.25 ≡ 0x42)
```

Semantics highlights (all per the draft): $x/0 \rightarrow \text{NaN}$ for every x including $\pm\text{Inf}$; $\text{Recip}(\pm\text{Inf}) = 0$; $0 \cdot \infty \rightarrow \text{NaN}$; a NaN operand generally propagates, except the *Number/*Finite extremum variants which prefer non-NaN (and finite) operands; trig of $\pm\text{Inf}$ is NaN; the π -scaled family reduces exactly mod 2 first.

4.7 Arrays, kernels, and sorting

Every operation has array methods — $\text{Op}(\text{fr}, \rho, A)$, $\text{Op}(\text{fr}, \rho, A, B)$, $\text{Op}(\text{fr}, \rho, A, B, C)$ — plus the generic `vmap / vmap!`.

(In that line `Op` stands for any operation name — Add, Exp, and so on. There is also a *type* called `Op`, introduced below, which turns an operation into a callable so `map` and `broadcast` work directly.)

```
julia> A = Binary8p4se.(randn(Xoshiro(7), 4) .* 2)
4-element Vector{Binary8p4se}:
 Binary8p4se(-0.875 ≡ 0xbe)
 Binary8p4se(4.0 ≡ 0x50)
 Binary8p4se(-3.25 ≡ 0xcd)
 Binary8p4se(2.25 ≡ 0x49)

julia> Exp(Binary8p4se, RNE_SN, A)
4-element Vector{Binary8p4se}:
 Binary8p4se(0.40625 ≡ 0x35)
 Binary8p4se(56.0 ≡ 0x6e)
 Binary8p4se(0.0390625 ≡ 0x1a)
 Binary8p4se(9.0 ≡ 0x59)
```

For pure (non-stochastic) specs, unary and binary array calls run as **table gathers**: the first call builds and caches a 256-byte (unary) or 64 KiB (binary) result table for that exact (`op`, `formats`, `$\rho$`) specialization, and every later element costs a single lookup — measured at 0.27 ns/element unary, 0.5 ns/element binary.

Ternary operations (FMA, FAA, Clamp) ride the same gather whenever the three operand bitwidths keep the table affordable, tiered by $K_1 + K_2 + K_3$:

- **Eager** (up to 256 KiB; every all- $K \leq 6$ signature): built on the first array call, like unary/binary.
- **Adaptive** (up to 2 MiB; the $K = 7$ band): built only once a signature has processed enough elements to earn its build, in a byte-bounded, LRU-evicted cache.

- **Compute** ($K = 8$; a 16 MiB table stops being a cache win): the scalar pipeline runs per element, optionally threaded for long arrays.

Every table entry — eager or adaptive — is built through the scalar path, so it is bit-identical by construction. Stochastic calls always run the scalar pipeline per element (each element draws its own random bits).

Inspect or reset the cache — unary/binary and ternary alike — with `table_bytes()` and `empty_tables!()`. The ternary policy thresholds and the threading cutoff are internal Refs (`SmallFloats.TERNARY_EAGER_BITS` and neighbors in `tables.jl` and `kernels.jl`) for the rare case you need to tune them.

Sorting is special-cased: values compare through integer order keys, and vectors of Binary sort with an **$O(n)$ counting sort** installed as the default algorithm — about $8\times$ the stock comparison sort at 64 K elements, `rev=true` included. `sort(A)` just works; NaN sorts last (first under `rev=true`), matching Base's conventions. `TotalOrder(x, y)` exposes the draft's total order directly (single NaN largest).

map, broadcast, and the Op callable

`vmap!` is the in-place, table-aware kernel and is what you want when you have a destination to fill. When you want Julia's own array verbs, bind the operation, result format and projection into an `Op` and use it as a function:

```
julia> A = Binary8p4se.([1.5, 0.25]); B = Binary8p4se.([0.25, 1.5]);
```

```
julia> map(Op(:Add, Binary8p4se, RNE_SN), A, B)
```

```
2-element Vector{Binary8p4se}:
```

```
Binary8p4se(1.75 ≡ 0x46)
```

```
Binary8p4se(1.75 ≡ 0x46)
```

```
julia> Op(:Exp, Binary8p4se, RNE_SN).(A)
```

```
2-element Vector{Binary8p4se}:
```

```
Binary8p4se(4.5 ≡ 0x51)
```

```
Binary8p4se(1.25 ≡ 0x42)
```

`Op` is a singleton type — the operation, format and projection all live in its type parameters — so a call specializes exactly as `Add(Binary8p4se, RNE_SN, x, y)` does and allocates nothing beyond the array map itself builds.

4.8 Blocks and scaled operations

A `Block{B,FS,FE}` is a scale in format FS and B elements in format FE; its represented values are `scale × element` (the MX-style scheme). Construct directly or quantize a tuple with the draft's algorithm, which picks the scale from the maximum finite `|element|` and projects each element against it:

```
julia> b = Block(Binary8pluf(4.0), Binary8p4se(1.5), Binary8p4se(-0.75),
```

```
Binary8p4se(2.0), Binary8p4se(0.5))
```

```
Block{4, Binary8pluf, Binary8p4se}(Binary8pluf(4.0 ≡ 0x82), (...))
```

```
julia> ConvertFromBlock(Binary8p3se, RNE_SN, b) # decode scale×elem, project
```

```
(Binary8p3se(6.0 ≡ 0x4a), Binary8p3se(-3.0 ≡ 0xc6), Binary8p3se(8.0 ≡ 0x4c), Binary8p3se(2.0 ≡ 0x44))
```

```
julia> BlockReduceAdd(Binary8p4se, RNE_SN, b) # exact Σ scale×xi, then project
```

```
Binary8p4se(13.0 ≡ 0x5d)
```

```
julia> ConvertToBlockMaxAbsFinite(Binary8pluf, Binary8p4se, RNE_SN, RNE_SN,
```

```
(Binary8p4se(100.0), Binary8p4se(-12.0), Binary8p4se(0.5), Binary8p4se(3.0)))
```

```
Block{4, Binary8pluf, Binary8p4se}(Binary8pluf(64.0 ≡ 0x86), (...))
```

Every scalar operation lifts to blocks (`BlockAdd`, `BlockExp`, ..., taking blocks and a result scale) and to the scaled form `ScaledOp` (`ScaledAdd(fr, p, s1, x1, s2, x2)` — block size 1). Reductions: `BlockReduceAdd`, `BlockReduceMultiply`, and `BlockDotProduct(fr, p, bx, by)`, whose lane products and accumulation are **exact** before the single final projection — there is no hidden intermediate rounding. `ConvertToBlock(fs, fr, p, xs, s)` quantizes against a scale you supply. `BlockVector` stores many same-shape blocks in structure-of-arrays layout.

4.9 Packed storage

For memory-bound work, `PackedVector{F}` stores code points at `bitwidth(F)` bits per element (a `Binary5p2se` vector at 5 bits/element instead of 8):

```
julia> v = Binary5p2se.(rand(Xoshiro(3), 6) .* 4);

julia> pv = PackedVector(v); sizeof(pv.data)
8                # 6 × 5 bits = 30 bits → one 64-bit word

julia> pv[3] == v[3]
true
```

It is a full `AbstractVector{F}` (indexing, collect, iteration), and `vmap` accepts it directly, unpacking cache-friendly tiles internally. The rule is *store packed, compute unpacked*: there is deliberately no in-place packed arithmetic.

4.10 Conformance and κ -approximate implementations

`conformance()` returns the live declaration — formats, operations, mode vocabulary, the table specializations actually instantiated this session, and every registered approximation. `conformance_report()` prints it; `conformance_dict()` returns a plain nested `Dict` for JSON/TOML serialization.

The default API is bit-exact, always. If you *want* a faster inexact kernel, register it — and the registry measures your honesty:

```
julia> ftz = ftz_variant(:Exp, Binary8p4se, Binary8p4se, RNE_SF); # Annex example

julia> impl = register_approx!(:my_fast_exp, :Exp, Binary8p4se, (Binary8p4se,),
                             RNE_SF, ftz);

julia> kappa(:my_fast_exp)    # measured max code-point deviation, verified exhaustively
4.0
```

Declaring κ smaller than the measured deviation throws; NaN-mismatching implementations must be acknowledged with an explicit $\kappa = \text{NaN}$. Retrieve with `approx(:my_fast_exp)`, list with `list_approx()`, and measure anything yourself with `measure_kappa / codedistance`.

4.11 Performance guidance

The abstract format as an array element type

`Binary{K,P, $\Sigma$ , $\Delta$ }` is **abstract**, so an array declared with it has a non-isbits element type and boxes every element:

```
julia> isbitstype(eltype(Vector{Binary{8,4,true,true}}(undef, 3)))
false

julia> isbitstype(eltype(Vector{Binary8p4se}(undef, 3)))
true
```

Nothing errors — every operation still returns the right answer, because the package normalizes the abstract format at each constructor and entry point. You simply lose the entire performance story, silently. That makes this the sharper edge of the `===` distinction: the type comparison is visibly false, while this one costs you and says nothing.

`similar(a, T)` normalizes the request; `Vector{T}(undef, n)` cannot be intercepted. Use the alias, or `format(K, P,  $\Sigma$ ,  $\Delta$ )` for runtime parameters.

- **Pass format types statically.** Through const bindings, type parameters, or function arguments, every entry point fully specializes (scalar `Add` $\approx$ 18–26 ns, project $\approx$ 13 ns, zero allocations). A format type read from a **non-const global** forces Julia's dynamic dispatch on every call ($\sim$ 1 μ s for keyword calls); one function barrier `f(::Type{T}, ...)` where `{T}` restores full speed.
- **The convenience forms are free at the initial default.** `x + y`, `Exp(x)`, and `T(2.1)` read the session default through a speculation guard, so they are allocation-free and concretely inferred while it holds its initial value. After you change the default they cost one dynamic dispatch per call; name `p` explicitly (`Add(T, p, x, y)` with a const `p`) in hot code that must be insensitive to the session default.
- **Bulk work belongs in array calls.** The table-gather kernels are $\sim$ 50 $\times$ the scalar path; the first call per specialization pays a one-time build ($\approx$ 0.4 ms unary, tens of ms for 8 $\times$ 8 binary tables, up to a few ms for a 2 MiB ternary table).
- **Ternary array calls scale with bitwidth.** $K \leq 6$ operand formats table eagerly ($\sim$ 35 $\times$ the scalar loop); $K = 7$ tables adaptively once a signature is hot; $K = 8$ runs the compute kernel, optionally threaded for long arrays ($\sim$ 4 $\times$ at 4 threads) — no action needed, the array call picks the right path.
- **Stochastic array calls** run the scalar pipeline per element; pass an explicit `rng` for reproducibility.
- **Memory:** `PackedVector` for storage; `BlockVector` for many blocks.
- The reproducible benchmark suite lives at `benchmarking/benchmarking.jl` and generates a full markdown report for your machine.

4.12 Environment switches

`ENV["SmallFloats_Float128"] = "disable"` (set before using `SmallFloats`) disables the internal `Float128` fast paths in favor of pure MPFR — results are bit-identical (tested); only build/oracle speed changes.

Chapter 5

User Examples

Worked, runnable examples. Every output shown was captured from a real session; seeds are fixed so you can reproduce them exactly.

5.1 Basic

One value, every rounding mode

The value 2.30078125 sits between the Binary8p4se grid points 2.25 and 2.5 (ulp = 0.25 in [2, 4]):

```
using SmallFloats

for μ in (NearestTiesToEven(), NearestTiesToAway(), TowardPositive(),
        TowardNegative(), TowardZero(), ToOdd())
    v = Convert(Binary8p4se, ProjSpec(μ, SatNone()), 2.30078125)
    println(rpad(string(typeof(μ)), 20), " → ", decode(v))
end
```

NearestTiesToEven	→ 2.25
NearestTiesToAway	→ 2.25
TowardPositive	→ 2.5
TowardNegative	→ 2.25
TowardZero	→ 2.25
ToOdd	→ 2.25

(ToOdd keeps 2.25 because its significand is already odd; try 2.05 to see it move.)

Enumerating a whole format

Small formats are small enough to look at in full — 16 code points for Binary4p2se, sorted by the total order (single NaN last):

```
decode.(sort(Binary4p2se.(0x00:0x0f))) # broadcast the code-point constructor
```

```
[-Inf, -2.0, -1.5, -1.0, -0.75, -0.5, -0.25, 0.0,
 0.25, 0.5, 0.75, 1.0, 1.5, 2.0, Inf, NaN]
```

This is a good way to build intuition for a format before committing to it: you can see the subnormal spacing, the binade structure, and the dynamic range.

Saturation in one line each

```
julia> w, two = Binary8p4se(200.0), Binary8p4se(2.0); # MaxFinite is 224

julia> Multiply(Binary8p4se, RNE_SN, w, two)
Binary8p4se(Inf ≡ 0x7f)

julia> Multiply(Binary8p4se, RNE_SF, w, two)
Binary8p4se(224.0 ≡ 0x7e)
```

Random values under a chosen projection

rand and randn take a projection keyword on their scalar `::Type` forms. One underlying Float64 draw, landed on the format's grid four different ways — the seed fixes the draw, the projection decides the landing:

```
julia> rand(Xoshiro(2), Float64) # the underlying uniform draw
0.0022505868897625403

julia> rand(Xoshiro(2), Binary8p4se) # default: floor (stays in [0,1))
Binary8p4se(0.001953125 ≡ 0x02)

julia> rand(Xoshiro(2), Binary8p4se; projection = RTP_SN) # ceiling
Binary8p4se(0.0029296875 ≡ 0x03)

julia> rand(Xoshiro(2), Binary8p4se; projection = RNE_SN) # nearest
Binary8p4se(0.001953125 ≡ 0x02)

julia> rand(Xoshiro(2), Binary8p4se; projection = RSA_SN(8)) # stochastic
Binary8p4se(0.001953125 ≡ 0x02)
```

A stochastic projection draws its random bits from the *same* rng as the uniform draw, so seeded streams stay reproducible. Note the default is the contract-keeper: a nearest or upward projection can return exactly 1.0 (mass near the top rounds up), which is why floor is the default.

The same keyword on randn — here the draw is -1.9757...:

```
julia> randn(Xoshiro(6), Binary8p4se) # default: nearest + SatFinite
Binary8p4se(-2.0 ≡ 0xc8)

julia> randn(Xoshiro(6), Binary8p4se; projection = RTZ_SF) # toward zero
Binary8p4se(-1.875 ≡ 0xc7)
```

Where the saturation half of the default earns its keep: Binary3p1se has MaxFinite = 1.0, so normal tail draws overflow constantly. Seed 360 draws $z = 3.326\dots$:

```
julia> randn(Xoshiro(360), Binary3p1se) # SatFinite clamps the tail
Binary3p1se(1.0 ≡ 0x02)

julia> randn(Xoshiro(360), Binary3p1se; projection = RNE_SN)
Binary3p1se(Inf ≡ 0x03) # opt out, get the overflow
```

The array and ! forms (rand(T, dims), randn!(A), ...) always use the defaults; for arrays under another projection, draw scalars: [rand(rng, T; projection = p) for _ in 1:n].

5.2 General AI

Log-odds belief updating — and where tiny formats bite reasoning

Classic probabilistic reasoning: accumulate independent evidence as log-likelihood ratios. A sensor with hit rate 0.85 and false-alarm rate 0.30 contributes $\log(0.85/0.30) \approx 1.0415$ per alarm. In Binary8p3se that datum is 1.0 — and the running belief stalls exactly the way the gradient accumulator does in the Deep Learning section:

```
using SmallFloats

function logodds_demo(nalarms)
    llr = Binary8p3se(log(0.85 / 0.30))    # quantized evidence weight
    acc = Binary8p3se(0.0)
    for _ in 1:nalarms
        acc = Add(Binary8p3se, RNE_SN, acc, llr)
    end
    post(l) = 1 / (1 + exp(-l))
    exact = nalarms * log(0.85 / 0.30)
    (decode(llr), round(exact; digits=2), decode(acc),
     round(post(exact); digits=5), round(post(decode(acc)); digits=5))
end
logodds_demo(12)
```

```
(1.0, 12.5, 8.0, 1.0, 0.99966)
# llr datum, exact log-odds, quantized log-odds, exact posterior, quantized posterior
```

The accumulator is wrong by a third (8.0 vs 12.5 — near 8 the ulp is 2.0, so $8 + 1$ rounds straight back to 8), yet the *decision* is untouched: both posteriors are ≈ 1 . Threshold decisions need order, not magnitude — which is why coarse formats work in reasoning pipelines far past the point where their arithmetic looks broken. When magnitude does matter, the cures are the same as for gradients: a wider accumulator format, or stochastic rounding.

Fuzzy inference in an unsigned format

Membership degrees live in $[0, 1]$ and are never negative — a job for an unsigned finite format. Binary8p6uf gives a $[0, 15.5]$ range with 1/32 resolution at 1; the classic fuzzy connectives are registry operations (Minimum = Gödel t-norm, Maximum = s-norm, Multiply = product t-norm):

```
F = Binary8p6uf
trap(x, a, b, c, d) = clamp(min((x - a) / (b - a), 1.0, (d - x) / (d - c)), 0.0, 1.0)
warm = F(trap(26.0, 15, 20, 24, 28))    # membership of 26 °C in "warm"
hot  = F(trap(26.0, 24, 30, 100, 101))   # ... and in "hot"

(decode(warm), decode(hot),
 decode(Minimum(F, RNE_SN, warm, hot)), # warm AND hot
 decode(Maximum(F, RNE_SN, warm, hot)),  # warm OR hot
 decode(Multiply(F, RNE_SF, warm, hot))) # product t-norm
```

```
(0.5, 0.3359375, 0.3359375, 0.5, 0.16796875)
```

Because every connective is a bit-exact registry op, a fuzzy rule base evaluated in Binary8p6uf is *reproducible across machines to the code point* — a property sampled float pipelines cannot promise.

Search heuristics: what quantized scores keep is the ordering

Game-tree and beam search consume evaluation scores only through comparisons. Quantizing 200 heuristic scores to 8 bits collapses them onto 78 distinct code points — yet the argmax survives, and sorting is exact (integer order keys, $O(n)$ counting sort, single NaN ordered last deterministically):

```
using Random
rng = Xoshiro(21)
scores = randn(rng, 200) .* 3
q = Binary8p4se.(scores)

(argmax(scores), argmax(decode.(q)), argmax(scores) == argmax(decode.(q)),
 decode.(sort(q; rev=true)[1:5]), length(unique(codepoint.(q))))
```

```
(140, 140, true, [8.0, 7.5, 6.5, 6.0, 6.0], 78)
```

The two 6.0s are the caveat: quantization creates *ties* the exact scores did not have. TotalOrder's deterministic tie behavior means the search expands the same node on every run — ties change which answer you get, not whether you can reproduce it.

5.3 Machine Learning

Quantizing a weight tensor and measuring the damage

```
using SmallFloats, Random, Statistics

rng = Xoshiro(42)
w = randn(rng, 10_000) .* 0.25          # typical trained-weight scale
q = Binary8p4se.(w)                    # project every weight
back = decode.(q)

mean((w .- back).^2), maximum(abs.(w .- back)), count(isinf, back) / length(back)
```

```
(4.41e-5, 0.0312, 0.0)                # MSE, max |error|, overflow fraction
```

The max error is exactly half an ulp of the largest binade the data reached — the worst case nearest rounding permits.

Picking a format: precision vs range

For the same Gaussian tensor, $K = 8$ formats trade significand bits against exponent range. RMSE doubles per lost significand bit; MaxFinite grows explosively:

```
rng = Xoshiro(7); x = randn(rng, 50_000)
for F in (Binary8p5se, Binary8p4se, Binary8p3se, Binary8p2se)
    back = [decode(Convert(F, RNE_SF, xi)) for xi in x]
    println(rpad(formatname(F), 12), " rmse = ", round(sqrt(mean((x .- back).^2)); sigdigits=3),
            " maxfinite = ", decode(MaxFiniteOf(F)))
end
```

```
Binary8p5se  rmse = 0.0133  maxfinite = 15.0
Binary8p4se  rmse = 0.0266  maxfinite = 224.0
Binary8p3se  rmse = 0.0528  maxfinite = 49152.0
Binary8p2se  rmse = 0.103   maxfinite = 2.147483648e9
```

For unit-scale data, spend bits on precision; buy range only when your data needs it (or get it from a block scale — see below).

Stochastic rounding is unbiased where nearest is not

Nearest rounding maps 0.30078125 to 0.3125 every single time — a systematic +0.0117 bias. Stochastic rounding is right *on average*:

```
target = 0.30078125          # v = 3/16 of an ulp above 0.296875
σ = RSA_SN(16)              # StochasticA{16} with SatNone
rng = Xoshiro(1)
m = mean(decode(Convert(Binary8p4se, σ, target; rng)) for _ in 1:200_000)
(decode(Binary8p4se(target)), m)
```

```
(0.3125, 0.300765)          # RNE result vs stochastic mean ≈ target
```

This is the property that makes stochastic rounding matter for accumulating many small contributions (gradients, activations statistics) in a low-precision format.

5.4 Deep Learning

MX-style block quantization — and the staging pitfall

Block formats pair a shared scale with narrow elements, tracking dynamic range that a bare element format cannot. Here each row of W lives at a different scale, spanning $\sim 2^{16}$ across the matrix:

```
using SmallFloats, Random, Statistics

rng = Xoshiro(3)
W = randn(rng, 64, 64) .* 2 .* (2.0 .^ (collect(0:63) ./ 4)) # per-row scales

function mx_rows(W, ::Type{FST}, ::Type{FS}, ::Type{FE}, ::Val{B}) where {FST,FS,FE,B}
    n, m = size(W)
    [begin
        # stage through a WIDE-RANGE format so nothing clamps before scaling
        seg = ntuple(k -> Convert(FST, RNE_SF, W[i, (j-1)*B + k]), Val(B))
        ConvertToBlockMaxAbsFinite(FS, FE, RNE_SN, RNE_SN, seg)
    end for i in 1:n, j in 1:m ÷ B]
end

blocks = mx_rows(W, Binary8p2se, Binary8pluf, Binary8p4se, Val(32))
recon = [let b = blocks[i, (j-1) ÷ 32 + 1]
    decode(b.s) * decode(b.x[(j-1) % 32 + 1])
end for i in 1:64, j in 1:64]
plain = [decode(Convert(Binary8p4se, RNE_SF, w)) for w in W]

relerr(A) = sqrt(mean(((W .- A) ./ max.(abs.(W), 1e-9)).^2))
relerr(plain), relerr(recon)
```

```
(0.616, 0.109)          # plain 8p4se clamps the large rows; MX tracks them
```

Stage wide, then block-quantize

ConvertToBlockMaxAbsFinite takes already-Binary elements. If you stage your Float64 data through the *element* format first, SatFinite clamps the large values **before** the block scale can absorb them, and blocks buy you nothing — we measured exactly that (0.617 vs 0.616) with Binary8p4se staging. Stage through a wide-range format (Binary8p2se above); the residual 0.109 here is the staging format's own precision, which bounds what any downstream scheme can keep.

At B = 32 with an 8-bit scale the storage cost is 8.25 bits/value.

Quantized dot products with one final rounding

BlockDotProduct computes every lane product and the accumulation *exactly*, then projects once:

```
rng = Xoshiro(9)
a64, b64 = randn(rng, 32), randn(rng, 32)
qb(v) = ConvertToBlockMaxAbsFinite(Binary8pluf, Binary8p4se, RNE_SN, RNE_SN,
    ntuple(i -> Convert(Binary8p4se, RNE_SF, v[i]), Val(32)))
dq = BlockDotProduct(Binary8p4se, RNE_SN, qb(a64), qb(b64))
(a64' b64, decode(dq))
```

```
(5.0634, 4.5)      # difference is input quantization only, never accumulation error
```

Why training loops like stochastic rounding: the swamping demo

Accumulate 400 gradient steps of 0.011 into a Binary8p3se accumulator. Under nearest rounding, the moment the accumulator dwarfs the increment, every add rounds back to the accumulator — it **stalls**. Stochastic rounding keeps absorbing the increments in expectation:

```
function accumulate_demo(nsteps, g)
    σ = RSA_SN(16)
    rng = Xoshiro(11)
    acc_rne = Binary8p3se(0.0); acc_sto = Binary8p3se(0.0)
    for _ in 1:nsteps
        acc_rne = Add(Binary8p3se, RNE_SN, acc_rne, Convert(Binary8p3se, RNE_SN, g))
        acc_sto = Add(Binary8p3se, σ, acc_sto, Convert(Binary8p3se, σ, g; rng); rng)
    end
    (nsteps * g, decode(acc_rne), decode(acc_sto))
end
accumulate_demo(400, 0.011)
```

```
(4.4, 0.125, 5.0)  # exact sum, RNE accumulator (stalled!), stochastic accumulator
```

The stochastic result is noisy (5.0 vs 4.4 on this seed) but unbiased; the RNE result is *wrong by 35×* and no amount of steps will fix it. In practice you keep a wider accumulator when you can — and use stochastic rounding when you can't.

Activation functions are 256-byte lookup tables

For pure specs, a unary activation over an 8-bit format is a *finite function* — the whole nonlinearity is one 256-byte table, built bit-exactly through the scalar path and then gathered per element. This is precisely the activation LUT you would burn into an accelerator:

```
using SmallFloats, Random
empty_tables!()
rng = Xoshiro(4)
pre = Binary8p4se.(randn(rng, 4096) .* 2.5)      # pre-activations,  $\pm 2.5\sigma$ 
act = Tanh(Binary8p4se, RNE_SN, pre)              # table-gather kernel

(table_bytes(),
 round(count(v -> abs(decode(v)) == 1.0, act) / 4096; digits=3),
 length(unique(codepoint.(act))),
 decode(NextLessThan(one(Binary8p4se))))
```

```
(256, 0.384, 100, 0.9375)
```

```
# LUT size; fraction saturated to  $\pm 1$ ; distinct output codes; last value below 1
```

Two things worth staring at. First, **saturation is severe at this input scale**: 38% of pre-activations land exactly on ± 1 — under RNE, tanh reaches 1 as soon as the true value rounds past the midpoint of the last gap, and the entire saturated tail becomes indistinguishable to the next layer. (Under TowardNegative the asymptote is honored instead: tanh of any finite positive input projects to 0.9375, never 1 — the projection mode is part of the activation design space.) Second, the choice of nonlinearity changes *information retention* in code points, not just shape:

```
actS = Softplus(Binary8p4se, RNE_SN, pre)
(length(unique(codepoint.(actS))), table_bytes())
```

```
(61, 512)      # softplus keeps 61 distinct codes here; two LUTs now cached
```

Auditing an activation for a format is this cheap: enumerate all 256 inputs, look at the output histogram, count what survives.

Packing a quantized model

```
n = 1_000_000
model = [rawvalue(Binary5p2se, UInt8(rand(0:31))) for _ in 1:n]
pv = PackedVector(model)
(sizeof(model), sizeof(pv.data))
```

```
(1000000, 625000) # 8 bits  $\rightarrow$  5 bits per value; indexing and vmap work directly on pv
```

Chapter 6

Julia Compatibility

Binary values speak ordinary Julia. The **Base register** (`src/juliacompat.jl`) maps every draft operation that has a Base counterpart onto its Base name, so generic Julia code — and your fingers — can use the spellings they already know. Every method there is exactly one same-format spec-register call under the session default projection; there is no third semantics: `x + y` is `Add(x, y)`, which is `Add(T, DefaultProjection(), x, y)`.

6.1 Using the overloads

```
julia> x, y, z = Binary8p4se(1.6), Binary8p4se(0.25), Binary8p4se(2.0);

julia> x + y                                     # Add
Binary8p4se(1.875 ≡ 0x47)

julia> exp(y)                                    # Exp — a 256-byte table lookup once warm
Binary8p4se(1.25 ≡ 0x42)

julia> atan(y, x)                               # ArcTan2, in Base's (y, x) argument order
Binary8p4se(0.15625 ≡ 0x2a)

julia> fma(x, y, z)                             # FMA — one rounding; muladd is the same op
Binary8p4se(2.5 ≡ 0x4a)

julia> sincos(y)                                # composite: (Sin(y), Cos(y)) componentwise
(Binary8p4se(0.25 ≡ 0x30), Binary8p4se(1.0 ≡ 0x40))
```

Because the veneers follow the session default projection, changing it changes them (see the session-defaults section of the [User Guide](#)):

```
julia> Binary8p4se(200.0) * Binary8p4se(2.0)    # RNE_SN: overflow → Inf
Binary8p4se(Inf ≡ 0x7f)

julia> DefaultProjection!(RTZ_SF);

julia> Binary8p4se(200.0) * Binary8p4se(2.0)    # now clamps to MaxFinite
Binary8p4se(224.0 ≡ 0x7e)

julia> DefaultProjection!(RNE_SN)               # restore: this setter is PROCESS-wide
(NearestTiesToEven, SatNone)
```

Code that must be insensitive to the session default names its projection: `Multiply(T, RNE_SN, x, y)`.

Restore what you set

DefaultProjection! mutates state shared by every module in the process, and nothing scopes it for you — it changes `T(x::Real)` construction too, so a later `Binary8p4se(1.1)` would truncate rather than round to nearest. Prefer `with_default_projection`, which restores on exit even if the body throws. Every example after this point assumes the pristine (`NearestTiesToEven`, `SatNone`), which is why the demonstration above puts it back.

6.2 The mapping

Base spelling	draft operation
<code>+ - * /, unary -</code>	Add, Subtract, Multiply, Divide, Negate
<code>abs, inv, sqrt</code>	Abs, Recip, Sqrt
<code>exp, exp2, expm1, log, log2, log1p</code>	Exp, Exp2, ExpMinusOne, Log, Log2, LogOnePlus
<code>sin, cos, tan, asin, acos, atan</code>	Sin, Cos, Tan, ArcSin, ArcCos, ArcTan
<code>sinh, cosh, tanh, asinh, acosh, atanh</code>	Sinh, Cosh, Tanh, ArcSinh, ArcCosh, ArcTanh
<code>sinpi, cospi, tanpi</code>	SinPi, CosPi, TanPi
<code>atan(y, x)</code>	ArcTan2 (Base's argument order)
<code>copysign, hypot</code>	CopySign, Hypot
<code>min, max</code>	Minimum, Maximum (NaN-propagating, exactly Base's float semantics)
<code>fma, muladd</code>	FMA (both: one rounding)
<code>clamp</code>	Clamp
<code>sincos, sincospi, minmax</code>	componentwise composites of the draft ops

The mapping is a declarative partition over the op lists, and the test suite asserts it is exhaustive: every operation is either mapped above or listed in `_NO_BASE_COUNTERPART`. Adding an operation to the registry forces an explicit decision here.

6.3 What is *not* mapped, and why

Draft operations with no Base spelling — call them by their draft names: `RSqrt`, `Softplus`, `ArcSinPi`/`ArcCosPi`/`ArcTanPi`/`ArcTan2Pi` (Base has only the `sinpi` family), the NaN-ignoring/magnitude/finite extremum families (`MinimumNumber`, `MaximumMagnitude`, `MinimumFinite`, ...; Base has no NaN-ignoring pair), and `FAA` (no Base fused add-add).

```
julia> min(Binary8p4se(NaN), x)           # Base semantics: NaN propagates
Binary8p4se(NaN == 0x80)

julia> MinimumNumber(Binary8p4se(NaN), x) # the NaN-ignoring draft op
Binary8p4se(1.625 == 0x45)
```

Mixed Binary formats — deliberately a promotion ... failed to change error, never a silent widening. Mixing formats is an explicit `Convert`:

```
julia> x + Convert(Binary8p4se, RNE_SN, Binary5p3sf(1.0))
Binary8p4se(2.5 == 0x4a)
```

Binary with ordinary numbers — promotes to the format's **promotion carrier** through the rules in `formats.jl`, so the result is an ordinary float, not a re-projected Binary.

For the 432 formats at rung 1 that carrier is `Float64`, which is what the example below shows. For the other 72 it is `BigFloat`: `Float64` cannot hold a `Binary16p1uf` datum, and promoting into it would return $\pm\text{Inf}$ from a perfectly finite value with no P3109 operation involved. `promotecarrier(T)` names the target.

```
julia> x + 2.0
3.625
```

Project it back explicitly when that is what you mean: `Convert(Binary8p4se, RNE_SN, decode(x) + 2.0)`.

6.4 The rest of the Base surface

Beyond arithmetic, Binary integrates where the other layers provide it: predicates and constants (`isnan`, `isfinite`, `signbit`, `zero`, `one`, `eps`, `typemin/typemax`, `floatmin/floatmax`), comparisons and `isless` on integer order keys, sort via an $O(n)$ counting sort, `nextfloat/prevfloat` as the draft Next operations, `codepoint`, and `rand/randn` (see the [User Guide](#)). All of it goes through the same projection engine and decode tables as the draft-named API.

The AbstractFloat contract

Binary `<`: `AbstractFloat`, and generic Julia code reaches for that interface without asking. The whole of it is supplied:

verb	behaviour
<code>hash</code> , <code>Base.decompose</code>	exact — a datum is $S \cdot 2^Q$, so Dict keys and Set elements work
<code>round</code> , <code>floor</code> , <code>ceil</code> , <code>trunc</code>	one rounding: the integer value is formed exactly on the carrier and projected once, under the session default's saturation
<code>round(x, r::RoundingMode)</code>	<code>RoundNearest</code> , <code>RoundNearestTiesAway</code> , <code>RoundUp</code> , <code>RoundDown</code> , <code>RoundToZero</code>
<code>exponent</code> , <code>significand</code> , <code>frexp</code>	exact; <code>DomainError</code> on zero and the non-finites, as in Base
<code>widen</code>	the format's promotion carrier — see the promotion study for why it is not another format
<code>reinterpret</code>	both directions, and the incoming one is checked against the representation invariant

```
julia> hash(Binary8p4se(1.5)) == hash(Binary8p4se(1.5))
true

julia> Base.decompose(Binary8p4se(1.5))           # 12 · 2-3 = 1.5, exactly
(12, -3, 1)

julia> floor(Binary8p4se(1.6)), ceil(Binary8p4se(1.1))
(Binary8p4se(1.0 ≡ 0x40), Binary8p4se(2.0 ≡ 0x48))

julia> reinterpret(Binary8p4se, 0x44)           # checked, unlike `rawvalue`
Binary8p4se(1.5 ≡ 0x44)
```

Refusals name themselves. Where a Base verb has no draft counterpart, the error says which and why rather than surfacing as a bare `MethodError` that cannot be told apart from an oversight:

```
julia> rem(Binary8p4se(1.5), Binary8p4se(2.0))
ERROR: ArgumentError: rem is not defined for Binary8p4se: the draft defines no remainder; ...
```

That covers `rem/mod`, `round` under `RoundFromZero` and `RoundNearestTiesUp` (no draft μ), and `significand` on the one format whose binade `[1, 2)` is truncated by its `Inf` encoding.

Chapter 7

Technical Guide

How SmallFloats.jl works inside: the encoding, the projection engine, the oracle and its correctness protocol, the performance layers, and the verification doctrine that holds it all together. Read the [User Guide](#) first; this page assumes its vocabulary.

7.1 Layer map

Source files load in dependency order, each layer speaking only downward:

layer	file	provides
formats	formats.jl	Binary{K,P,SGN,EXT}, the 504 named aliases, Group M queries
specs	projspec.jl	rounding/saturation singletons, ProjSpec{R,S}, the predefined spec grid
de-faults	defaults.jl	settable session defaults (DefaultType, DefaultProjection, ...) behind Refs; consumed via the speculation guard
codec	decode_ encode.jl	decode (generated tables + bit-composed compute), encode, order keys, counting sort, Class, Next ops
engine	project.jl	round_to_precision (mask-based Float64 core + generic core), saturate, project, project_interval
ops	ops_scalar.jl	result-kind protocol, apply_op, the operation registry, the spec register
compat	juliacompat.jl	the Base register: declarative op $\Rightarrow$ Base-function map
oracle	oracle.jl	ω -semantics for all 52 operations
tables	tables.jl	the pure-p result-table cache (unary/binary + bitwidth-gated ternary)
kernels	kernels.jl	Shape-A gathers, Shape-B scalar/threaded loops, vmap
blocks	blocks.jl	Block, block/scaled ops, exact reductions
packed	packed.jl	sub-byte PackedVector storage
approx	approx.jl	κ measurement/registry, conformance declaration
rand	rand.jl	Random-API hooks: uniform-[0,1] floor projection, clamped normal

7.2 Encoding and decoding

A value is its code point (UInt8). The bit layout is the draft's: sign (signed formats), biased exponent, trailing significand; one NaN at the negative-zero slot; no -0 ; $\pm\text{Inf}$ adjacent to the extremes in extended formats.

decode has **two shapes, selected by `decodepolicy(T)`** — a representation decision, not a semantic one.

At $K \leq 8$ it is a **@generated constant-tuple lookup**: per format, a 2^K -tuple of datums built once from the computational decode (so table and computation are correct by construction and asserted equivalent exhaustively). Constant inputs still fold — `maxfinite_datum(T)` is a compile-time constant — while runtime decode is a single indexed load (≈ 0.7 ns). Above $K = 8$ a 2^{16} -tuple is not a constant worth materializing, so the computational decode runs directly.

The computational decode assembles the datum **by bits** (normalize with `leading_zeros`, place exponent and mantissa fields, reinterpret) wherever the carrier is Float64.

Which carrier that is depends on the format's exponent bias, and is the second of the package's two axes. Float64's normal range holds every datum of the 432 formats at rung 1 — the property the bit-assembly rests on. It does not hold the other 72: Binary16pluf reaches 2^{32768} . Those decode to Float128 (rung 2) or to an exact dyadic carrier (rung 3), chosen by `datumcarrier(T)`, and every one of them is exact for the datums it accepts. So the invariant is not "Float64 is the exact carrier for all datums" — that was true only while $K \leq 8$ — but the stronger and still-checkable one:

every format's datum carrier is exact for that format's datums, and the choice is a dispatch decision derived from the exponent bias, never a caller's.

The suite checks it against an independent big-float transliteration of the draft (gate G6, exhaustively over the (datum, head) pairs), and gate G4 checks the consequence that matters: forcing a *wider* carrier never changes a code point, so the choice buys time and never an answer.

Ordering runs on **integer order keys**: a sign-magnitude fold into an unsigned type wide enough for the format's $2^K + 1$ keys (UInt16 at $K \leq 8$, wider above), monotone with the total order, NaN mapped to `typemax`. Same-format `TotalOrder`, `isless`, and the numeric comparisons are key comparisons (~ 1 ns); since a format has at most $2^K + 1$ distinct keys, vectors sort with an **$O(n)$ counting sort** installed via `Base.Sort.default` (forward and reverse orderings; anything exotic falls back to the stock algorithm).

7.3 The projection engine

`project(fr, p, X; R, sticky)` is the single write path into a code point:

RoundToPrecision → Saturate → Encode

RoundToPrecision produces a `Rounded(kind, sign, S, Q)` — an exact scaled significand S and exponent Q per the draft's $Q = \max(\lfloor \log_2 |X| \rfloor, 1-B) - P + 1$. Two implementations, proven equivalent exhaustively:

- the **generic core** (`_rtp_core`) works on any carrier (`Float64`, `Float128`, `BigFloat`) via exact power-of-two scaling and a fraction v compared against the mode's decision points;
- the **mask-based Float64 core** (`_rtp_f64`) extracts sign/exponent/significand fields directly, represents v as a 128-bit fixed-point integer with an OR-mask sticky for bits shifted out, and evaluates every mode — including the stochastic $\lfloor v \cdot 2^N \rfloor$ comparisons and RNITE ties — as integer field tests. Specials, zeros, and (Convert-only-reachable) subnormal `Float64` inputs bail to the generic core.

Symbolic sticky is how enclosures talk to the engine: `sticky` $\in \{-1, 0, +1\}$ declares the true value to be the carried value plus an infinitesimal of that sign. The engine folds it into every comparison; the delicate case — true value just *below* a representable dyadic — decrements into the previous binade and sets $v = 1^-$ (encoded in the mask core as an all-ones fixed-point fraction, which reproduces every predicate including the RNITE tie behavior). This is what lets directed modes land exactly on asymptotes: $\tanh \rightarrow 1^-$ under `TowardNegative` projects to `NextLessThan(1)`, automatically.

Saturate classifies against the format's range with two integer comparisons per side, then maps the classification through the draft's twenty saturation rows to `as-is` / `MaxFinite` / `MinFinite` / `±Inf` / `NaN`. The comparisons stay cheap because:

- the extremal magnitude in canonical (S, Q) form is a constant-folded function of the type parameters;
- the rounded value and the extremum share one lexicographic (Q, S) order (subnormals and the lowest normal binade share the same Q);
- signed formats use sign-magnitude symmetry, and unsigned underflow is just the sign bit;
- an internal HUGEQ sentinel represents "finite but astronomically large" (the $v = 1^-$ image of an infinite endpoint), so directed `SatNone` clamps it to `MaxFinite` correctly.

Encode is pure integer bit assembly, including the significand-carry renormalization and the subnormal/normal field split.

7.4 The oracle and the result-kind protocol

Every operation's defined result is computed by `weval`, which returns one of five result kinds; `apply_op` fast-splits the common one and finishes the rest:

kind	meaning	finished by
Float64	exact (specials; representable arithmetic)	direct project
Dyadic	exact at rung 3: an Int128 significand and an Int64 exponent, isbits	direct project (dyadic carrier)
Float128	exact by width analysis	direct project (Float128 carrier)
StickyF	wide-spread FMA/FAA tail: head value (Float64 or Float128) + tail sign	direct project with sticky set — no allocation
BigExactF	exact at a precision derived from the operands — bigprec, not a constant (wide-spread tail for Add; the near-impossible FAA distillation miss)	project on BigFloat
Enclose128	correctly-rounded Float128 bracket	sticky agreement, MPFR fallback
EncloseF	MPFR directed enclosure $f(\text{prec})$, optional Float128 pre-filter f_q , optional eager Float64 estimate y_d	three-stage: $y_d \rightarrow f_q \rightarrow$ interval protocol

Two **rigor classes** govern every non-Float64 path, and their arguments are never mixed:

Class R (unconditional) rests on two facts that hold without any accuracy assumption:

- **Width analysis.** Sums of decoded datums are *exactly representable* in Float128 whenever operand bits + exponent spread fit 113 bits — checked in advance by integer exponent arithmetic (Add at $\Delta E \leq 100$, FMA ≤ 92 , FAA span ≤ 98). Beyond the threshold, Add escalates to the exact MPFR path — at a precision **derived from the operands** by bigprec, not the retired 2200-bit constant, which was ample at $K \leq 8$ and insufficient for 50 of the 504 formats (gate G2 measures this) — while FMA/FAA take a non-allocating **sticky-head** shortcut: past that spread the smaller term is provably too small (by construction of the threshold) to affect anything but the tail direction of the larger one, so the result is StickyF(head, sign) — no BigFloat involved. FAA's three-term case runs a bounded Float128 2sum-distillation (Priest-style, ≤ 6 sweeps) to find that head/tail split directly; the residual BigExactF MPFR fallback exists only for the near-impossible case the distillation doesn't converge.
- **IEEE correct rounding.** Division and sqrt are correctly rounded at every binary width, by mandate. Divide/Recip therefore use the **Float64** quotient directly: it is within half an ulp of truth, so it serves as EncloseF's eager y_d estimate with no Float128 arithmetic on the path at all. Sqrt/RSqrt use the **Float128** CR result: an inexact nearest-CR value q brackets the truth in the *open* interval ($\text{prevfloat}(q)$, $\text{nextfloat}(q)$). Exactness itself is detected by an fma residual test.

Class E (envelope-conditional). Faithful-but-not-CR evaluations stand in for an enclosure only inside a generous envelope, resolved by the two-sided sticky gate. Two stages, cheapest first:

- *Float64 stage:* for operations whose Float64 libm is faithful (≤ 1 ulp $\approx 2^{-52}$ relative — the exp/log families, the hyperbolics, forward trig inside the $|x| \leq 10^{15}$ reduction window, atan, asinh, the softplus composition), the eager estimate y_d carries envelope $E = |y_d| \cdot 2^{-45}$, $\geq 2^7$ slacker than any faithful-libm error (measured margin on this machine: $\geq 2^{6.6}$).
- *Float128 stage:* libquadmath's estimate $y = f_q()$ carries $E = |y| \cdot 2^{-90}$, at least 2^{18} slacker than any published libquadmath bound.

In both stages, if the sticky-projected endpoints agree, that code point is the answer; if not, the next stage (ultimately the MPFR ladder) decides. An envelope failure can therefore only cost speed, never correctness — unless both endpoints agree *and* the envelope is wrong, which the differential-build tests rule out empirically by building every standard table twice (Float128-first and pure-MPFR) and byte-diffing, and which the exhaustive oracle cross-check re-verifies per operation against the 3072-bit protocol.

The interval protocol (`project_interval`) is the termination backbone: evaluate the MPFR enclosure at precision p ; if $lo == hi$ the value is exactly representable — project it; otherwise project lo with `sticky = +1` and hi with `sticky = -1`; agreement (projection is monotone at fixed R) yields the answer; disagreement doubles p , clamped to the `maxprec` ceiling (**derived from the format**, $\max(4096, \text{bigprec}(T) + 64)$ — a flat 4096 was ample at $K \leq 8$ and could not decide ArcCosPi at `Binary15p2ue` under a directed or stochastic rule, honored exactly even when it is not a power-of-two multiple of the 256-bit start). Termination requires that the enclosure not be chasing a value the grid can actually hit — which is why the π -scaled operations carry **Niven peels**: for dyadic arguments, $\tan(\pi r)$ takes rational values only at the quarter-integers (± 1) and $\text{atan2}/\pi$ only on the diagonals ($\pm \frac{1}{4}, \pm \frac{3}{4}$); those cases are answered exactly before any enclosure is built, and Niven's theorem proves the peel set complete.

7.5 Tables and kernels

For pure specs, unary and binary operations are **finite functions** — 256 or 65 536 entries — so the kernel layer materializes them once per $(op, \text{formats}, p)$ into a locked cache (`Dict{TableKey, Memory{UInt8}}`), double-checked locking, builds outside the lock) and serves every later array call as a gather: Shape-A, one load per element, measured 0.27 ns/elem unary and 0.5 ns/elem binary. Tables are built *through the scalar path*, so they inherit its bit-exactness; the suite asserts `table == scalar` over every entry.

Ternary (FMA, FAA, Clamp) is a finite function too — $2^{(K1+K2+K3)}$ entries — but that count spans four orders of magnitude across the 3–16 bitwidth range (512 B at $K=3$, 16 MiB at $K=8$), so one policy doesn't fit the whole range. A separate `TernaryKey` → `TernaryEntry` cache (`_ternary_table_for`, in `tables.jl`) tiers by Σ bitwidth:

- **Eager** (≤ 18 bits, all $K \leq 6$ combinations, ≤ 256 KiB): builds and caches on the first array call.
- **Adaptive** (≤ 21 bits, the $K = 7$ band, up to 2 MiB): accumulates a per-signature element count across calls and builds only once a signature has processed enough elements to amortize the build; a byte-bounded LRU eviction (`TERNARY_CACHE_BYTES`) guards against many hot signatures coexisting.
- **Compute** ($K = 8, 16+$ MiB — a table is never worth it): `vmap!` runs Shape-B, the fully specialized scalar pipeline per element, optionally split across `Threads.@threads` for long enough arrays (each ternary draw is independent under a fixed, non-stochastic p , so lanes cannot interact).

Every ternary table entry, eager or adaptive, is still built *through the scalar path* — the tiering changes when/whether the cache exists, never what it contains. Stochastic calls of any arity always take Shape-B, with the RNG resolved once per array rather than per element.

7.6 Blocks: exactness without a superaccumulator

`blockdecode` produces each lane's `scale × element` exactly (≤ 17 -bit significands in `Float64`). Reductions then apply **span filters**: one integer pass over lane exponents decides whether the whole sum (or dot product, with ≤ 33 -bit lane products) is exactly representable in `Float128`; if so, a plain `Float128` accumulation is the exact answer; if not, an exact big-float accumulation takes over. Either way there is exactly one projection, at the end. `ωBlockProject` follows the draft's special rows for the scale (`NaN`, `0`, `±Inf`) and divides each element result by the scale through its own cheapest-first CR-bracket / enclosure cascade (exact `Float64` quotient → CR `Float128` → bracket/pre-filter → MPFR interval), mirroring the scalar quotient group's rigor arguments.

7.7 The κ registry and conformance

κ is the maximum code-point distance (along the total order) between an implementation's result and the defined result, over inputs with finite defined results; any mismatch on non-finite defined results makes $\kappa = \text{NaN}$. Because inputs are enumerable, `register_approx!` measures κ exhaustively at registration and rejects understatement — a declared bound is a verified property, not a promise. `conformance()` assembles the declaration live from the operation registry, the table cache, and the approximation registry.

7.8 Verification doctrine

The value sets are small enough that sampling is never necessary, so the suite enumerates — ≈ 8.9 M assertions in all:

- formats against an independent draft transliteration (14 679);
- ordering over all 2.5 M same-format pairs plus Next-op edge tables (7.6 M);
- every unary operation on every input against a 3072-bit protocol run; divide and the ternaries exhaustively;
- stochastic R-sweeps with directed-asymptote pins;
- table $\equiv$ scalar over every entry, including the ternary tiers (eager and adaptive) against the scalar path;
- the sticky-head FMA/FAA escalation against the MPFR reference across every rounding-mode family and adversarial cancellation cases;
- blocks against a from-scratch reference composition;
- Float128 carrier $\equiv$ Float64 carrier, and Float128-first $\equiv$ MPFR differential builds;
- the mask rounding core $\equiv$ the generic core over datums and reachable sums/products at boundary stochastic budgets $N \in \{45, 60\}$;
- packed round-trips, and κ /conformance behavior.

Deterministic **specialization regressions** (concrete inferred return types at the public entry points, zero warm-path allocation) stand in for timing assertions.

7.9 Benchmark doctrine

Recorded after two measurement post-mortems: a benchmark closure over any non-const global measures Julia's dispatch machinery, not the code under test — and it distorts *ratios*, not just absolutes, because dispatch cost varies with call shape (a dynamic keyword call costs $\sim 1 \mu\text{s}$; a dynamic positional call far less; six unresolved interior sites cost six times one).

The rules the shipped benchmarking/benchmarking.jl enforces structurally:

- format types enter as type parameters, never as globals;
- operands come from untimed setup;
- functions retrieved reflectively pass through argument barriers to specialize;
- a preflight aborts the run if warm scalar paths allocate — including the wide-spread FMA/FAA sticky-head path.

The table-build section reports both the cold build (cache evicted per sample) and the steady-state warm cache hit, since callers amortize the former through the latter. Vary one binding per variant; verify specialization before believing a number.

7.10 Deliberate limitations

No implicit cross-format arithmetic (promotion is to Float64, explicitly). No in-place packed arithmetic. Threading is opt-in and narrow: only the untabled ternary ($K = 8$) compute kernel threads, and only above a size cutoff and when `Threads.nthreads() > 1`; every other kernel is single-threaded. Irrational/Rational inputs to `Convert` are rejected rather than double-rounded silently. The Float128 machinery never changes results — disabling it (`SmallFloats_Float128=disable`) is a tested no-op semantically.

Chapter 8

Technical Examples

Internals-level recipes: introspecting the pipeline, verifying custom code against the oracle, measuring κ , and benchmarking without fooling yourself. Outputs are captured from real sessions. Several examples use unexported internals (`SmallFloats.round_to_precision`, `SmallFloats.project`, ...); those are stable enough to learn from but are not covered by semantic-versioning guarantees.

8.1 Basic

Watching RoundToPrecision work

`round_to_precision` returns the draft's exact (sign, S, Q) decomposition before saturation ever sees it:

```
using SmallFloats
using SmallFloats: round_to_precision

r = round_to_precision(4, 8, NearestTiesToEven(), 2.30078125, 0, 0) # P=4, bias=8
(r.sign, r.S, r.Q, r.S * 2.0^r.Q)
```

```
(1, 9, -2, 2.25)          # S·2^Q = 9/4: the nearest P=4 value
```

Symbolic sticky: projecting "just below" a number

The engine accepts `sticky` $\in \{-1, 0, +1\}$ meaning "the true value is the carrier plus an infinitesimal of this sign". This is how enclosure endpoints and asymptotes ($\tanh \rightarrow 1^-$) are projected exactly:

```
using SmallFloats: project
project(Binary8p4se, ProjSpec(TowardNegative(), SatNone()), 1.0; sticky=-1)
```

```
Binary8p4se(0.9375  $\equiv$  0x3f)    # == NextLessThan(1.0): the engine crossed the binade
```

Tables are the scalar path, memoized

```
using SmallFloats: get_table
empty_tables!()
tbl = get_table(:Exp, Binary8p4se, Binary8p4se, RNE_SN)
(tbl[Int(0x45) + 1],
 codepoint(Exp(Binary8p4se, RNE_SN, rawvalue(Binary8p4se, 0x45))),
 table_bytes())
```

```
(0x52, 0x52, 256)          # table entry ≡ scalar result; one 256-byte table cached
```

Order keys

Ordering is integer arithmetic: a sign-magnitude fold, monotone with the total order, NaN at the top. This is what comparisons and the O(n) counting sort run on:

```
using SmallFloats: order_key
[(v, order_key(v)) for v in (Binary8p4se(-1.0), Binary8p4se(0.0),
                             Binary8p4se(1.0), Binary8p4se(NaN))]
```

```
-1.0 → 64      0.0 → 129      1.0 → 193      NaN → 65535
```

8.2 General AI

Ranking safety: an exhaustive monotonicity audit of score conversion

Decision and ranking systems consume scores through *order*. If scores are produced in one format and compared in another, the conversion must be monotone — and formats are small enough to prove it by enumeration rather than trust it. Walk every Binary8p3se datum in total order and check that its Binary8p4se image never goes backward on the integer order keys:

```
using SmallFloats
using SmallFloats: order_key

function monotone_conversion(::Type{From}, ::Type{To}) where {From, To}
    prev = nothing
    for v in sort(From.(0x00:UInt8(2^bitwidth(From) - 1)))
        isnan(decode(v)) && continue
        g = Convert(To, RNE_SN, v)
        prev !== nothing && order_key(g) < order_key(prev) && return false
        prev = g
    end
    true
end
monotone_conversion(Binary8p3se, Binary8p4se)
```

```
true
```

Run this for every (From, To, p) triple your system actually uses; it is a few hundred integer comparisons per triple. A ranking pipeline whose conversions all pass this audit cannot invert a preference by changing formats — a guarantee no amount of spot-testing provides.

k-safe decision margins for approximate evaluators

Search under a compute budget often wants a cheap, approximate evaluation function. The κ registry turns "how approximate?" into a *measured* code-point bound — and code-point bounds compose into a decision rule: **if two defined evaluations differ by more than 2κ code points along the total order, the approximate evaluator cannot invert their comparison.** Verify the rule exhaustively for a $\kappa = 2$ evaluator:

```
using SmallFloats: order_key

fast(x) = (r = Exp(Binary8p4se, RNE_SN, x);
           isfinite(decode(r)) ? NextGreaterThan(NextGreaterThan(r)) : r)
κ, exhaustive = measure_kappa(fast, :Exp, Binary8p4se, (Binary8p4se,), RNE_SN)

function margin_audit(fast, κ)
    codes = [rawvalue(Binary8p4se, UInt8(c)) for c in 0:255]
    safe = violations = 0
    for a in codes, b in codes
        da, db = Exp(Binary8p4se, RNE_SN, a), Exp(Binary8p4se, RNE_SN, b)
        (isnan(decode(da)) || isnan(decode(db))) && continue
        codedistance(da, db) > 2κ || continue
        safe += 1
        (order_key(fast(a)) < order_key(fast(b))) ==
            (order_key(da) < order_key(db)) || (violations += 1)
    end
    (safe, violations)
end
(κ, exhaustive, margin_audit(fast, κ)...)

(2.0, true, 49522, 0)
```

49,522 operand pairs clear the 2κ margin, and the approximate evaluator agrees with the defined ordering on every one of them — zero violations, exhaustively. Inside the margin, comparisons are genuinely undecidable at this κ ; a search can treat sub-margin comparisons as ties to expand, or escalate those few nodes to the exact evaluator. Either way the pruning is *provably* sound, with κ measured at registration rather than promised.

8.3 Machine Learning

Verifying a quantizer exhaustively against an independent reference

Formats are small enough that "test on a few points" is never necessary. Here every in-range datum of Binary8p3se is projected into Binary8p4se and checked against a brute-force nearest search under 256-bit arithmetic (distance agreement; the draft's tie rule is the projection engine's job and is pinned elsewhere in the suite):

```
using SmallFloats

function ref_nearest_distance(::Type{T}, x) where {T}
    fins = [v for v in (rawvalue(T, UInt8(c)) for c in 0:255) if isfinite(decode(v))]
    minimum(abs(setprecision(() -> BigFloat(decode(v)) - BigFloat(x), BigFloat, 256))
            for v in fins)
end

function verify_quantizer()
    ok = true
    for c in 0x00:0xff
        x = decode(rawvalue(Binary8p3se, c))
        (isfinite(x) && abs(x) <= decode(MaxFiniteOf(Binary8p4se))) || continue
        got = Convert(Binary8p4se, RNE_SN, x)
        ok &= abs(decode(got) - x) == Float64(ref_nearest_distance(Binary8p4se, x))
    end
    ok
end

verify_quantizer()
```

```
true
```

This pattern — enumerate the inputs, compare against an independently written reference — is how the entire package is tested, and it is available to *your* quantization code at trivial cost.

The κ workflow, including the part where it says no

Register an approximate kernel and the registry measures it; understate the bound and it refuses:

```
step2(x) = (r = Exp(Binary8p4se, RNE_SN, x);
            isfinite(decode(r)) ? NextGreaterThan(NextGreaterThan(r)) : r)

measure_kappa(step2, :Exp, Binary8p4se, (Binary8p4se,), RNE_SN)
```

```
(2.0, true)           #  $\kappa$  = 2 code points, verified over all 256 inputs
```

```
register_approx!(:cheater, :Exp, Binary8p4se, (Binary8p4se,), RNE_SN, step2;  $\kappa$ =1)
```

```
ERROR: ArgumentError: declared  $\kappa$  = 1.0 understates measured  $\kappa$  = 2.0 – registration rejected
```

Exporting the conformance declaration

```
d = conformance_dict()           # plain nested Dict{String,Any}
(d["package"], length(d["formats"]), length(d["operations"]),
 [s["op"] for s in d["cached_specializations"]])
```

Serialize `d` with any JSON/TOML writer to attach a machine-readable conformance statement to experiment artifacts.

8.4 Deep Learning

Proving your fused kernel exact: BlockDotProduct vs 512-bit truth

The block layer promises *one* projection with exact lane arithmetic. Trust, then verify — against big-float truth, over random blocks with mixed scales and an honest special-value mix:

```
using SmallFloats, Random

rng = Xoshiro(5)
mk() = Block(Binary8p1uf(2.0^rand(rng, -3:3)),
             ntuple(_ -> rawvalue(Binary8p4se, UInt8(rand(rng, 0:255))), 32))

function verify_dots(trials)
    for _ in 1:trials
        bx, by = mk(), mk()
        lx = [decode(bx.s) * decode(v) for v in bx.x]
        ly = [decode(by.s) * decode(v) for v in by.x]
        (any(!isfinite, lx) || any(!isfinite, ly)) && continue
        truth = setprecision(() -> sum(BigFloat(lx[i]) * BigFloat(ly[i]) for i in 1:32),
                                     BigFloat, 512)
        got = BlockDotProduct(Binary8p4se, RNE_SN, bx, by)
        ref = setprecision(() -> SmallFloats.project(Binary8p4se, RNE_SN, truth), BigFloat, 512)
        codepoint(got) == codepoint(ref) || return false
    end
    true
end
verify_dots(200)
```

```
true
```

The same shape verifies any custom fused kernel you write: compose the truth in BigFloat, project once, compare code points.

Stochastic rounding, audited: the full-R sweep

Every stochastic projection is a deterministic function of the draw R , so its *distribution* is checkable exactly. For $x = 2 + 3/64$ in Binary8p4se the fraction is $v = 3/16$ of an ulp, so StochasticA{4} must round up for exactly 3 of the 16 draws:

```
σ4 = RSA_SN(4) # = ProjSpec(StochasticA{4}(), SatNone())
x = 2.0 + 3/64
count(decode(SmallFloats.project(Binary8p4se, σ4, x; R)) == 2.25 for R in 0:15)
```

```
3
```

Sweeping R like this turns "is my stochastic pipeline unbiased?" from a statistical question into an exhaustive one — the pattern the shipped test suite uses.

An accelerator-style activation, κ -measured: hard-tanh vs Tanh

Hardware activation units often ship piecewise approximations. The κ machinery makes the substitution honest: measure the approximation's worst code-point deviation from the defined activation, exhaustively, and register it under that measured bound. Hard-tanh — `clamp(x, -1, 1)` — as a stand-in for Tanh:

```
one4 = Binary8p4se(1.0)
hardtanh(x) = Clamp(Binary8p4se, RNE_SN, x, Negate(one4), one4)

κ, exhaustive = measure_kappa(hardtanh, :Tanh, Binary8p4se, (Binary8p4se,), RNE_SN)
(κ, exhaustive)
```

```
(4.0, true) # worst deviation: 4 code points, verified on all 256 inputs
```

Where is it worst? At $x = 1.0$: hard-tanh returns 1.0 while the defined `tanh(1.0)` is 0.75 — four code points along the total order. The registration round-trip, including the conformance surface:

```
impl = register_approx! (:hardtanh_act, :Tanh, Binary8p4se, (Binary8p4se,),
                        RNE_SN, hardtanh; κ=4)
(kappa(:hardtanh_act), kappa_measured(impl), :hardtanh_act in list_approx())
```

```
(4.0, 4.0, true) # declared = measured; conformance_report() now lists it
```

(`unregister_approx! (:hardtanh_act)` removes it.) Declaring $\kappa=3$ would be *rejected* — understatement is impossible by construction. The decision of whether a $\kappa = 4$ activation is acceptable now belongs to your accuracy budget, not to hope: combined with the margin rule from the General AI section, any two pre-activations whose defined Tanh images sit more than 8 code points apart keep their order under hard-tanh, provably.

Benchmarking without measuring the dispatcher

The package's benchmark doctrine, in one snippet (needs the benchmarking/ environment for Chairmarks). Format types enter as **type parameters**; operands come from Chairmarks' *untimed* setup; and if you retrieve functions reflectively (`getfield(SmallFloats, op)`), pass them through an argument barrier so they specialize:

```
using Chairmarks, SmallFloats, Random
using Statistics: median

function bench_add(::Type{T}) where {T<:Binary} # T: type parameter, not a global
    pool = [rawvalue(T, rand(UInt8)) for _ in 1:4096]
    @be (rand(pool), rand(pool)) (t -> Add(T, RNE_SN, t[1], t[2]))(t)
end

b = bench_add(Binary8p4se)
(round(median(b).time * 1e9; digits=1), median(b).allocs)
```

```
(16.3, 0.0) # ns per full scalar Add, zero allocations
```

The 60x trap

The same call with `T` read from a non-const global measures $\sim 1 \mu\text{s}$ — Julia's dynamic keyword dispatch, not this package. Two project post-mortems trace to exactly this mistake; the shipped `benchmarking/benchmarking.jl` asserts specialization (zero warm-path allocation) before it believes any number, and so should yours.

Chapter 9

SmallFloats.jl benchmark report

Generated: 2026-07-30 12:32 UTC · Julia 1.12.6 · alderlake (32 logical CPUs, 1 Julia thread) · Float128 paths: enabled · Chairmarks 1.3.1

Reference format for per-operation tables: Binary8p4se under (NearestTiesToEven, SatNone). Every table names its operand class: the scalar-operation tables appear in four variants — all code points (NaN and $\pm\text{Inf}$ sampled), NaN excluded, finite-only, and per-operation in-domain — and every other sampled table uses the all-code-points pool, identified in its note. Times are per call; minima first, medians alongside. Methodology per the recorded benchmark doctrine: type-parameterized barriers, untimed setup, specialization preflight.

9.1 Core primitives

The decode/compare/step/classify layer plus the projection engine. Operands: all code points — NaN and $\pm\text{Inf}$ sampled.

operation	min	median	allocs
decode	1.6 ns	1.6 ns	0
order_key	1.6 ns	1.8 ns	0
$x < y$	1.8 ns	1.8 ns	0
TotalOrder	2.1 ns	2.1 ns	0
Class	1.6 ns	2.5 ns	0
NextGreaterThan	1.8 ns	2.1 ns	0
project (RNE·SatNone)	2.2 ns	6.7 ns	0
project (StochasticA[8], R drawn)	2.1 ns	7.1 ns	0

9.2 Scalar operations

Each arity is measured under four operand classes — safe args, no NaN/Inf, no NaN, and all code points — so operand-class effects (NaN fast rows, $\pm\text{Inf}$ special rows) are separable.

Unary (30)

Safe args

Finite operands within each operation's safe domain — the fully unmasked per-operation scalar cost. The argument-restricted ops (Sqrt, RSqrt, Log, Log2, LogOnePlus, Recip, Divide, ArcSin, ArcCos, ArcCosh, ArcTanh) draw from explicit per-argument safe-domain predicates; every other op uses finite operand tuples whose defined result is not NaN (oracle-derived). Sorted by median.

operation	min	median	allocs
Abs	4.9 ns	6.7 ns	0
Negate	4.8 ns	6.9 ns	0
Recip	7.9 ns	18.0 ns	0
Sqrt	5.7 ns	18.7 ns	0
RSqrt	9.6 ns	20.7 ns	0
Cosh	7.3 ns	21.4 ns	0
Sin	6.2 ns	21.7 ns	0
Cos	7.3 ns	21.7 ns	0
Exp2	7.3 ns	22.4 ns	0
Exp	7.4 ns	22.5 ns	0
ArcSin	5.7 ns	23.2 ns	0
ExpMinusOne	5.3 ns	23.2 ns	0
Tanh	5.8 ns	23.2 ns	0
ArcCos	5.7 ns	24.3 ns	0
ArcSinPi	5.5 ns	25.5 ns	0
Sinh	5.7 ns	25.6 ns	0
Log	5.5 ns	25.7 ns	0
Log2	5.9 ns	26.6 ns	0
Tan	5.3 ns	26.6 ns	0
LogOnePlus	5.4 ns	26.7 ns	0
ArcCosPi	6.1 ns	26.8 ns	0
CosPi	6.9 ns	27.8 ns	0
SinPi	5.6 ns	28.1 ns	0
ArcTan	5.2 ns	28.4 ns	0
ArcTanPi	5.1 ns	31.0 ns	0
ArcTanh	6.2 ns	31.1 ns	0
TanPi	6.2 ns	31.1 ns	0
ArcCosh	5.5 ns	33.3 ns	0
Softplus	13.8 ns	33.3 ns	0
ArcSinh	6.2 ns	33.4 ns	0

No NaN, Inf args

Operands exclude NaN and $\pm\text{Inf}$; finite datums only (zeros and subnormals kept). Domain-restricted ops still take NaN fast rows on out-of-domain finite operands. Sorted by median.

operation	min	median	allocs
Sqrt	1.9 ns	2.1 ns	0
ArcCosh	2.1 ns	2.1 ns	0
ArcSinPi	1.9 ns	2.2 ns	0
Abs	4.3 ns	6.7 ns	0
Negate	4.8 ns	6.9 ns	0
Log2	2.1 ns	8.6 ns	0
RSqrt	2.1 ns	9.5 ns	0
Recip	1.9 ns	17.8 ns	0
Cos	7.1 ns	21.2 ns	0
Sin	5.4 ns	21.5 ns	0
Cosh	7.0 ns	21.6 ns	0
Exp2	7.3 ns	22.3 ns	0
Exp	7.0 ns	22.4 ns	0
ExpMinusOne	4.5 ns	22.8 ns	0
ArcSin	2.3 ns	22.8 ns	0
Tanh	5.7 ns	23.1 ns	0
ArcCos	2.5 ns	23.8 ns	0
Log	2.3 ns	25.3 ns	0
Sinh	5.1 ns	25.6 ns	0
LogOnePlus	1.9 ns	26.3 ns	0
ArcCosPi	2.1 ns	26.5 ns	0
Tan	6.1 ns	26.6 ns	0
CosPi	6.4 ns	27.4 ns	0
SinPi	5.2 ns	27.7 ns	0
ArcTan	5.5 ns	28.3 ns	0
ArcTanh	2.1 ns	28.7 ns	0
TanPi	5.3 ns	30.8 ns	0
ArcTanPi	4.8 ns	30.8 ns	0
Softplus	14.3 ns	33.5 ns	0
ArcSinh	5.0 ns	33.6 ns	0

No NaN args

Operands exclude the NaN code point; $\pm\text{Inf}$ and every finite datum are sampled. Sorted by median.

operation	min	median	allocs
ArcCosh	2.1 ns	2.1 ns	0
RSqrt	2.1 ns	2.2 ns	0
Log	2.3 ns	2.5 ns	0
ArcSin	2.5 ns	2.6 ns	0
Sqrt	1.9 ns	5.1 ns	0
Abs	4.4 ns	6.7 ns	0
Negate	4.4 ns	6.8 ns	0
ArcSinPi	2.0 ns	7.3 ns	0
Log2	2.1 ns	8.6 ns	0
Recip	1.9 ns	17.8 ns	0
Cosh	5.0 ns	21.2 ns	0
Sin	2.1 ns	21.4 ns	0
Cos	1.9 ns	21.5 ns	0
Exp2	5.5 ns	22.3 ns	0
Exp	5.0 ns	22.3 ns	0
Tanh	5.7 ns	23.0 ns	0
ExpMinusOne	4.6 ns	23.1 ns	0
ArcCos	2.6 ns	23.8 ns	0
Sinh	5.1 ns	25.6 ns	0
LogOnePlus	1.9 ns	26.2 ns	0
ArcCosPi	2.1 ns	26.6 ns	0
Tan	2.0 ns	27.2 ns	0
CosPi	2.5 ns	27.5 ns	0
SinPi	2.1 ns	27.7 ns	0
ArcTan	5.7 ns	28.3 ns	0
ArcTanh	2.1 ns	28.6 ns	0
ArcTanPi	4.7 ns	30.8 ns	0
TanPi	2.1 ns	30.8 ns	0
Softplus	4.8 ns	33.5 ns	0
ArcSinh	4.5 ns	33.6 ns	0

All code points

Operands drawn uniformly over ALL code points — NaN and $\pm\text{Inf}$ are sampled; medians of domain-restricted ops are diluted by instant NaN rows. Sorted by median. Transcendental rows mix special-row fast returns with enclosure-path evaluations, so these are *scalar-path* costs; bulk unary work routes through 256-byte tables (see Array kernels).

operation	min	median	allocs
Sqrt	1.9 ns	2.0 ns	0
ArcCosh	2.1 ns	2.1 ns	0
ArcSinPi	1.9 ns	2.1 ns	0
RSqrt	2.1 ns	2.1 ns	0
Log	2.3 ns	3.8 ns	0
Log2	2.1 ns	5.5 ns	0
ArcSin	2.1 ns	5.8 ns	0
ArcTanh	2.1 ns	6.6 ns	0
Abs	2.0 ns	6.7 ns	0
Negate	2.1 ns	6.8 ns	0
Recip	1.8 ns	17.8 ns	0
Cosh	2.3 ns	21.0 ns	0
Cos	1.9 ns	21.3 ns	0
Sin	2.1 ns	21.5 ns	0
Exp2	2.1 ns	22.3 ns	0
Exp	2.0 ns	22.3 ns	0
ExpMinusOne	2.0 ns	22.7 ns	0
Tanh	2.3 ns	23.1 ns	0
ArcCos	2.3 ns	23.7 ns	0
Sinh	2.3 ns	25.5 ns	0
LogOnePlus	1.8 ns	26.4 ns	0
Tan	2.0 ns	26.5 ns	0
ArcCosPi	1.9 ns	26.6 ns	0
CosPi	2.3 ns	27.5 ns	0
SinPi	2.0 ns	27.7 ns	0
ArcTan	2.2 ns	28.2 ns	0
ArcTanPi	2.1 ns	30.8 ns	0
TanPi	1.9 ns	30.8 ns	0
Softplus	2.0 ns	33.5 ns	0
ArcSinh	2.1 ns	33.6 ns	0

Binary (18)

Safe args

Finite operands within each operation's safe domain — the fully unmasked per-operation scalar cost. The argument-restricted ops (Sqrt, RSqrt, Log, Log2, LogOnePlus, Recip, Divide, ArcSin, ArcCos, ArcCosh, ArcTanh) draw from explicit per-argument safe-domain predicates; every other op uses finite operand tuples whose defined result is not NaN (oracle-derived). Sorted by median.

operation	min	median	allocs
MinimumMagnitude	4.9 ns	7.1 ns	0
MaximumMagnitude	6.9 ns	7.1 ns	0
CopySign	4.8 ns	7.2 ns	0
Maximum	4.5 ns	7.2 ns	0
Minimum	4.6 ns	7.2 ns	0
MaximumFinite	4.8 ns	7.2 ns	0
MinimumFinite	4.9 ns	7.3 ns	0
MinimumNumber	4.8 ns	7.3 ns	0
MaximumNumber	5.0 ns	7.3 ns	0
MaximumMagnitudeNumber	7.1 ns	7.3 ns	0
MinimumMagnitudeNumber	4.8 ns	7.3 ns	0
Multiply	4.9 ns	7.4 ns	0
Add	7.1 ns	9.0 ns	0
Subtract	7.4 ns	9.2 ns	0
Divide	6.4 ns	18.3 ns	0
Hypot	7.8 ns	28.0 ns	0
ArcTan2	8.0 ns	32.7 ns	0
ArcTan2Pi	7.2 ns	35.3 ns	0

No NaN, Inf args

Operands exclude NaN and $\pm\text{Inf}$; finite datums only (zeros and subnormals kept). Domain-restricted ops still take NaN fast rows on out-of-domain finite operands. Sorted by median.

operation	min	median	allocs
MinimumMagnitude	4.8 ns	7.1 ns	0
MaximumMagnitude	6.9 ns	7.1 ns	0
Minimum	4.7 ns	7.2 ns	0
Maximum	4.6 ns	7.2 ns	0
CopySign	4.7 ns	7.2 ns	0
MinimumFinite	5.0 ns	7.2 ns	0
MaximumFinite	5.1 ns	7.2 ns	0
MinimumNumber	6.1 ns	7.3 ns	0
MaximumNumber	4.8 ns	7.3 ns	0
MinimumMagnitudeNumber	4.8 ns	7.3 ns	0
MaximumMagnitudeNumber	7.2 ns	7.3 ns	0
Multiply	4.9 ns	7.4 ns	0
Add	7.1 ns	9.0 ns	0
Subtract	7.9 ns	9.5 ns	0
Divide	2.6 ns	18.3 ns	0
Hypot	8.2 ns	28.0 ns	0
ArcTan2	9.0 ns	32.7 ns	0
ArcTan2Pi	7.1 ns	35.4 ns	0

No NaN args

Operands exclude the NaN code point; $\pm\text{Inf}$ and every finite datum are sampled. Sorted by median.

operation	min	median	allocs
MinimumMagnitude	4.6 ns	7.1 ns	0
MaximumMagnitude	5.0 ns	7.1 ns	0
CopySign	5.0 ns	7.2 ns	0
Minimum	4.6 ns	7.2 ns	0
Maximum	4.6 ns	7.2 ns	0
MaximumFinite	5.0 ns	7.2 ns	0
MinimumFinite	5.0 ns	7.2 ns	0
MinimumNumber	5.1 ns	7.3 ns	0
MaximumNumber	4.6 ns	7.3 ns	0
MinimumMagnitudeNumber	4.6 ns	7.3 ns	0
MaximumMagnitudeNumber	5.1 ns	7.3 ns	0
Multiply	5.0 ns	7.4 ns	0
Add	6.3 ns	9.0 ns	0
Subtract	7.2 ns	9.5 ns	0
Divide	2.6 ns	18.3 ns	0
Hypot	6.1 ns	28.0 ns	0
ArcTan2	7.1 ns	32.7 ns	0
ArcTan2Pi	6.7 ns	35.4 ns	0

All code points

Operands drawn uniformly over ALL code points — NaN and $\pm\text{Inf}$ are sampled; medians of domain-restricted ops are diluted by instant NaN rows. Sorted by median.

operation	min	median	allocs
MinimumMagnitude	2.1 ns	7.1 ns	0
MaximumMagnitude	1.9 ns	7.1 ns	0
CopySign	2.1 ns	7.2 ns	0
Minimum	2.1 ns	7.2 ns	0
Maximum	1.8 ns	7.2 ns	0
MaximumFinite	5.6 ns	7.2 ns	0
MinimumFinite	5.0 ns	7.2 ns	0
MinimumNumber	5.1 ns	7.3 ns	0
MaximumNumber	4.6 ns	7.3 ns	0
MinimumMagnitudeNumber	4.6 ns	7.3 ns	0
MaximumMagnitudeNumber	5.0 ns	7.4 ns	0
Multiply	2.1 ns	7.4 ns	0
Add	3.7 ns	9.0 ns	0
Subtract	4.6 ns	9.5 ns	0
Divide	2.5 ns	18.3 ns	0
Hypot	3.0 ns	28.0 ns	0
ArcTan2	4.6 ns	32.7 ns	0
ArcTan2Pi	3.8 ns	35.3 ns	0

Ternary (3)

Safe args

Finite operands within each operation's safe domain — the fully unmasked per-operation scalar cost. The argument-restricted ops (Sqrt, RSqrt, Log, Log2, LogOnePlus, Recip, Divide, ArcSin, ArcCos, ArcCosh, ArcTanh) draw from explicit per-argument safe-domain predicates; every other op uses finite operand tuples whose defined result is not NaN (oracle-derived). Sorted by median.

operation	min	median	allocs
Clamp	5.1 ns	8.0 ns	0
FMA	8.1 ns	9.6 ns	0
FAA	8.3 ns	9.9 ns	0

No NaN, Inf args

Operands exclude NaN and $\pm\text{Inf}$; finite datums only (zeros and subnormals kept). Domain-restricted ops still take NaN fast rows on out-of-domain finite operands. Sorted by median.

operation	min	median	allocs
Clamp	5.5 ns	8.0 ns	0
FMA	8.3 ns	9.7 ns	0
FAA	8.1 ns	9.7 ns	0

No NaN args

Operands exclude the NaN code point; $\pm\text{Inf}$ and every finite datum are sampled. Sorted by median.

operation	min	median	allocs
Clamp	5.2 ns	8.0 ns	0
FMA	6.9 ns	9.7 ns	0
FAA	7.1 ns	9.7 ns	0

All code points

Operands drawn uniformly over ALL code points — NaN and $\pm\text{Inf}$ are sampled; medians of domain-restricted ops are diluted by instant NaN rows. Sorted by median.

operation	min	median	allocs
Clamp	2.1 ns	8.0 ns	0
FMA	3.7 ns	9.7 ns	0
FAA	4.1 ns	9.7 ns	0

9.3 Sensitivity studies

How the scalar costs move with the format and with the projection specification.

Format sensitivity

Same three binary ops across formats; Binary8pluf exercises the wide-exponent-spread escalations, small-K formats the tiny-table regime. Operands: all code points — NaN and $\pm\text{Inf}$ sampled.

operation	min	median	allocs
Add(Binary8p4se)	3.7 ns	9.0 ns	0
Divide(Binary8p4se)	2.5 ns	18.3 ns	0
Multiply(Binary8p4se)	2.1 ns	7.4 ns	0
Add(Binary8p3sf)	3.9 ns	8.7 ns	0
Divide(Binary8p3sf)	2.6 ns	16.4 ns	0
Multiply(Binary8p3sf)	2.1 ns	7.2 ns	0
Add(Binary8pluf)	4.7 ns	74.0 ns	0
Divide(Binary8pluf)	2.6 ns	8.4 ns	0
Multiply(Binary8pluf)	2.1 ns	6.9 ns	0
Add(Binary5p2se)	4.0 ns	9.1 ns	0
Divide(Binary5p2se)	2.5 ns	9.0 ns	0
Multiply(Binary5p2se)	1.9 ns	7.4 ns	0
Add(Binary3plse)	4.0 ns	7.6 ns	0
Divide(Binary3plse)	2.6 ns	6.3 ns	0
Multiply(Binary3plse)	2.1 ns	5.9 ns	0

Projection by rounding/saturation mode

project(Binary8p4se, p , x) over the mode vocabulary (stochastic budgets $N = 8$). Operands: all code points — NaN and $\pm\text{Inf}$ sampled.

operation	min	median	allocs
NearestTiesToEven	2.0 ns	6.7 ns	0
NearestTiesToAway	2.8 ns	7.7 ns	0
TowardPositive	3.0 ns	6.0 ns	0
TowardNegative	2.6 ns	5.9 ns	0
TowardZero	2.5 ns	5.3 ns	0
ToOdd	2.0 ns	6.6 ns	0
StochasticA[8]	2.0 ns	7.0 ns	0
StochasticB[8]	2.5 ns	7.1 ns	0
StochasticC[8]	2.3 ns	7.1 ns	0
RNE · SatFinite	2.1 ns	7.6 ns	0
RNE · SatPropagate	2.1 ns	7.2 ns	0

9.4 Kernels and storage

Array-shaped work: the table-gather and compute kernels, the ternary bitwidth policy, sorting, table builds, blocks, and packed/converted storage.

Array kernels (vmap)

Warm caches: table specializations prebuilt, so table rows measure the gather; scalar-loop rows measure the full compute pipeline per element. The ternary row here is Binary8p4se (K=8, always the compute path); see the next section for how the ternary bitwidth policy behaves across K. Operands: all code points — NaN and $\pm\text{Inf}$ sampled.

operation	min	median	allocs	per element
vmap unary (table gather), n=65536	8.56 μs	8.71 μs	0	0.13 ns/elem — 7.52 Gelem/s
vmap binary (table gather), n=65536	16.45 μs	16.81 μs	0	0.26 ns/elem — 3.9 Gelem/s
vmap ternary (scalar loop), n=65536	1.01 ms	1.02 ms	0	15.54 ns/elem — 0.06 Gelem/s
vmap binary stochastic (scalar loop), n=65536	646.57 μs	655.78 μs	0	10.01 ns/elem — 0.1 Gelem/s
vmap unary through PackedVector, n=65536	76.97 μs	90.29 μs	773	1.38 ns/elem — 0.73 Gelem/s

Ternary bitwidth tiers (FMA/FAA)

FMA/FAA/Clamp are total functions on $2^{(K1+K2+K3)}$ code points, but that count spans 512 B (K=3) to 16 MiB (K=8), so the array kernel tables small operand formats eagerly, tables mid-size ones adaptively (after enough elements amortize the build; not shown here — see the adaptive-cache gate in `test/ternary_opt.jl`), and always runs the scalar compute kernel at K=8, threaded above a size cutoff when `Threads.nthreads() > 1`. Each tier's optimized row is paired with a scalar-loop baseline (policy Refs forced off around the measurement, restored after) so the win is visible per tier; this process has 1 Julia thread. No threaded/sequential comparison below — rerun with `julia -t N (N > 1)` to see it. Same reference format, p, and operand pool discipline as Array kernels above.

operation	min	median	allocs	per element
FMA K=4 (eager table), n=65536	19.58 μs	20.46 μs	0	0.31 ns/elem — 3.2 Gelem/s
FMA K=4 (eager table), scalar-loop baseline, n=65536	967.88 μs	974.03 μs	0	14.86 ns/elem — 0.07 Gelem/s
FMA K=6 (eager table), n=65536	24.03 μs	24.34 μs	0	0.37 ns/elem — 2.69 Gelem/s
FMA K=6 (eager table), scalar-loop baseline, n=65536	1.0 ms	1.01 ms	0	15.46 ns/elem — 0.06 Gelem/s
FMA K=8 (compute), n=65536	990.69 μs	1.0 ms	0	15.31 ns/elem — 0.07 Gelem/s
FMA K=8 (compute), scalar-loop baseline, n=65536	990.11 μs	1.0 ms	0	15.27 ns/elem — 0.07 Gelem/s

Sorting (64 K values)

Counting sort is installed as the default algorithm for Binary vectors. Operands: all code points — NaN and $\pm\text{Inf}$ sampled.

operation	min	median	allocs
sort! (counting sort via defalg), n=65536	155.36 μs	157.4 μs	2
sort! (stock comparison sort), n=65536	1.36 ms	1.37 ms	3
sort! rev=true (counting sort), n=65536	155.48 μs	157.95 μs	2

Table builds (oracle + projection, Float128-first)

Cold cache per sample (empty_tables! in untimed setup); JIT pre-warmed. The warm-hit column is the steady-state cost of get_table when the specialization is already cached (min / median). Table entries enumerate every code point by construction (NaN and $\pm\text{Inf}$ included).

operation	min	median	allocs	warm hit
Exp(8p4se) (256 entries)	30.31 μs	31.16 μs	257	65.8 ns / 67.4 ns
Tanh(8p4se) (256 entries)	29.49 μs	30.93 μs	257	65.4 ns / 67.5 ns
Add(8p4se $\times$ 8p4se) (64 K entries)	13.77 ms	13.93 ms	458754	65.0 ns / 66.6 ns
Divide(8p4se $\times$ 8p4se) (64 K)	14.02 ms	14.14 ms	458754	65.1 ns / 66.6 ns
Add(8p1uf $\times$ 8p1uf) (64 K, wide-spread)	27.08 ms	29.5 ms	995304	65.8 ns / 67.8 ns

Block and scaled operations

Elements Binary8p4se, scales Binary8p1uf, B = 32. Operands: all code points — NaN and $\pm\text{Inf}$ sampled.

operation	min	median	allocs	per lane
BlockAdd (B=32)	963.7 ns	1.17 μs	35	36.65 ns/lane
BlockDotProduct $\rightarrow$ 8p4se (B=32)	211.3 ns	265.3 ns	3	8.29 ns/lane
BlockReduceAdd $\rightarrow$ 8p4se (B=32)	43.5 ns	489.4 ns	0	15.29 ns/lane
ConvertToBlockMaxAbsFinite (B=32)	3.79 μs	3.95 μs	111	123.3 ns/lane

Conversions and packed storage

Operands: all code points — NaN and $\pm\text{Inf}$ sampled.

operation	min	median	allocs	per element
T(::UInt8) code-point constructor	1.8 ns	1.8 ns	0	
rawvalue (unchecked kernel route)	1.8 ns	1.8 ns	0	
T(::Float64) numeric constructor (projects)	3.2 ns	7.7 ns	0	
Convert 8p4se $\rightarrow$ 8p3se (scalar)	2.1 ns	6.7 ns	0	
Float64 $\rightarrow$ 8p4se (project)	2.0 ns	6.7 ns	0	
PackedVector pack, n=65536	33.47 μs	35.43 μs	3	0.54 ns/elem
PackedVector unpack (collect), n=65536	28.86 μs	29.05 μs	3	0.44 ns/elem

All numbers from this machine/run; absolute values vary by host. Regenerate with `julia --project=benchmark benchmark/benchmarking.jl`.

Chapter 10

Adding Operations

How to add a new operation to `SmallFloats.jl` so that it inherits everything the existing catalog has: bit-exact defined results, the full projection-mode vocabulary, generated scalar/array/block surfaces, table caching, exhaustive test coverage, benchmark pickup, and a truthful conformance declaration. Read the [Technical Guide](#) first — this page assumes its vocabulary (the result-kind protocol, the two rigor classes, the interval protocol, the yd envelope stage).

The architecture is registry-driven: for the supported arities, **most of the surface is generated**. Adding an operation is mostly (1) one registry line and (2) one rigorous `weval` method — plus the *duties* that keep the package's guarantees true. Every subsection below states the duties explicitly; they are not optional.

10.1 Universal duties (every new operation)

1. **Special rows are spec, not optimization.** Write out the NaN/±Inf/zero rows explicitly in `weval`, in draft style, before any general evaluation. Mark any behavior the draft text under-determines with `[interp]`.
2. **Result-kind protocol.** `weval` must return one of `Float64` | `Float128` | `StickyF` | `BigExactF` | `EncloseF` | `Enclose128F`, with `Float64/Float128` reserved for *provably exact* results and `StickyF` for exact-head-plus-tail-direction results whose soundness bound is discharged at the emitting site (see the FMA/FAA wide-spread escalations). Keep the return union narrow (ideally `{Float64, EncloseF}`) — a wide union defeats inference and costs allocations (measured: a 4-type union turned a 50 ns op into 240 ns with 2 allocations).
3. **Termination (the Niven duty).** `project_interval` terminates only if the enclosure is not chasing a value the projection grid can actually hit. Any input whose *exact* result is a representable dyadic must be **peeled or detected** before an enclosure is built. For transcendental ops this is usually a finite set of special rows (prove completeness, as the π -family does with Niven's theorem); for algebraic ops it is an exactness test.
4. **Envelope justification.** If you supply an eager yd estimate, you owe an error bound: the true value must lie within $|yd| \cdot 2^{-45}$ (`_F64_RELEXP`). Acceptable bases, strongest first: IEEE-CR hardware ops ($+$, $\sqrt{}$: $\leq \frac{1}{2}$ ulp, unconditional); short compositions of CR ops ($\leq \sim 1.5$ ulp); faithful libm (≤ 1 ulp, an empirical-but-published claim); short faithful compositions with condition number ≤ 1 ($\leq \sim 3$ ulp). **Measure it:** sample your estimator against ≥ 256 -bit MPFR truth over the op's domain, including the treacherous regions (near zeros of the result, domain boundaries), and demand $\geq 2^6$ slack under 2^{-45} . The same duty applies to a `Float128` fq estimator against its 2^{-90} envelope. Beware **argument-error amplification**: an estimate computed as $f(g(x))$ where g rounds (e.g. $\sin(\pi \cdot r)$ in `Float64`) can be catastrophically worse near zeros of f than a native result-domain evaluation ($\sin \pi(r)$) — measure the composition you actually ship.
5. **Ladder validity.** The MPFR closure must return a genuine directed enclosure. Whole-expression set rounding is valid only when the composition is monotone in every intermediate (state the argument in a comment, as `Softplus` does); otherwise round each step in the correct direction explicitly (as `_mpfr_rsqr` does for the anti-monotone $1/\sqrt{x}$).

6. **Gates.** Run the full suite. The unary exhaustive xhp gate and the coverage check pick up registry ops automatically; extend the hand-curated lists where applicable (monotonicity ops, binary exhaustive sweeps). Then regenerate the benchmark report — `bench_scalar_ops` iterates the registry lists, so your op appears in all four operand-class tables automatically; **add a `_SAFE_DOMAINS` entry** in `benchmarking/benchmarking.jl` if the op is argument-restricted, or the `safe-args` table will use the oracle-derived fallback pool.

10.2 Scalar operations

The registry lives in `src/ops_scalar.jl`. Adding a name to `_UNARY_OPS` / `_BINARY_OPS` / `_TERNARY_OPS` causes, with no further work:

- registration in `OP_REGISTRY` (group `:B/:C/:A` per the loop defaults — override with an explicit `register_op!` if the default group is wrong);
- the generated spec-register method `Op(fr, p, operands...; rng, R)` and the same-format convenience `Op(x...)`;
- generated array methods routing through `vmap` (Shape-A tables for pure `p` — always at arity ≤ 2 , bitwidth-gated at arity 3 — Shape-B scalar/threaded loops otherwise);
- generated `BlockOp` and `ScaledOp` variants (`src/blocks.jl`);
- the export (`src/SmallFloats.jl` loops the registry);
- table cacheability, conformance listing, and test-harness coverage.

What is *not* generated is the mathematics: `weval` in `src/oracle.jl`.

Adding a unary operation

Worked example: **Cbrt** (cube root), chosen because it exercises the hardest duty — exactness detection for an algebraic op.

Step 1 — registry (`src/ops_scalar.jl`): append `:Cbrt` to `_UNARY_OPS`. The loop registers it as group `:B` with arity 1.

Step 2 — ω -semantics (`src/oracle.jl`), with every duty discharged.

Type the row `::AbstractFloat`, not `::Float64`

An enclosure-ladder row is reached at **every rung**. `_LADDER_OPS` is derived by subtraction in `oracle.jl` — whatever is neither an exact selection, nor exact arithmetic, nor `Convert` — so a new registry operation automatically gets `_wf128`, `_wexact` and `_wdyadic` rows that forward to *this* method with `Float128`, `BigFloat` or converted-from-Dyadic operands.

A row typed `::Float64` therefore works for the 432 formats at rung 1 and raises a `MethodError` for the other 72. The package types 22 of its own ladder rows `::AbstractFloat` for exactly this reason; the 6 that are `::Float64` are the exact-arithmetic rows, which have per-head siblings written separately.

Gate G10 catches the omission — it calls every registry operation at every rung and asserts only that the call returns — and G4 asserts that `_EXACT_SELECTION`, `_EXACT_ARITH`, `_LADDER_OPS` and `Convert` partition `OP_REGISTRY` exactly, so an operation cannot be classified into none of them and silently lose its wide-rung rows.

```

# Exact-cube detection (termination duty):  $x = \pm m \cdot 2^{(3k)}$  with  $m$  an odd perfect
# cube  $\Rightarrow \text{cbrt}(x)$  is a Float64-exact dyadic that project_interval must never
# chase. Oracle-path cost is irrelevant; correctness of the peel is everything.
function _exact_cbrt(x::Float64)
    fr, e = frexp(abs(x))          #  $|x| = fr \cdot 2^e$ ,  $fr \in [0.5, 1)$ 
    m = Int64(ldexp(fr, 53)); e -= 53  #  $|x| = m \cdot 2^e$  exactly
    tz = trailing_zeros(m); m >= tz; e += tz
    e % 3 == 0 || return nothing
    r = round(Int64, cbrt(Float64(m)))
    for c in max(r - 1, 1):(r + 1)    # cbrt is faithful  $\Rightarrow$  true root  $\in r \pm 1$ 
        Int128(c)^3 == Int128(m) && return flpsign(ldexp(Float64(c), e ÷ 3), x)
    end
    nothing
end

function weval(::Val{:Cbrt}, x::AbstractFloat)    # ::AbstractFloat, NOT ::Float64
    isnan(x) && return NaN                    # special rows first: they are spec
    isinf(x) && return x                      #  $\text{cbrt}(\pm\infty) = \pm\infty$ 
    iszero(x) && return 0.0
    c = _exact_cbrt(x)
    c === nothing || return c                # exact dyadic result – peeled
    # yd: Float64 cbrt is faithful ( $\leq 1$  ulp  $\approx 2^{-52}$ ; measure it)  $\Rightarrow \geq 2^7$  slack
    # under the  $2^{-45}$  envelope. fq: libquadmath cbrtq under the  $2^{-90}$  envelope.
    # Ladder: cbrt is monotone  $\Rightarrow$  whole-expression directed rounding encloses.
    _encll(cbrt, x; fq=() -> cbrt(Float128(x)), yd=cbrt(x))
end

```

Step 3 — optional surface: a Base veneer (`Base.cbrt(x::Binary) = Cbrt(x)` next to the others in `ops_scalar.jl`), a docstring, and — since `cbrt` is monotone — its symbol in the test suite's monotonicity list.

Step 4 — validate: envelope measurement for `cbrt` (include tiny and huge x ; `cbrt` has no dangerous regions — the result is never near zero except at zero, which is peeled), then the full suite: the exhaustive `xhp` gate now checks `Cbrt` on every code point of the reference format under four modes against the 3072-bit protocol, automatically.

Adding a binary operation

Worked example: **LogAddExp** — $\log(e^x + e^y)$ — chosen because it exercises composed estimators and the ladder-monotonicity argument.

Step 1 — registry: append `:LogAddExp` to `_BINARY_OPS` (default group `:C`; the special five arithmetic names get `:A`).

Step 2 — ω -semantics. One generic evaluator serves all three precisions — Float64 (yd), Float128 (fq), and BigFloat (the ladder):

```
# Stable form:  $h + \log p(\exp(l - h))$ ,  $h = \max$ ,  $l = \min$ . Every step is monotone
# nondecreasing in  $(x, y)$ , so whole-expression directed rounding in _mpfr2 is a
# valid enclosure (the same argument Softplus records).
_logaddexp(a, b) = (h = max(a, b); l = min(a, b); h + loglp(exp(l - h)))

function weval(::Val{:LogAddExp}, x::AbstractFloat, y::AbstractFloat)
    (isnan(x) | isnan(y)) && return NaN
    (x == Inf || y == Inf) && return Inf          #  $\infty$  dominates
    x == -Inf && return y                        #  $\log(0 + e^y) = y$ , exactly
    y == -Inf && return x
    # yd: two faithful libm calls plus additions, condition  $\leq 1$  throughout
    # (result  $\geq \max(x, y)$ )  $\Rightarrow \leq \sim 3$  ulp  $\approx 2^{-50} - 2^5$  slack; measure it, including
    #  $x \approx y$  (where  $\log p(\exp(0))$  dominates) and widely separated operands.
    _encl2(_logaddexp, x, y;
        fq=() -> _logaddexp(Float128(x), Float128(y)),
        yd=_logaddexp(x, y))
end
```

Termination note: can `logaddexp(x, y)` be exactly dyadic for dyadic operands with the special rows peeled? $\log(e^x + e^y) = d$ requires $e^x + e^y = e^d$ — by Lindemann–Weierstrass this forces relations that dyadic arguments cannot satisfy except on the peeled rows, so no further peel is needed; **record that argument in a comment**. When you cannot prove such a statement for your op, hunt for exact cases the way `_exact_cbrt` does.

Step 3 — tests: binary ops are not in the unary exhaustive gate; add a divide-style exhaustive sweep (256×256 operand pairs against a ladder-only reference, a handful of modes) to the suite, and an envelope-sanity loop entry if you introduced a new libm dependence.

Adding a ternary operation

Worked example: **FMS** — fused multiply-subtract, $x \cdot y - z$ — chosen because the cheapest correct implementation is *delegation*, and delegation is a first-class technique here.

Step 1 — registry: append `:FMS` to `_TERNARY_OPS` (group `:A` by the loop). The generated array method routes through the same ternary bitwidth policy every other ternary op uses (`tables.jl`'s `_ternary_table_for`): small operand formats table (eagerly or adaptively), $K = 8$ runs the — optionally threaded — Shape-B scalar loop, and stochastic p always draws per element. No op-specific work needed; the policy keys on formats and p , not on which op it is.

Step 2 — ω -semantics by delegation to FMA's already-verified analysis (width thresholds, `_twosum` exactness, wide-spread fallback):

```
#  $x \cdot y - z \equiv \text{FMA}(x, y, -z)$ . Negation of  $z$  follows the Subtract convention:
# preserve NaN, and map  $-0$  to  $+0$  so the single-zero datum model is respected.
weval(::Val{:FMS}, x::Float64, y::Float64, z::Float64) =
    weval(Val{:FMA}, x, y, isnan(z) ? z : (iszero(z) ? 0.0 : -z))
```

Delegation inherits FMA's result kinds, its exactness thresholds, *and* its test pedigree — but verify the sign algebra of your reduction on the special rows (here: $\text{FMS}(\infty, 1, \infty)$ must be NaN, which the delegation produces because $\text{FMA}(\infty, 1, -\infty)$ is $\infty - \infty$). Add the delegated op to the ternary exhaustive sweep in the suite.

Adding a quaternary operation

Arity 4 is **not yet plumbed** — the generation loops, kernels, and block layer handle arities 1–3. Adding a quaternary op is therefore two tasks: extend the plumbing (once), then add the op. Worked example: **FMMA** — $x \cdot y + z \cdot w$, a two-term dot product.

Plumbing (one-time), four sites:

1. `src/ops_scalar.jl` — register manually after the arity loops (`register_op!{:FMMA, 4, :A}`), and add an `op.arity == 4` branch to the spec-register generation loop, mirroring the ternary branch with a fourth operand:

```
# ...continuing the existing `if op.arity == ...` cascade:
if op.arity == 4
    @eval begin
        @inline function $name(fr::Type{<:Binary}, p::ProjSpec,
                               x::Binary, y::Binary, z::Binary, w::Binary;
                               rng::MaybeRNG=nothing, R::Union{Nothing,Int}=nothing)
            apply_op($V(), fr, p, _drawR(p, rng, R),
                    decode(x), decode(y), decode(z), decode(w))
        end
        @inline $name(x::T, y::T, z::T, w::T; kw...) where {T<:Binary} =
            $name(T, default_projspec(T), x, y, z, w; kw...)
    end
end
```

(`apply_op` and `weval` are already variadic — no change needed there.)

2. `src/kernels.jl` — a four-array `vmap!` method. The plain Shape-B loop (copy the *body* of the ternary method's compute branch, add operand D) is enough to get correct results; the ternary method's table-tiering and threading are a bitwidth-driven optimization layered on top (`tables.jl`'s `_ternary_table_for` and its LRU cache), not required plumbing — extend to arity 4 only if a table win at that arity is worth the 2^4 (4K) growth. Also add an arity-4 branch in the registry-generated array-surface loop and the matching `rng`-threading overload (mirroring the three-array one) so stochastic `p` dispatches correctly.
3. `src/blocks.jl` — either an arity-4 branch in the Block/Scaled generation loop (four operand blocks, four blockdecodes) or an explicit skip (`op.arity > 3 && continue`) if a block form is not wanted; be deliberate.
4. `src/approx.jl` — `conformance_report` prints arities with `for a in 1:3`; widen to `1:4`.

The `op` itself follows the FAA/FMA width-analysis playbook exactly:

```
const _DE_FMMA = 92      # two exact ≤17-bit products: 18 + ΔE ≤ 113, margin 3

function weval(::Val{:FMMA}, x::Float64, y::Float64, z::Float64, w::Float64)
    (isnan(x) | isnan(y) | isnan(z) | isnan(w)) && return NaN
    (((iszero(x) && isinf(y)) || (isinf(x) && iszero(y))) ||
     ((iszero(z) && isinf(w)) || (isinf(z) && iszero(w)))) && return NaN
    p = x * y; q = z * w                                # exact: ≤8-bit significands
    if isinf(p) || isinf(q)
        (isinf(p) && isinf(q) && p != q) && return NaN
        return isinf(p) ? p : q
    end
    s, e = _twosum(p, q)
    e == 0.0 && return iszero(s) ? 0.0 : s                # exact in Float64
    (_fl28() && !iszero(p) && !iszero(q) && _expdiff(p, q) <= _DE_FMMA) &&
        return Float128(p) + Float128(q)                # exact by width
    BigExactF() -> setprecision() ->
        BigFloat(x) * BigFloat(y) + BigFloat(z) * BigFloat(w),
        BigFloat, bigprec_prod(x, y, z, w))              # DERIVED, never a constant
end
```

Derive the precision from the operands — never a constant

This escalation used to read `_BIGP`, a 2200-bit constant. It was ample for every format that existed at $K \leq 8$ and **insufficient for 50 of the 504**, and it did not announce that: it returned a plausible wrong number, which gate G2 was written red to record (1//1 where the exact answer is 2//1, the operand returned unchanged because a residual 30 000 binades below the rounding position fell off the accumulator).

`bigprec(F...)` derives the precision from the formats; `bigprec(x...)` and `bigprec_prod(x...)` derive it from the operands in flight. Gate G2 asserts the derivation is sufficient across the whole grid, so a new operation that uses it inherits that proof. A new constant would need its own.

Testing duty is heavier at arity 4: the full cross-product is 2^{32} tuples, so the suite entry must be *sampld* (seeded, ladder-referenced, $\geq 10^5$ tuples across several modes) plus hand-picked exhaustion of the special-row algebra and the width-threshold boundary ($\Delta E \in \{91, 92, 93\}$ constructions). Document the sampling in the test, since it breaks the suite's "enumerate, never sample" norm — that exception must be visible, not silent.

10.3 Block operations

The block layer (`src/blocks.jl`) implements the draft's elementwise schema **once** — `blockdecode` → `ω-op` `lanewise` → `blockproject` — and generates every registry op's `BlockOp/ScaledOp` from it. The reductions are hand-written on a second shared pattern. Which subsection you need depends on whether your operation fits one of those two patterns.

Adding a scaled block operation

If the scalar operation is (or can be) a **registry op, the scaled form is free**: registering `:Cbrt` above already produced `ScaledCbrt(fr, p, s, x)` and `BlockCbrt(fr, p, b, sr)` with the draft's §5.4 semantics — decode `scale × element` exactly per lane, apply the `ω-op`, `BlockProject` against the result `scale` through the division cascade. **Prefer this route**; it is the package's non-divergence mechanism.

Write a scaled op by hand only when it is *not* an elementwise image of a registry op. The primitive to compose with is `_bp_element(fr, p, R, res, Sdat)` — it owns the draft's scale-special rows (NaN/0/±Inf scale) and the entire division-by-scale rigor cascade. Example, a scaled affine $s_1 \cdot x + s_2 \cdot y$ projected against a *unit* result scale:

```
function ScaledAffine(fr::Type{<:Binary}, p::ProjSpec,
    s1::Binary, x::Binary, s2::Binary, y::Binary;
    rng::MaybeRNG=nothing)
    # Each (scale, element) pair is a TWO-FACTOR MONOMIAL and gets its own head
    # from the two formats — the same rule `blockdecode` applies to a block.
    h1 = rung(Val{(:Multiply)}, typeof(s1), typeof(x))
    h2 = rung(Val{(:Multiply)}, typeof(s2), typeof(y))
    a = weval(h1, Val{(:Multiply)}, lift(h1, decode(s1)), lift(h1, decode(x))):: carriertype(h1)
    b = weval(h2, Val{(:Multiply)}, lift(h2, decode(s2)), lift(h2, decode(y))):: carriertype(h2)
    # The two products may land on DIFFERENT carriers, so the add runs at their
    # join with each operand lifted into it.
    h = _joinheads(a, b)
    res = weval(h, Val{(:Add)}, lift(h, a), lift(h, b))
    _bp_element(fr, p, _drawR(p, rng, nothing), res, 1.0)
end
```

The assertion is `::carriertype(h)`, and writing `::Float64` there is a real defect rather than a stylistic choice. It is true for the 432 formats at `rung 1` and an outright `TypeError` for the other 72, whose lanes decode to `Float128` or to the exact dyadic carrier. The generated `Scaled*` surface in `blocks.jl` carried exactly that mistake and it is recorded as §11 M44 in the execution log — the guidance here used to teach it.

What the assertion *is* load-bearing for survives the correction: Multiply is closed on the carrier at every rung (gate G4 pins this), so the row cannot escalate to a lazy BigExactF here, and saying so is what lets `_joinheads` below see carrier values only. JET finds the missing statement before any test can.

The Add result may be any kind in the protocol — `_bp_element` finishes all of them. Dividing by a scale of 1.0 is the identity fast path.

Adding an elementwise block operation

Same schema, whole-block granularity. The three obligations: decode operand blocks with `blockdecode` (exact), evaluate lanewise with `weval` (never with ad-hoc Float64 math), and finish with `blockproject` against the supplied result scale (never with per-lane project — the scale-division rows belong to `blockproject`). Example — `BlockRelu`, an elementwise $\max(x, 0)$ that reuses the verified `MaximumNumber` semantics:

```
"""BlockRelu(fr, p, b, sr): lanewise max(xi, 0) — NaN lanes propagate NaN per
MaximumNumber's number-preferring rows only when both operands are NaN, i.e.
never here; state your lane semantics this explicitly for any new op."""
function BlockRelu(fr::Type{<:Binary}, p::ProjSpec, b::Block{B}, sr::Binary;
                  rng::MaybeRNG=nothing) where {B}
    X = blockdecode(b)
    Z = ntuple(i -> weval(Val{MaximumNumber}, X[i], 0.0), Val{B})
    blockproject(fr, p, sr, Z; rng)
end
```

Add the name to the block-surface export list and to `conformance()`'s `blocknames` extension vector so the declaration stays truthful. Test against a from-scratch reference composition (the suite's existing block tests show the pattern) — element format × scale format × B, exhaustively where feasible.

Adding a reductive block operation

Reductions do **not** fit the elementwise schema; they follow the second pattern, visible in `BlockReduceAdd`/`BlockDotProduct`: (1) resolve the ∞ /NaN **fold algebra** on Float64 classifications first; (2) an integer **span filter** decides whether the whole reduction is exactly representable in Float128 — if so, plain Float128 accumulation *is* the exact answer; (3) otherwise a BigExactF exact accumulator at provably sufficient precision; (4) exactly **one** projection, at the very end, via `_finish`. Never round intermediates.

Worked example — **BlockSumOfSquares**, $\sum x_i^2$:

```

"""BlockSumOfSquares(fr, p, b): project( $\sum$  decode-lane.2) – one rounding total.
Fold algebra: any NaN lane  $\Rightarrow$  NaN; else any  $\pm$ Inf lane  $\Rightarrow$  +Inf (squares cannot
cancel – no opposite-infinity NaN case exists, unlike ReduceAdd); else exact."""
function BlockSumOfSquares(fr::Type{<:Binary}, p::ProjSpec, b::Block{B};
                           rng::MaybeRNG=nothing, R::Union{Nothing,Int}=nothing) where {B}
    X = blockdecode(b)
    res = if any(isnan, X)
           NaN
        elseif any(isinf, X)
           Inf
        elseif all(iszero, X)
           0.0
        else
            # span filter: lanes carry  $\leq 17$ -bit significands  $\Rightarrow$  squares  $\leq 34$  bits and
            # a square's exponent is  $2 \cdot (\text{lane exponent})$ ; the sum is exact in Float128
            # when  $34 + 2 \cdot \text{span} + \lceil \log_2 B \rceil + 1 \leq 113 \Rightarrow 2 \cdot \text{span} + \lceil \log_2 B \rceil \leq 78$ .
            if _f128() && 2 * _expspan(X) + _log2ceil(B) <= 78
                acc = Float128{0}
                for v in X
                    q = Float128(v)
                    acc += q * q                # exact squares, exact sum
                end
                acc
            else
                BigExactF{() -> setprecision(BigFloat, _REDPREC)} do
                    acc = BigFloat{0}
                    for v in X
                        acc += BigFloat(v)^2    # exact at _REDPREC
                    end
                    acc
                end()
            end
        end
    _finish(fr, p, _drawR(p, rng, R), res)
end

```

The two lines that demand your care in any new reduction are the **fold algebra** (derive it from the draft's reduce definition on the extended reals — here the absence of an opposite-infinities NaN case is a *theorem about squares*, stated in the docstring) and the **span-filter inequality** (derive it from operand widths, write the derivation in the comment, and add boundary-condition tests at the threshold, exactly as the suite does for the existing `_DE_*` constants). Everything else — the exact accumulator, the single `_finish`, stochastic R plumbing — is pattern.

10.4 After any addition: the closing checklist

- Full test suite green, including your new sweep entries.
- Envelope measurements recorded for any new yd/fq estimator.
- Benchmark report regenerated; `_SAFE_DOMAINS` entry added if the op is argument-restricted; sanity-check the op's row in the safe-args table.
- Docstring on the public name; `[interp]` markers where you interpreted.
- `conformance_report()` shows the op (automatic for registry ops; manual block additions extend blocknames).

Chapter 11

External Reference

Docstrings for the documented **public** surface — the exported names. The full export list is far larger (400+ names, including the 120 exported format aliases — the other 384 are opt-in via using `SmallFloats.Formats` — the predefined projection-spec grid, and the generated operation and block registers); names without docstrings are covered descriptively in the [User Guide](#); unexported machinery is in the [Internal Reference](#).

`SmallFloats.SmallFloats` - Module.

`SmallFloats`

A conforming, performance-oriented Julia implementation of the IEEE P3109 draft standard for arithmetic formats for machine learning (bitwidths 3–16).

Bit-exact defined results on every default path; the projection engine (`RoundToPrecision` → `Saturate` → `Encode`) is the single write path into a code point; approximate fast paths exist only behind the explicit `κ` registry (`register_approx!` / `approx`), never substituted silently. See `conformance()` for the live declaration and `SmallFloats.draft_revision()` for the implemented draft.

Performance note

Pass format types through `const` bindings, type parameters, or function arguments. All exported entry points fully specialize when the format type is statically known (measured: a complete scalar `Add` ≈ 26 ns, `project` ≈ 13 ns, zero allocations). Calling through a **non-const global** format type forces dynamic dispatch on every call — measured at ~1 μs per scalar keyword call — which is Julia semantics, not package overhead; a single function barrier (`f(::Type{T}, args...)` where `{T}`) restores full speed. The test suite pins the specialization properties (concrete return types, zero warm-path allocation) as deterministic regressions.

`SmallFloats.RNA_SF` - Constant.

(`NearestTiesToAway`, `SatFinite`).

`SmallFloats.RNA_SN` - Constant.

(`NearestTiesToAway`, `SatNone`).

`SmallFloats.RNA_SP` - Constant.

(`NearestTiesToAway`, `SatPropagate`).

`SmallFloats.RNE_SF` - Constant.

(`NearestTiesToEven`, `SatFinite`).

`SmallFloats.RNE_SN` - Constant.

(`NearestTiesToEven`, `SatNone`) — the package-wide default p.

`SmallFloats.RNE_SP` - Constant.

(`NearestTiesToEven`, `SatPropagate`).

`SmallFloats.RTN_SF` - Constant.

(`TowardNegative`, `SatFinite`).

`SmallFloats.RTN_SN` - Constant.

(`TowardNegative`, `SatNone`).

`SmallFloats.RTN_SP` - Constant.

(`TowardNegative`, `SatPropagate`).

`SmallFloats.RTO_SF` - Constant.

(`ToOdd`, `SatFinite`).

`SmallFloats.RTO_SN` - Constant.

(`ToOdd`, `SatNone`).

`SmallFloats.RTO_SP` - Constant.

(`ToOdd`, `SatPropagate`).

`SmallFloats.RTP_SF` - Constant.

(`TowardPositive`, `SatFinite`).

`SmallFloats.RTP_SN` - Constant.

(`TowardPositive`, `SatNone`).

`SmallFloats.RTP_SP` - Constant.

(`TowardPositive`, `SatPropagate`).

`SmallFloats.RTZ_SF` - Constant.

(`TowardZero`, `SatFinite`).

`SmallFloats.RTZ_SN` - Constant.

(`TowardZero`, `SatNone`).

`SmallFloats.RTZ_SP` - Constant.

(`TowardZero`, `SatPropagate`).

`SmallFloats.ApproxImpl` - Type.

A registered κ -approximate implementation of one operation specialization.

SmallFloats.Binary - Type.

```
Binary{K,P,SGN,EXT} <: AbstractFloat
```

A P3109 **format**: bitwidth $K \in 3:16$, precision P , signedness $SGN::\text{Bool}$ ($\text{true} = \text{Signed}$), domain $EXT::\text{Bool}$ ($\text{true} = \text{Extended}$, i.e. the datum set includes infinities).

Binary is **abstract**, and that is the draft's own distinction: a format is "a datum set and an encoding" (D1 §3.1), while the width of the machine word carrying a code point is a representation choice below the standard's level of description. Code8 is that representation. The parameterized name is the *format*; the draft-named alias is its *representation*:

```
Binary{8,4,true,true} # the format - abstract
Binary8p4se           # Code8{8,4,true,true}, its representation - concrete
```

Consequently `Binary8p4se == Binary{8,4,true,true}` is **false** and `Binary8p4se <: Binary{8,4,true,true}` is **true**. Use the alias, or `format(K, P,  $\Sigma$ ,  $\Delta$ )`, wherever a concrete type is needed — an array element type, a similar, a constructor target.

The code point occupies the low K bits of the storage unit; the high bits are maintained zero as a representation invariant. Construct raw code points with `T(c::Unsigned)` (validated code-point construction), `rawvalue(T, c)` (unchecked kernel route), or `Code8{...}(Val(:code), x)`; construct from numeric values with `T(x::Real)` (default projection spec) or `Convert`. Unsigned is the one argument-type class meaning *code point*, at **every** width — `T(0x02)` is code point 2 whether `T`'s unit is a `UInt8` or a `UInt16`; all other `Reals` mean *value*.

There are $4K - 2$ formats at each K ($P \in 1:K \times \text{signed/unsigned} \times \text{finite/extended}$, less the two $P == K$ signed combinations, which have no exponent bits), so $K_{\text{MIN}}:K_{\text{MAX}} = 504$ in total.

SmallFloats.Block - Type.

```
Block{B, FS<:Binary, FE<:Binary}
```

A P3109 block (`s, [x1 ... xB]`): scale factor in format `FS`, $B \geq 1$ elements in format `FE` (draft §5). `isbits`; blocks live in registers/stack.

SmallFloats.BlockVector - Type.

```
BlockVector{B,FS,FE} <: AbstractVector{Block{B,FS,FE}}
```

Structure-of-arrays storage: `scales::Vector{FS}`, `elems::Matrix{FE}` ($B \times n$, column-major so each block's elements are one contiguous column).

SmallFloats.FPClass - Type.

Eight-way datum classification returned by `Class` (draft §4.13.1).

SmallFloats.NearestTiesToAway - Type.

Round to nearest; ties away from zero (draft §4.7.1).

SmallFloats.NearestTiesToEven - Type.

Round to nearest; ties to the even code point (IEEE default; draft §4.7.1).

SmallFloats.Op - Type.

```
Op(name::Symbol, FR::Type{<:Binary}, p::ProjSpec) -> Op
```

A registry operation bound to a result format and a projection, as a **callable** — so the idiomatic array verbs work directly:

```
map(Op(:Add, Binary8p4se, RNE_SN), A, B)
Op(:Exp, Binary8p4se, RNE_SN).(A)
```

Op is a singleton type: the operation, format and projection all live in the type parameters, so calls specialize exactly as Add(Binary8p4se, RNE_SN, x, y) does. Use **vmap!** when you have a destination array to fill — it is the in-place, table-aware kernel and this type is a thin front for it.

SmallFloats.PackedVector - Type.

```
PackedVector{F<:Binary} <: AbstractVector{F}
```

Bit-packed storage of F code points at bitwidth(F) bits per element. Construct with PackedVector(A::AbstractVector{F}); recover with Vector(pv) or collect(pv). Memory: $\lceil n \cdot K / 64 \rceil$ words vs $n \cdot \text{sizeof}(\text{codeunit_type}(F))$ unpacked.

It saves nothing when $K == 8 \cdot \text{sizeof}(\text{codeunit_type}(F))$ — $K = 8$ and $K = 16$ — where the packed layout is bit-for-bit the unpacked one and every access pays the shift/mask/splice for no return. `packing_saves(F)` says so; see its docstring for why that is a query rather than an error.

SmallFloats.ProjSpec - Type.

```
ProjSpec{R<:RoundingMode3109, S<:SaturationMode}
```

A projection specification $p = (\text{rounding mode}, \text{saturation mode})$, draft §4.2. Zero-size; construct as ProjSpec(NearestTiesToEven(), SatNone()) or via the exported constants. Kernels specialize on its type.

SmallFloats.StochasticA - Type.

Stochastic rounding, variant A of draft §4.7.4: $\text{RoundAway} \iff \lfloor v \cdot 2^N \rfloor + R \geq 2^N$.

SmallFloats.StochasticB - Type.

Stochastic rounding, variant B: $\text{RoundAway} \iff \lfloor v \cdot 2^{(N+1)} \rfloor + (2R+1) \geq 2^{(N+1)}$.

SmallFloats.StochasticC - Type.

Stochastic rounding, variant C: $\text{RoundAway} \iff \text{RNITE}(v \cdot 2^N) + R \geq 2^N$.

SmallFloats.ToOdd - Type.

Round inexact results to the nearest *odd* code point (draft §4.7.3; sticky-friendly).

SmallFloats.TowardNegative - Type.

Directed rounding toward $-\infty$ (draft §4.7.2).

SmallFloats.TowardPositive - Type.

Directed rounding toward $+\infty$ (draft §4.7.2).

`SmallFloats.TowardZero` – Type.

Directed rounding toward zero — truncation (draft §4.7.2).

`SmallFloats.BlockDotProduct` – Method.

`BlockDotProduct` (draft §5.3.2). Lane products can carry 64 significant bits, so the products and their sum are formed in the exact accumulator, with the ∞/NaN fold algebra resolved on the `Float64` classifications first.

`SmallFloats.BlockReduceAdd` – Method.

`BlockReduceAdd` (draft §5.3.1): `project(reduce( $\omega$ Add, [0, X...]))`.

`SmallFloats.BlockReduceMultiply` – Method.

`BlockReduceMultiply` (draft §5.3.1): `project(reduce( $\omega$ Multiply, [1, X...]))`.

`SmallFloats.Class` – Method.

`Class(v) -> FPClass` (draft §4.13.1): classify `v` as `NaN`, $\pm\text{Inf}$, $\pm\text{normal}$, $\pm\text{subnormal}$, or zero.

`SmallFloats.Convert` – Method.

```
Convert(fr, p, A: AbstractArray{<: Union{Float16, Float32, Float64, BFloat16}}; rng) -> Array{fr}
```

Bulk ingestion: exact widening per element, then projection under `p`. Unlike the `Binary-array Convert` (a `Shape-A` table gather), external inputs are not enumerable, so this is a `Shape-B` loop; stochastic `p` draws per element.

`SmallFloats.Convert` – Method.

```
Convert(fr, p, x) -> fr
```

Draft §4.9 `Convert(fx, fr, p)`. Accepts `Binary`, IEEE binary16/32/64 floats (widened exactly to the `Float64` carrier), `Float128` (projected directly), `Integer` (exact via a sufficiently wide `BigFloat`), and `BigFloat` (projected directly; the caller warrants the value is exact).

`SmallFloats.ConvertFromBlock` – Method.

`ConvertFromBlock` (§5.2.1): `blockdecode`, then project each element (no scale division).

`SmallFloats.ConvertToBlock` – Method.

`ConvertToBlock` (§5.2.2): `decode` elements, `BlockProject` against the supplied scale.

`SmallFloats.ConvertToBlockMaxAbsFinite` – Method.

`ConvertToBlockMaxAbsFinite` (§5.2.3): `S = reduce( $\omega$ MaximumFinite, [NaN, |X1|...])`, project `S` under `ps`, then `BlockProject` the elements under `p`.

`SmallFloats.DefaultProjection` – Method.

```
DefaultProjection() -> ProjSpec
DefaultProjection!(p::ProjSpec) -> p
DefaultProjection!(m, s) -> ProjSpec
```

The session's default projection specification. Initialized to `(DefaultRoundingMode(), DefaultSaturationMode()) = RNE_SN`. Setting it directly decomposes `p` into `DefaultRoundingMode` and `DefaultSaturationMode`, so the three stay coherent in both directions.

SmallFloats.DefaultRNG – Method.

```
DefaultRNG() -> Type{<:AbstractRNG} | AbstractRNG  
DefaultRNG!(rng) -> rng
```

The session's default random-number generator for stochastic rounding. Initialized to the Xoshiro *type* (a fresh generator per use); may be set to an AbstractRNG instance for a reproducible stream.

SmallFloats.DefaultRbits – Method.

```
DefaultRbits() -> Int  
DefaultRbits!(n::Int) -> n
```

The session's default random-bit budget N for the stochastic rounding families (StochasticA{N}, StochasticB{N}, StochasticC{N}). Initialized to 8; must satisfy $1 \leq N \leq 60$.

SmallFloats.DefaultReturnType – Method.

```
DefaultReturnType() -> Type{<:Binary}  
DefaultReturnType!(T::Type{<:Binary}) -> T
```

The session's default *result* format — the format an operation projects into when the caller does not name one. Initialized to Binary8p2se. Independent of DefaultType, which is the default *operand* format.

SmallFloats.DefaultRoundingMode – Method.

```
DefaultRoundingMode() -> RoundingMode3109  
DefaultRoundingMode!(m) -> m
```

The session's default rounding mode. Initialized to NearestTiesToEven(). Setting it rebuilds [DefaultProjection](#) from the new mode and the current [DefaultSaturationMode](#). Accepts an instance or a fully-parameterized type (NearestTiesToAway, StochasticA{8}, ...).

SmallFloats.DefaultSaturationMode – Method.

```
DefaultSaturationMode() -> SaturationMode  
DefaultSaturationMode!(s) -> s
```

The session's default saturation mode. Initialized to SatNone(). Setting it rebuilds [DefaultProjection](#) from the current [DefaultRoundingMode](#) and the new mode. Accepts an instance or a type.

SmallFloats.NextGreaterThan – Method.

NextGreaterThan(v) (draft §4.16): the least datum greater than v in the total order — one step up the code lattice, with NaN $\rightarrow$ NaN and MaxFinite/+Inf $\rightarrow$ NaN at the top. Base.nextfloat on Binary is this operation.

SmallFloats.NextLessThan – Method.

NextLessThan(v) (draft §4.16): the greatest datum less than v in the total order — one step down the code lattice, with NaN $\rightarrow$ NaN and MinFinite/-Inf $\rightarrow$ NaN at the bottom. Base.prevfloat on Binary is this operation.

SmallFloats.RoundOf – Function.

Draft-named accessor: RoundOf(p) — the rounding mode of p (§4.2).

SmallFloats.SatOf – Function.

Draft-named accessor: SatOf(p) — the saturation mode of p (§4.2).

`SmallFloats.TotalOrder` – Method.

`TotalOrder(fx,fy)` (draft §4.12.1): $x \leq y$ in the total order (single NaN largest). Same-format comparisons run on the integer order key (bitops plan K1) — proven equivalent to the decode order exhaustively over all pairs; cross-format keys are not comparable, so mixed formats keep the decode path.

`SmallFloats.approx` – Method.

Retrieve a registered approximate implementation (callable via `.fn`).

`SmallFloats.bitwidth` – Method.

Format bitwidth K (3–8).

`SmallFloats.codedistance` – Method.

Code-point distance along the total order between two values of one format.

`SmallFloats.codeunit_type` – Method.

The storage unit of a format's code point: `UInt8` for $K \leq 8$, `UInt16` above.

`SmallFloats.conformance` – Method.

```
conformance() -> ConformanceDeclaration
```

The package's draft-§4.6-style conformance declaration, derived live from the operation registry, the table cache (the specializations actually instantiated), and the approximate-implementation registry. Serialize with `conformance_dict` or render with `conformance_report`.

`SmallFloats.conformance_dict` – Function.

Nested `Dict{String,Any}` form of the declaration — serializable by any JSON/TOML writer.

`SmallFloats.conformance_report` – Function.

Human-readable conformance report.

`SmallFloats.decode!` – Method.

```
decode!(dest::AbstractArray{Float32}, A::AbstractArray{<:Binary}) -> dest  
decode!(dest::AbstractArray{Float64}, A::AbstractArray{<:Binary}) -> dest
```

Exact bulk `ωDecode` into an external float array.

Exactness is the whole contract here — it is what distinguishes `decode!` from `Float32.(A)`, which rounds like any other Julia conversion — so the destination element type is **gated on the format**, not assumed. `datumsexact(X, F)` decides it, and a format whose datums do not all fit X raises rather than silently rounding a bulk array. Every $K \leq 8$ format passes both gates, so this is a new refusal only for formats that did not exist before the $K \leq 16$ extension.

Where the gate passes and $K \leq \text{KSPLIT}$, the implementation is unchanged: a `Shape-A` gather of the (narrowed) constant decode table.

`SmallFloats.decode` - Method.

```
decode(v: Binary) -> datumcarrier(typeof(v))
```

ω Decode (draft §4.7.2). **Exact for every datum of every format**, which is why the return type is the format's carrier rather than a fixed one: `Float64` where the bias allows (432 of the 504 formats), `Float128` to `B = 8192`, `BigFloat` above. `Float64(v)` is the way to ask for a `Float64` and accept the rounding.

Two implementations, selected by `decodepolicy` — dispatch on the representation, not a branch on `K` (invariant 9, §2 R-F):

- `TableDecode` ($K \leq \text{KSPLIT}$): a 2^K -entry constant-tuple lookup (bitops plan $K2$), generated once from `_decode_compute`, so the two are correct by construction and asserted equivalent exhaustively; constant inputs fold.
- `ComputeDecode` ($K > \text{KSPLIT}$): `_decode_compute` directly. No table — 65 536 entries is what invariant 10 exists to forbid.

`SmallFloats.empty_tables!` - Method.

Drop every cached table and adaptive counter (they rebuild lazily on next use).

`SmallFloats.expbias` - Method.

Exponent bias: $2^{(K-P-1)}$ signed, $2^{(K-P)}$ unsigned.

`SmallFloats.expsbitwidth` - Method.

Width of the exponent field in bits: $(K - \text{signbit}) - (P - 1)$.

`SmallFloats.f32_exact` - Method.

```
f32_exact(op, f1, f2) -> Bool
```

True iff `op` $\in$ (`:Add`, `:Subtract`, `:Multiply`) on `Float32`-decoded operands is exact for every finite pair of (`f1`, `f2`) datums — checked once by exhaustive enumeration against a 300-bit `BigFloat` oracle, then cached. When true, a `Float32` intermediate is an exact carrier under every projection mode.

`SmallFloats.f32_impl` - Method.

```
f32_impl(op, fr, p) -> fn
```

The `decode32` $\rightarrow$ guarded-`op32` $\rightarrow$ project kernel for `op(...  $\rightarrow$  fr, p)`, shaped for `register_approx!`. Nearest-ties `p` only: directed modes after a nearest-rounded `Float32` compute measure $\kappa \geq 1$ (and $\kappa = \text{NaN}$ at saturation boundaries), and stochastic `p` is rejected by κ measurement itself.

`SmallFloats.format` - Method.

```
format(K, P,  $\Sigma$ ,  $\Delta$ ) -> Type
```

The concrete representation type for a format, from **runtime** parameters — `format(8, 4, true, true) == Binary8p4se`. The programmatic route to a type that can be an array element or a constructor target, and the supported replacement for spelling `Binary{K,P, $\Sigma$ , $\Delta$ }` where a concrete type is meant.

Deliberately a `Dict` lookup: this path is reflection, it is not performance-sensitive, and it must not pretend to be. Where the parameters are static, the aliases and `reptype` are the zero-cost routes.

`SmallFloats.formatname` - Method.

`formatname(T)` — the draft §3.2 name of a format type.

`SmallFloats.ftz_variant` - Method.

```
ftz_variant(op, fr, fl, p) -> fn
```

Build the Annex-style approximate unary implementation: compute the defined result, then flush subnormal results to zero or `MinNormalOf(fr)`, whichever is nearer, ties toward zero (magnitude-symmetric for signed formats). Returned `fn` is suitable for `register_approx!`; for `fr` with precision P its κ is $2^{(P-2)}$ when $P \geq 2$ (largest flushed-to-zero subnormal) and 0 when $P = 1$ (no subnormals exist).

`SmallFloats.isextended` - Method.

Whether the format's domain is Extended (datum set includes infinities).

`SmallFloats.issigned` - Method.

Whether the format is Signed (has a sign bit and negative datums).

`SmallFloats.list_approx` - Method.

Sorted names of all registered κ -approximate implementations.

`SmallFloats.measure_kappa` - Method.

```
measure_kappa(fn, op, fr, argformats, p; max_exhaustive=2^22, rng, samples=2^20)
-> ( $\kappa$ : Float64, exhaustive: Bool)
```

Measure κ of `fn(x::argformats[1], ...)::fr` against the defined results of draft operation `op` under `p`. Enumerates the full input cross-product when it has at most `max_exhaustive` points (always true for arity ≤ 2); otherwise measures on samples uniform draws and reports `exhaustive = false`.

`SmallFloats.nrandbits` - Method.

Number of random bits N consumed per projected element; 0 for pure modes.

`SmallFloats.projmode` - Method.

Translate a `Base.RoundingMode` to its `RoundingMode3109` counterpart (identity on the latter). Used only at API boundaries. Internals always carry `RoundingMode3109`.

`SmallFloats.rawvalue` - Method.

Unsafe raw constructor used by kernels after invariants are established.

Accepts an abstract format and normalizes through `reptype`; the type is passed *as an argument* to `_rawvalue` rather than assigned to a local, so it becomes a type parameter of the callee instead of a value the compiler has to fold (technique T6). Kernel call sites pass concrete types and pay nothing.

`SmallFloats.register_approx!` - Method.

```
register_approx!(name, op, fr, argformats, p, fn;  $\kappa$ =nothing, kwargs...) -> ApproxImpl
```

Register `fn` as a named κ -approximate implementation of `op(argformats → fr, p)`. κ is measured by enumeration (see `measure_kappa`); a declared κ smaller than the measured value is **rejected**. Omitting κ declares the measured value. NaN- κ implementations (non-finite mismatches) may be registered only by declaring κ =NaN.

SmallFloats.register_f32! - Method.

```
register_f32!(op, fr, argformats, ρ; κ=nothing) -> ApproxImpl
```

Build, measure, and register the Float32 kernel for $op(argformats \rightarrow fr, \rho)$ under the canonical name $f32/op/args \rightarrow fr/rm_sm$. Registration measures κ by exhaustive enumeration and fails loudly on understatement — that is the point. Measured baseline (2026-07-24): $\kappa = 0$ for all 120 same-format signatures $\times \{Add, Subtract, Multiply, Divide, Sqrt\} \times RNE_{\{SatNone, SatFinite, SatPropagate\}}$.

SmallFloats.reptype - Method.

```
reptype(F) -> Type
```

The concrete representation of a format. The identity on a concrete input, and the $Binary\{K,P,S,E\} \rightarrow Code8|Code16$ map on an abstract one.

Julia will **not** exploit "this abstract type has exactly one subtype": there are no sealed types, inference cannot narrow an abstract element type to its unique subtype, and nothing stops a third party declaring another $<: Binary$. So `reptype` carries real weight rather than being belt-and-braces — it is how a method that holds only the format parameters reaches a constructible type.

The `Val` barrier is deliberate (invariant 9). The obvious spelling, $K \leq KSPLIT ? Code8\{\dots\} : Code16\{\dots\}$, returns a `Type`, so its inferred return type is a small Union unless the compiler folds the comparison. It usually will; "usually" cannot underwrite a correctness-critical selection that also gates a zero-allocation guarantee. Behind a `Val`, each `_rep` method has exactly one concrete return type and there is nothing to fold. Gate G9 checks it per format.

SmallFloats.table_bytes - Method.

Total bytes currently held by every table cache.

SmallFloats.table_count - Method.

Number of cached unary/binary specializations, both code units.

SmallFloats.table_policy - Method.

```
table_policy(op, fr, f1[, f2[, f3]], ρ) -> (; shape, entries, bytes, reason)
```

Which shape an array call on this signature will take, and why — `:A` for the table gather, `:B` for the per-element scalar path.

§4 Stage 5 item 6: "this signature is running the compute kernel" should be *discoverable*, not inferred from a benchmark that came out slower than expected. Reads the same predicates the kernels do, so it cannot drift from them.

SmallFloats.ternary_count - Method.

Number of cached ternary specializations, both code units.

SmallFloats.trailingsigbits - Method.

Trailing-significand width $P - 1$ (the stored fraction bits).

SmallFloats.unregister_approx! - Method.

Remove a registered κ -approximate implementation (no-op if absent).

SmallFloats.vmap! - Function.

```
vmap!(dest, Val(op), fr, p, A[, B[, C]]; rng) -> dest  
vmap(op, fr, p, A...; rng)
```

Elementwise draft operation over arrays of Binary values, projecting into `fr` under `p`. Pure `p` runs the Shape-A gather whenever a table exists or the ternary bitwidth policy grants one; stochastic specializations (and untabled ternary signatures) run the scalar path per element (each element drawing its own `R`).

SmallFloats.vmap - Method.

```
vmap(op, fr, p, pv::PackedVector; rng=nothing) -> Vector{fr}
```

Elementwise `op` over packed storage: unpack a 256-element tile into a byte scratch, run the ordinary byte kernel (`vmap!`), emit, repeat. Never computes on packed words directly — the deliberate compute-unpacked/store-packed boundary.

SmallFloats.with_default_projection - Function.

```
with_default_type(f, args...) -> f(DefaultType(), args...)  
with_default_returntype(f, args...) -> f(DefaultReturnType(), args...)  
with_default_projection(f, args...) -> f(DefaultProjection(), args...)
```

Call `f` with the named session default as its first argument — the supported way to *consume* a default. While the default still holds its initial value (the overwhelmingly common case) the call is statically compiled against that constant: no dynamic dispatch, `f` fully specialized. After the default is changed, the call crosses a function barrier instead: one dynamic dispatch, everything inside fully specialized.

Allocation contract: the combinator's return type unions both paths. When `f`'s result type does not depend on the default — e.g. `with_default_projection` with the formats fixed by the caller — the union is concrete and the call is **zero-allocation with a concretely inferred result**. When the result's type *is* the default (`with_default_type((T, x) -> T(x), ...)`), the value is computed on the specialized path but boxes once at escape — the irreducible cost of a runtime-chosen type.

```
julia> with_default_type((T, x) -> T(x), 1.5)  
Binary8p2se(1.5 ≡ 0x41)  
  
julia> with_default_projection((p, x, y) -> Add(Binary8p4se, p, x, y),  
                               Binary8p4se(1.5), Binary8p4se(0.25))  
Binary8p4se(1.75 ≡ 0x46)
```

SmallFloats.with_default_returntype - Function.

```
with_default_type(f, args...)      -> f(DefaultType(), args...)
with_default_returntype(f, args...) -> f(DefaultReturnType(), args...)
with_default_projection(f, args...) -> f(DefaultProjection(), args...)
```

Call `f` with the named session default as its first argument — the supported way to *consume* a default. While the default still holds its initial value (the overwhelmingly common case) the call is statically compiled against that constant: no dynamic dispatch, `f` fully specialized. After the default is changed, the call crosses a function barrier instead: one dynamic dispatch, everything inside fully specialized.

Allocation contract: the combinator's return type unions both paths. When `f`'s result type does not depend on the default — e.g. `with_default_projection` with the formats fixed by the caller — the union is concrete and the call is **zero-allocation with a concretely inferred result**. When the result's type *is* the default (`with_default_type((T, x) -> T(x), ...)`), the value is computed on the specialized path but boxes once at escape — the irreducible cost of a runtime-chosen type.

```
julia> with_default_type((T, x) -> T(x), 1.5)
Binary8p2se(1.5 ≡ 0x41)

julia> with_default_projection((p, x, y) -> Add(Binary8p4se, p, x, y),
                               Binary8p4se(1.5), Binary8p4se(0.25))
Binary8p4se(1.75 ≡ 0x46)
```

SmallFloats.with_default_type - Method.

```
with_default_type(f, args...)      -> f(DefaultType(), args...)
with_default_returntype(f, args...) -> f(DefaultReturnType(), args...)
with_default_projection(f, args...) -> f(DefaultProjection(), args...)
```

Call `f` with the named session default as its first argument — the supported way to *consume* a default. While the default still holds its initial value (the overwhelmingly common case) the call is statically compiled against that constant: no dynamic dispatch, `f` fully specialized. After the default is changed, the call crosses a function barrier instead: one dynamic dispatch, everything inside fully specialized.

Allocation contract: the combinator's return type unions both paths. When `f`'s result type does not depend on the default — e.g. `with_default_projection` with the formats fixed by the caller — the union is concrete and the call is **zero-allocation with a concretely inferred result**. When the result's type *is* the default (`with_default_type((T, x) -> T(x), ...)`), the value is computed on the specialized path but boxes once at escape — the irreducible cost of a runtime-chosen type.

```
julia> with_default_type((T, x) -> T(x), 1.5)
Binary8p2se(1.5 ≡ 0x41)

julia> with_default_projection((p, x, y) -> Add(Binary8p4se, p, x, y),
                               Binary8p4se(1.5), Binary8p4se(0.25))
Binary8p4se(1.75 ≡ 0x46)
```

Chapter 12

Internal Reference

Unexported machinery documented in-source and referenced by the [Technical Guide](#) and [Technical Examples](#). Public exported names are in the [External Reference](#).

`SmallFloats.DEFAULT_RBITS` – Constant.

Default random-bit budget used by no-argument stochastic projection constructors.

`SmallFloats.ExactExternalFloat` – Type.

External float element types whose widening to the Float64 carrier is exact.

`SmallFloats.KMAX` – Constant.

Highest bitwidth this package implements.

`SmallFloats.KMIN` – Constant.

Lowest bitwidth the draft defines ($K = 3$: sign, one exponent bit, one significand bit).

`SmallFloats.KSPLIT` – Constant.

Largest K whose code point fits a UInt8 — the Code8/Code16 seam.

`SmallFloats.MaybeR` – Type.

An explicit stochastic draw R , or nothing to draw one from the rng.

`SmallFloats.TABLE_EAGER_BITS` – Constant.

Largest table the package will *build*, as $\log_2(\text{entries})$. $16 = 65\,536$ entries.

A bound on **time**, not memory: a table costs one scalar trip per entry, at a measured $\sim 1\,\mu\text{s}/\text{entry}$, so this band caps first-call latency at $\sim 0.1\,\text{s}$ while 2^{24} entries would be tens of seconds — regardless of how comfortably 32 MiB fits in RAM.

16 is not a guess — it is exactly the largest table the $K \leq 8$ grid already builds (a binary 8×8 table, 65 536 entries), so **no $K \leq 8$ decision changes** and G5 stays byte-identical by construction. It also admits the case that matters most at wide K : a unary table for a $K = 16$ format is 2^{16} entries, the same count, now 128 KiB instead of 64 KiB because the entries are UInt16.

`SmallFloats.TABLE_MAX_BITS` – Constant.

Hard ceiling on a single table, as $\log_2(\text{bytes})$ — the byte budget invariant 10 requires. $25 = 32\,\text{MiB}$.

Compared **as bits**. Computing $1 \ll \Sigma K$ in order to decide whether $1 \ll \Sigma K$ is too large is exactly the bug the budget exists to prevent: at $\Sigma K = 64$ the shift wraps to zero and the impossible allocation looks free.

`SmallFloats.TERNARY_ADAPTIVE_BITS` - Constant.

$K_1+K_2+K_3$ up to which a ternary table may be built adaptively once the signature has processed `TERNARY_BUILD_ELEMS[]` elements (default 21 bits = 2 MiB — the $K=7$ band). Above this, ternary ops always run the compute kernel.

`SmallFloats.TERNARY_BUILD_ELEMS` - Constant.

Cumulative element count at which an adaptive-band signature earns its table.

`SmallFloats.TERNARY_CACHE_BYTES` - Constant.

Byte budget for the ternary cache; least-recently-used tables evict first.

`SmallFloats.TERNARY_EAGER_BITS` - Constant.

$K_1+K_2+K_3$ up to which a ternary table is built eagerly on first array call (default 18 bits = 256 KiB — covers every all- $K \leq 6$ signature).

`SmallFloats.THREADED_KERNELS` - Constant.

Master switch for threaded ternary compute loops.

`SmallFloats.THREAD_MIN_ELEMS` - Constant.

Minimum element count before the pure-p ternary compute loop threads.

`Core.Type` - Method.

```
(T::Type{<:Binary})(c::Unsigned) -> T
```

Construct a value from its **code point** (validated: *c* must satisfy the representation invariant for the format's *K*, or an `ArgumentError` is thrown; the check folds away when *K* equals the storage width and for constant codes). `Binary5p3sf(0x08) == Binary5p3sf(1.0)`.

`Unsigned` is the argument-type *class* with code-point semantics, at every width — every other `Real` (including signed `Integers`) constructs by projecting the numeric value. `Binary8p4se(0x02)` is code point 2 (the second-smallest subnormal), while `Binary8p4se(2)` is the number 2.0. Widening an `Unsigned` is lossless, so the only way a wrong-width argument can be wrong is by being out of code range — which throws. `rawvalue(T, c)` remains the unchecked kernel-internal route.

`SmallFloats.BigExactF` - Type.

Deferred exact result: `f()` returns the value as an exact `BigFloat`.

`SmallFloats.Code16` - Type.

```
Code16{K,P,SGN,EXT} <: Binary{K,P,SGN,EXT}
```

The two-byte representation — every format with $KSPLIT < K \leq KMAX$. Differs from its `Code8` sibling only in the width of the word carrying the code point; all semantics live in the `Binary` parameters.

`SmallFloats.Code8` - Type.

```
Code8{K,P,SGN,EXT} <: Binary{K,P,SGN,EXT}
```

The one-byte representation of a `P3109` format — every format with $K \leq 8$. Differs from its `Code16` sibling only in the width of the word carrying the code point; all semantics live in the `Binary` parameters.

`SmallFloats.ComputeDecode` – Type.

$2^K > 256$: compute from the code point; a 65 536-element tuple literal is not a table, it is a compile-time hazard (invariant 10).

`SmallFloats.DecodePolicy` – Type.

Decode strategy for a format: a 2^K -entry constant tuple, or computed.

`SmallFloats.Enclose128F` – Type.

Correctly-rounded Float128 bracket (plan Site C, Class R): IEEE mandates correct rounding for Float128 /, sqrt, fma, so a nearest-CR result of a value known inexact brackets the truth in the *open* interval $(lo, hi) = (prevfloat(q), nextfloat(q))$ with no envelope assumption. f is the MPFR fallback for (near-impossible at $P \leq 8$) grid-straddling brackets.

`SmallFloats.EncloseF` – Type.

Deferred enclosure, resolved by `_finish` through up to three stages, cheapest first:

1. yd — an *eager* Float64 estimate ($NaN \Rightarrow$ absent) whose libm evaluation is faithful (≤ 1 ulp); the true value is taken to lie within $|yd| \cdot 2^{-45}$ (`_F64_RELEXP`), $\geq 2^7$ slacker than any faithful-libm error.
2. fq — a zero-argument Float128 estimator (plan Site D, Class E; nothing $\Rightarrow$ absent): true value within $|y| \cdot 2^{-90}$ of $y = fq()$ (`_F128_RELEXP`), a bound $\geq 2^{18}$ slacker than any published libquadmath transcendental error, discharged empirically by the differential-build tests.
3. $f(prec)$ — the rigorous MPFR ladder: directed $(lo, hi)::NTuple\{2, BigFloat\}$ strictly bracketing the true value.

Each estimate stage returns only when the two-sided sticky projection of its envelope agrees on a single code point; any disagreement (or an absent/degenerate estimate) falls through to the next stage. Stage 3 always decides.

`SmallFloats.Head` – Type.

Head

Evaluation-carrier selector: a singleton tag, so carrier choice is a method lookup rather than a value the compiler must fold. Rungs are ordered `HeadF64 < HeadF128 < HeadExact` by carrier width.

`SmallFloats.HeadExact` – Type.

Rung 3 — unbounded exponent. `BigFloat` until Stage 7, Dyadic after; the tag is the seam that makes that swap a one-line change (gate G7 pins that the two agree).

`SmallFloats.HeadF128` – Type.

Rung 2 — Float128 arithmetic (the existing escalation machinery, promoted to entry point). Allocation-free: Float128 is a bitstype.

`SmallFloats.HeadF64` – Type.

Rung 1 — Float64 arithmetic. Every $K \leq 8$ format and 432 of the 504.

`SmallFloats.Rounded` – Type.

Result of `ωRoundToPrecision` in canonical integer form: a kind tag plus $sign \cdot S \cdot 2^Q$ for finite results (design §5.2). The saturation stage consumes this without ever re-touching the carrier value.

`SmallFloats.StickyF` - Type.

Sticky-head exact result (wide-spread tail, Float128 revision follow-on): the true value is $v + \text{sgn} \cdot \varepsilon$ for an infinitesimal $\varepsilon > 0$ — v carries every bit the projection can consume and $\text{sgn} \in \{-1, +1\}$ the direction of the neglected tail. Sound because v is either exactly on a rounding threshold of the target grid (where sticky decides, for every mode including stochastic sub-grids) or strictly farther from the nearest threshold than the tail magnitude; the emitting sites in `oracle.jl` discharge that bound (operand significands ≤ 17 bits, target grids $\leq P-1+N \leq 67$ fractional bits, spreads $>$ the DE^* thresholds). Non-allocating: replaces the BigFloat escalation for FMA/FAA.

`SmallFloats.TableDecode` - Type.

$2^K \leq 256$: the generated constant tuple folds and costs one load.

`SmallFloats.TableKey` - Type.

Value key for the table cache: `op` + format parameters + `p` parameters (all values, so the cache itself is type-stable and non-allocating to query).

`SmallFloats.TernaryKey` - Type.

Value key for a ternary table: `op` + result + three operand formats + `p`.

`SmallFloats._bp_element` - Method.

One element of `ωBlockProject`: the draft's S-special rows, then `ωDivide` + `ωProject`.

Fast-path cascade after the special rows, ordered cheapest-first by result kind: exact Float64 quotient → CR Float128 quotient (exact or one-ulp bracket) → CR-divided Enclose128F bracket → fq-composition envelope (2^{89}) — each resolved by the two-sided sticky gate; any miss falls through to the rigorous MPFR interval at the bottom.

`SmallFloats._check_tabulable` - Method.

Stochastic `p` is a distribution over `R`, never a table (design §5.4).

`SmallFloats._comparable` - Method.

Numeric comparison is defined only when neither operand is NaN — every `==/</>=` below returns false otherwise, per IEEE unorderedness.

`SmallFloats._extexprange` - Method.

(least subnormal exponent, greatest normal exponent) of an external float.

`SmallFloats._resolve_rng` - Method.

Resolve a caller-supplied `rng`, falling back to the task-local default. Call sites hoist this out of loops so array kernels resolve once per call, not per element.

`SmallFloats._rng_for` - Method.

The `rng` an operation under `p` will actually draw from — nothing for pure `p`, which must never touch RNG state.

`SmallFloats._rungleindex` - Method.

Per-format `rung` index from the exponent bias. A pure `Int` function of the type parameters, so `Val(_rungleindex(F))` is a compile-time constant.

Stated on `2B` rather than `B` so it is visibly the $n = 2$ case of `_rungleindex_span` and not an independent constant pair. $B \leq 512 \iff 2B \leq 1024$ and $B \leq 8192 \iff 2B \leq 16384$, so the 432/64/8 partition is unchanged — G9 enumerates it.

SmallFloats._rungindex_span - Method.

Rung index for a monomial whose factor biases sum to ΣB . This is the primitive form of the boundary rule stated above; the per-format answer is its $n = 2$ same-format case.

SmallFloats._ternary_table_for - Method.

```
_ternary_table_for(op, fr, f1, f2, f3, p, nelems) -> Union{Nothing, Memory{UInt8}}
```

The kernel-facing policy gate, called once per array operation with the call's element count. Eager band ($\sum K_i \leq \text{TERNARY_EAGER_BITS}[]$): fetch/build now. Adaptive band ($\leq \text{TERNARY_ADAPTIVE_BITS}[]$): return the cached table if present, otherwise accumulate nelems against the signature and build only once it has earned `TERNARY_BUILD_ELEMS[]`. Beyond the adaptive band (the $K=8$ territory): always nothing — the compute kernel is the right tradeoff there.

SmallFloats._unitmask - Method.

```
_unitmask(U, K) -> U
```

The low- K -bits mask of storage unit U , built by **complement** — shift down from `typemax`, never up from one.

Julia defines a shift at or beyond a type's width as zero, so `UInt8(1) << 8` and `UInt16(1) << 16` are both 0, and 2^K computed in a unit of width K is 0. The format grid **always contains a format whose K equals its storage width** ($K = 8$ on `UInt8`, $K = 16$ on `UInt16`), so that case is structural, not an edge. Here the shift amount is $\text{width} - K \in [0, \text{width}-1]$, in which the pathological amount is unreachable *by construction*, at every K , for every unit — including a future `Code32`.

SmallFloats._unsupported - Method.

Refuse a Base verb with a reason. Absence and refusal must not look alike.

SmallFloats.apply_op - Method.

`apply_op(Val(op), fr, p, R, xs...)` — evaluate op 's w -semantics on decoded operands and project the result into fr under p (with random bits R).

The evaluation carrier comes from the **operands**, not from fr . That is `carriers.jl`'s stated rule — the carrier constrains what `weval` may form, never what may be projected into a format, because `round_to_precision` works in (P, B) integer space and is exact for any exact input. Stage 3 selected on `rung(fr)` instead, as a deliberate stand-in while decode still refused $K \geq 9$; keeping that would have refused `Add(Binary16p2se, p, x8, y8)`, whose operands are ordinary `Float64` datums and whose exact sum needs no wide carrier at all.

Completeness of the operand-only rule rests on one premise, stated because Stage 3's version of it is what M28 was: the spec register's maximum factor count is **2**, and the `rung` boundaries are stated on $2B$ precisely so that a datum on carrier C guarantees its square is also on C . Beyond two factors — `ScaledMultiply`'s four — the monomial bound stops following from the operand types and `rung(op, Fs...)` must be used with the formats in hand. That is why `blocks.jl` calls the join and this does not.

SmallFloats.bigprec_prod - Method.

```
bigprec_prod(xs...) -> Int
```

`bigprec` for an expression whose leading term is the **product** $x_1 \cdot x_2$ and whose remaining terms are added to it — the FMA shape. The product's exponent is $e_1 + e_2$, which lies outside the range of the operands' own exponents, so the sum method would understate the spread. Its significand is $P_1 + P_2$ bits wide.

`SmallFloats.blockdecode` – Method.

ω BlockDecode (draft §5.1.1): per lane ω Multiply(`decode(s)`, `decode(xi)`), exact on the carrier the block's two formats jointly require.

The carrier is keyed on `rung(Val(:Multiply), FS, FE)` — **not** on either format's own rung, and that distinction is the whole content of this method. A block's scale and element formats are independent, and what has to fit is their *product*: `carriers.jl`'s rule is $\Sigma B_i \leq e_{\max}$, so two formats that are each rung 1 at $2B \leq 1024$ can compose into a lane product needing $B_S + B_E$ well past it. Asserting `carriertype(rung(FE))` would be wrong in the worst way available here — the overflow produces $\pm\text{Inf}$, a plausible special value, rather than an error.

The assertion stays because it is load-bearing for more than documentation: it is what keeps the returned `NTuple` concretely typed, and with it the reductions downstream allocation-free. It now names a carrier computed from both formats instead of a constant.

`SmallFloats.blockproject` – Method.

ω BlockProject (draft §5.1.2) over a tuple of result kinds, scale supplied as a value.

`SmallFloats.codemask` – Method.

The low-K-bits mask of a format's storage unit (representation invariant 3): the code point occupies the low K bits, the high bits are maintained zero.

`SmallFloats.datumcarrier` – Method.

The exact carrier for the *datums* of F: what `decode(v::F)` returns.

`SmallFloats.datumsexact` – Method.

```
datumsexact(X, F) -> Bool
```

true when **every** datum of format F is exactly representable in the external float type X.

Three conditions, one per way a conversion can lose information:

- $P \leq \text{precision}(X)$ — the significand fits;
- the largest finite datum's exponent is within X's normal range. That exponent is $B - 1$ for a signed format and 0 for an unsigned one, which is not symmetry-breaking pedantry: an unsigned format spends its whole exponent field below 1, so `Binary16p1uf`'s datums run down to 2^{-32769} while its largest is 0.5;
- $2 - P - B \geq X$'s least subnormal exponent — and the smallest datum being a *subnormal* of X is fine, because a datum is $\text{sig} \cdot 2^{(2-P-B)}$ with integer `sig`, so once the step is a multiple of X's subnormal step every datum lands on X's grid exactly.

`SmallFloats.decodepolicy` – Method.

Decode strategy: a $2^K \leq 256$ constant tuple, or computed from the code.

`SmallFloats.encode` – Method.

```
encode(T, sign, S, Q) -> UInt8 (private; design §3.3)
```

ω Encode from canonical integer form: `value = sign · S · 2Q`, with $S \in 0:2^P$ ($S == 2^P$ is the next-binade carry, draft §4.7.4 NOTE 4). Precondition: the value is in the datum set of T (guaranteed by `RoundToPrecision ◦ Saturate`).

SmallFloats.get_table - Function.

```
get_table(op, fr, f1, [f2,] p) -> Memory{codeunit_type(fr)}
```

Fetch (building and caching on first use) the complete result table for the pure- p specialization $op(fr; f1[, f2])$ under p : entry $c + 1$ (unary) or $(c1 \ll K2) + c2 + 1$ (binary) holds the result code point for those operand code points. Throws for stochastic p , and throws when the table would exceed the byte budget — its contract is to return a table, so it says so rather than returning nothing. Callers that can fall back want `table_for`.

@noinline by design: kernels call this once per array operation and index the returned table in their hot loop.

SmallFloats.get_table - Method.

```
get_table(op, fr, f1, f2, f3, p) -> Memory{UInt8}
```

Ternary form: entry $((c1 \ll K2 \mid c2) \ll K3) + c3 + 1$ holds the result code point. Builds unconditionally (policy lives in `_ternary_table_for`); pure p only.

SmallFloats.joinhead - Method.

The wider of two heads. The carrier join is a max over this order, which is what lets block and scaled operations compose without a second rule.

SmallFloats.nan_order_key - Method.

```
nan_order_key(F)
```

The order key reserved for the single NaN — above every finite key and $\pm\text{Inf}$.

Per format, not a constant, and the key type is `orderkeytype(F)` rather than `UInt16`. The reason is arithmetic, not tidiness: a format has 2^K code points mapping to keys $1 \dots 2^K$, so `UInt16` runs out at exactly $K = 16$ — `UInt16(c) + UInt16(1)` for $c = 0xffff$ wraps **silently to 0**, putting the largest datum below the smallest. `orderkeytype` returns `UInt32` for `Code16`, which has room for the keys and for a sentinel strictly above all of them.

(§4 Stage 4 item 3, pulled forward into Stage 3 — §11 M16. The rest of Stage 3 makes wide formats fail loudly; a wraparound in the total order is the one $K \geq 9$ defect that would have been silent, so it is closed here rather than carried for a commit.)

SmallFloats.orderkeytype - Method.

Unsigned type wide enough for a format's $2^K + 1$ total-order keys.

SmallFloats.packing_saves – Method.

```
packing_saves(F) -> Bool
```

Whether PackedVector{F} is smaller than Vector{F} — false exactly when bitwidth(F) equals the storage unit's width, i.e. $K = 8$ and $K = 16$.

Why this is a query and not a refusal. §4 Stage 4 item 5 called for PackedVector to refuse loudly at $K = 16$, on the grounds that packing is the identity there and a silent identity invites benchmark confusion. The grounds are right and the boundary is not: packing is *equally* the identity at $K = 8$, which has shipped since before the extension and is enumerated by G5's packed section over all 120 formats. A rule that fires on one of two identical cases is not a rule, and the alternative — refusing both — would be a breaking change justified by tidiness.

So the fact is exposed instead of enforced. Benchmarks and the docs read it; PackedVector at those widths stays correct, stays supported, and stays pointless. (§11 M21.)

SmallFloats.project – Method.

```
project(T, p, X; R=0, sticky=0) -> T
```

Project a closed-extended-real carrier value into format T under p. $X::\text{Float64}$ must be exact (the Float64 fast path); $X::\text{BigFloat}$ likewise (or an interval endpoint with $\text{sticky} \in \{-1, +1\}$). R is the stochastic draw.

SmallFloats.promotecarrier – Method.

The public promotion target for F against external numeric types. Always a Julia AbstractFloat, never the internal exact carrier.

SmallFloats.rung – Method.

```
rung(op::Val, Fs::Type{<:Binary}...) -> Head
```

The carrier for evaluating op over operands of formats Fs. A **join of two lower bounds**, and both are necessary:

- the **monomial bound** — op's exact result is a sum of monomials in at most $\text{opfactors}(\text{op})$ datum factors, so it needs $\text{_runglex_span}(\text{nf} \cdot \text{maxB})$ ($\text{nf} \cdot \text{maxB}$ rather than ΣB_i over the actual operands: it dominates every monomial shape, including x^2 from Hypot, which ΣB_i does not, and it coincides with the tight bound in the same-format case that is virtually all traffic);
- the **operand bound** — $\text{decode}(v::F_i)$ has already produced a value on $\text{datumcarrier}(F_i)$, so the head can never be narrower than any operand's own rung regardless of what the monomial needs.

Omitting the second is the subtle error. Abs has one factor, so the monomial bound alone would pick rung 1 for a $B = 1024$ format — whose datums arrive as Float128, for which $\text{weval}::\text{HeadF64}, \dots$ has no row. The join makes lift's missing narrowing method a consequence rather than a hazard: the head dominates every operand, so nothing ever needs narrowing.

For the same format at two factors this agrees with $\text{rung}(F)$ exactly, which is what keeps the 432/64/8 partition and every $K \leq 8$ result unchanged.

SmallFloats.rung – Method.

```
rung(F) -> Head
```

The starting carrier for evaluating on datums of format F alone. For an operation over several formats use $\text{rung}(\text{op}, \text{Fs} \dots)$, which additionally accounts for the operation's factor count — a ScaledMultiply of two Group A formats can need a wider carrier than either operand does.

`SmallFloats.rungindex` – Method.

Rung index 1/2/3 of a head — for ordering and for the carrier join.

`SmallFloats.table_for` – Method.

```
table_for(op, fr, f1[, f2], p) -> Union{Nothing,Memory{codeunit_type(fr)}}
```

The kernel-facing gate for unary and binary specializations: the table if policy grants one, nothing if the caller should run the scalar path per element.

Separate from `get_table` on purpose — **one name per question**. `get_table`'s contract is "return the table", so it throws when it cannot honour that, and that is what the suite, conformance and the precompile workload want. `table_for` asks the different question "should there be a table at all", and answers nothing without prejudice. Its Union return costs nothing because kernels call it once per array operation and branch outside the loop.

`SmallFloats.table_keys` – Method.

Keys of every cached unary/binary specialization, in a deterministic order.

`SmallFloats.tablebits` – Method.

$\log_2$ of the bytes a table over Fs with results in fr would occupy.

`SmallFloats.tablecache` – Method.

The unary/binary table cache holding results of format fr.

`SmallFloats.ternarycache` – Method.

The ternary table cache holding results of format fr.

`SmallFloats.unpack_tile!` – Method.

Unpack `pv[first:first+len-1]` into `buf` (byte scratch); tile-granular kernel entry.

`SmallFloats.within_byte_budget` – Method.

Would this table fit the byte budget? (Memory, not build time.)